MT5864

Advanced Group Theory

MRQ

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University of St Andrews

Contents

Introduction

Groups are important mathematical objects because they encode the symmetry that might be found in all mathematical structures. The purpose of this lecture course is to take their study further beyond what was covered in *MT4003 Groups*. We will build on that prerequisite to cover ideas and material that are used throughout the advanced study of group theory. The techniques developed in this course are used to handle and classify groups. They are employed throughout research into these algebraic structures.

The structure of these notes is as follows:

Chapter 1 – Revision: We review basic concepts that were introduced in *MT2505 Abstract Algebra* and *MT4003 Groups*: subgroup, normal subgroup, quotient group and homomorphism, and give a statement of Sylow's Theorem. (A few small, but probably new, results will be proved that are used regularly in this module.)

Chapter 2 – Group Actions: We expand upon the material on group actions from *MT4003*. We will explain how using a group to induce permutations of a set enables us to establish structural properties about the group itself.

Chapter 3 – Series: Suppose that a group G has a collection of subgroups

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1}$$

where G_i is a normal subgroup of G_{i-1} for each $i \in \{1, 2, \dots, n\}$ and where the subgroups are arranged in the chain as shown above. The idea here is that information about the quotient groups G_0/G_1, G_1/G_2, …, G_{n-1}/G_n will yield information about the group G. One specific example of such a series is the *composition series*, where the quotients G_{i-1}/G_i are simple groups. We shall prove the Jordan–Hölder Theorem which shows that the simple factors in a composition series are uniquely determined by the group G.

Chapter 4 – Semidirect Products: We discuss how groups may be constructed and in particular some ways in which the above decomposition into a series may be reversed. We present the semidirect product (which can be viewed as a generalization of the direct product encountered in *MT4003*) and show how this can be used to classify some finite groups of particular orders.

Chapters 5 & 6 – Soluble and Nilpotent Groups: In the next two chapters, we consider two classes of groups that are defined via specific types of series. In both cases the factors in the series are abelian and so they can be viewed as having a fairly restricted structure. It is this that makes them more tractible for study. We shall show that a finite soluble group has *Hall subgroups*, which are generalizations of Sylow subgroups. Every p-group is nilpotent and we shall observe that they are the prototypical example of nilpotent groups.

Chapter 7 – Primitive Permutation Groups: In this chapter we return to the theme of group actions and consider the structure of permutation groups (that is, subgroups of symmetric groups). Cayley's Theorem was proved in *MT4003* using group actions: every group isomorphic to a permutation group. What this means is that the answer to the question "What groups occur as subgroups of symmetric groups?" is "All groups!" However, we shall observe that much more can be said if we exploit the nature of the action. In particular, we shall observe that primitive permutation groups can be viewed as building blocks from which (finite) permutation groups are constructed and we shall describe the structure of primitive permutation groups in the O'Nan–Scott Theorem at the end of the course. This is an important theorem that is used throughout current research in finite group theory.

Chapter 1

Review/Revision

In this first section we shall primarily recall definitions and results from earlier lecture courses. During the lectures, proofs of results that have been previously met will often be omitted (though these notes will contain many of them). We delay recalling examples of groups until the early part of Chapter 2 since the most natural way that groups arise is via their actions.

Definition 1.1 A *group* G is a set with a binary operation (usually written multiplicatively) such that

(i) the binary operation is *associative*: $x(yz) = (xy)z$ for all $x, y, z \in G$;

(ii) there is an *identity element* 1 in G: $x1 = 1x = x$ for all $x \in G$;

(iii) each element x in G possesses an *inverse* x^{-1}: $xx^{-1} = x^{-1}x = 1$.

Comments:

(i) This definition makes no explicit reference to 'closure' as an axiom. The reason for this is that this condition is actually built into the definition of a binary operation. A binary operation takes two elements of our group and produces an element *in the group*, and so closure is automatically achieved.

(ii) Associativity ensures that any product $x_1 x_2 \dots x_n$ of n elements from a group is uniquely determined irrespective of how it is bracketed. As a consequence, we shall typically omit brackets from such products.

(iii) It was observed in *MT2505* that the identity element in a group is unique and that each element has a unique inverse. Consequently, we shall usually speak of *the identity* in a group G and *the inverse* of an element $x \in G$.

(iv) We define powers x^n where $x \in G$ and $n \in \mathbb{Z}$ by $x^0 = 1$,

$$x^n = \underbrace{xx \dots x}_{n \text{ times}}$$

and $x^{-n} = (x^{-1})^n$ for every positive integer $n \geqslant 1$. The standard power laws hold in group where the binary operation is multiplicatively written:

$$x^{m+n} = x^m x^n \qquad \text{and} \qquad x^{mn} = (x^m)^n$$

for $x \in G$ and $m, n \in \mathbb{Z}$. In general, group elements do not commute so, for example, we cannot easily expand $(xy)^n$. We can, however, expand the following inverse:

$$(xy)^{-1} = y^{-1}x^{-1}.$$

PROOF (OMITTED IN LECTURES):

$$(y^{-1}x^{-1})(xy) = y^{-1}x^{-1}xy = y^{-1}1y = y^{-1}y = 1,$$

so multiplying on the right by the inverse of xy yields $y^{-1}x^{-1} = (xy)^{-1}$. $\square$

For completeness, let us record the term used for groups where all the elements present do commute:

Definition 1.2 A group G is called *abelian* if all its elements *commute*; that is, if

$$xy = yx \qquad \text{for all } x, y \in G.$$

Subgroups

Although one is initially tempted to attack groups by examining their elements in detail, this turns out not to be terribly fruitful. Even an only moderately sized group is unyielding to consideration of its multiplication table. Instead one needs to find some sort of "structure" to study and this is provided by subgroups and homomorphisms (and, particularly related to the latter, quotient groups).

A subgroup of a group is a subset which is itself a group under the multiplication inherited from the larger group. Thus:

Definition 1.3 A subset H of a group G is a *subgroup* of G if

(i) H is non-empty,

(ii) $xy \in H$ and $x^{-1} \in H$ for all $x, y \in H$.

We write $H \leqslant G$ to indicate that H is a subgroup of G. If G is a group, the set containing the identity element (which we shall denote by $\mathbf{1}$) and the whole group G are always subgroups. We shall usually be interested in finding other subgroups of a group. (We mention in passing that the above conditions for a subset to be a subgroup are not the only ones that can be used, but they are sufficient for our needs and easily memorable.)

The identity element of G lies in every subgroup, so it is easy to see that the conditions of Definition 1.3 are inherited by intersections. Therefore:

Lemma 1.4 *If $\{ H_i \mid i \in I \}$ is a collection of subgroups of a group G, then $\bigcap_{i \in I} H_i$ is also a subgroup of G.*

PROOF (OMITTED IN LECTURES): Since $1 \in H_i$ for all i, it follows that $\bigcap_{i \in I} H_i \neq \varnothing$. Now let $x, y \in \bigcap_{i \in I} H_i$. Then for each i, $x, y \in H_i$, so $xy \in H_i$ and $x^{-1} \in H_i$ since $H_i \leqslant G$. We deduce that $xy \in \bigcap_{i \in I} H_i$ and $x^{-1} \in \bigcap_{i \in I} H_i$. Thus the intersection is a subgroup. $\square$

In general, the union of a family of subgroups of a group is not itself a subgroup. This is not a disaster, however, as the following construction provides a way around this.

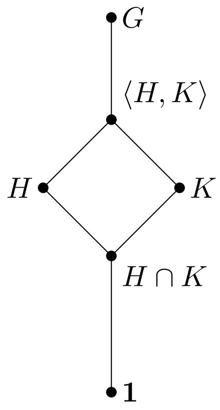


Figure 1.1: The arrangement of two subgroups H and K, their intersection and the subgroup $\langle H, K \rangle$ that they generate in the subgroup diagram.

Definition 1.5 Let G be a group and X be a subset of G. The *subgroup of G generated by X* is denoted by $\langle X \rangle$ and is defined to be the intersection of all subgroups of G which contain X:

$$\langle X \rangle = \bigcap \{ H \mid X \subseteq H \leqslant G \}$$

Lemma 1.4 ensures that $\langle X \rangle$ is a subgroup of G. It is the smallest subgroup of G containing X (in the sense that it is contained in all other such subgroups; that is, if H is any subgroup of G containing X then $\langle X \rangle \leqslant H$).

Lemma 1.6 *Let G be a group and X be a subset of G. Then*

$$\langle X \rangle = \{ x_1^{\varepsilon_1} x_2^{\varepsilon_2} \dots x_n^{\varepsilon_n} \mid n \geqslant 0, \;\; x_i \in X, \;\; \varepsilon_i = \pm 1 \text{ for all } i \}.$$

Thus $\langle X \rangle$ consists of all products of elements of X and their inverses.

Proof (Omitted in lectures): Let S denote the set on the right-hand side. Since $\langle X \rangle$ is a subgroup (by Lemma 1.4) and by definition it contains X, we deduce that $\langle X \rangle$ must contain all products of elements of X and their inverses. Thus $S \subseteq \langle X \rangle$.

On the other hand, S is non-empty (for example, it contains the empty product (where $n = 0$) which by convention is taken to be the identity element 1), it contains all elements of X (the case $n = 1$ and $\varepsilon_1 = 1$), is clearly closed under products and

$$(x_1^{\varepsilon_1} x_2^{\varepsilon_2} \dots x_n^{\varepsilon_n})^{-1} = x_n^{-\varepsilon_n} x_{n-1}^{-\varepsilon_{n-1}} \dots x_1^{-\varepsilon_1} \in S.$$

Hence S is a subgroup of G. The fact that $\langle X \rangle$ is the smallest subgroup containing X now gives $\langle X \rangle \leqslant S$ and we deduce the equality claimed in the lemma. $\square$

Suppose that H and K are subgroups of a group G. Then the intersection $H \cap K$ is the largest subgroup of G that is contained in both H and K. Our observations above tells us that the subgroup $\langle H, K \rangle$, generated by H and K, is the smallest subgroup of G that contains both H and K. We can represent the arrangement of these subgroups by the diagram in Figure 1.1.

We shall frequently wish to understand how the subgroups of a group relate to each other. Diagrams of the above sort are useful to describe this. We represent each subgroup by a node and use an upward (often sloping) edge to denote inclusion.

Cosets

Subgroups enforce a rigid structure on a group: specifically, a group is the disjoint union of the cosets of any particular subgroup. Accordingly we need the following definition.

Definition 1.7 Let G be a group, H be a subgroup of G and x be an element of G. The (*right*) *coset* of H with *representative* x is the following subset of G:

$$Hx = \{ hx \mid h \in H \}$$

We can equally well define what is meant by a left coset, but we shall work almost exclusively with right cosets. For the latter reason we shall simply use the term 'coset' to always mean 'right coset'.

Theorem 1.8 *Let G be a group and H be a subgroup of G.*

(i) *If $x, y \in G$, then $Hx = Hy$ if and only if $xy^{-1} \in H$.*

(ii) *Any two cosets of H are either equal or are disjoint: if $x, y \in G$, then either $Hx = Hy$ or $Hx \cap Hy = \varnothing$.*

(iii) *G is the disjoint union of the cosets of H.*

(iv) *If $x \in G$, the map $h \mapsto hx$ is a bijection from H to the coset Hx.*

Proof (Omitted in lectures): (i) Suppose $Hx = Hy$. Then $x = 1x \in Hx = Hy$, so $x = hy$ for some $h \in H$. Thus $xy^{-1} = h \in H$.

Conversely if $xy^{-1} \in H$, then $hx = h(xy^{-1})y \in Hy$ for all $h \in H$, so $Hx \subseteq Hy$. Also $hy = hyx^{-1}x = h(xy^{-1})^{-1}x \in Hx$ for all $h \in H$, so $Hy \subseteq Hx$. Thus $Hx = Hy$ under this assumption.

(ii) Suppose that $Hx \cap Hy \neq \varnothing$. Then there exists $z \in Hx \cap Hy$, say $z = hx = ky$ for some $h, k \in H$. Then $xy^{-1} = h^{-1}k \in H$ and we deduce $Hx = Hy$ by (i).

(iii) If $x \in G$, then $x = 1x \in Hx$. Hence the union of all the (right) cosets of H is the whole of G. Part (ii) ensures this is a disjoint union.

(iv) By definition of the coset Hx, the map $h \mapsto hx$ is a surjective map from H to Hx. Suppose $hx = kx$ for some $h, k \in H$. Then multiplying on the right by x^{-1} yields $h = k$. Thus this map is also injective, so it is a bijection, as claimed. □

Write $|G : H|$ for the number of cosets of H in G and call this the *index* of H in G. The previous result tells us that our group G is the disjoint union of $|G : H|$ cosets of H and each of these contain $|H|$ elements. Hence:

Theorem 1.9 (Lagrange's Theorem) *Let G be a group and H be a subgroup of G. Then*

$$|G| = |G : H| \cdot |H|.$$

In particular, if H is a subgroup of a finite group G, then the order of H divides the order of G. □

The following fact about indices of subgroups is frequently used:

Lemma 1.10 *Let H and K be subgroups of a group G with $K \leqslant H \leqslant G$. Then*

$$|G : K| = |G : H| \cdot |H : K|.$$

For finite groups, this is easily deduced from Lagrange's Theorem:

$$|G:K| = \frac{|G|}{|K|} = \frac{|G|}{|H|} \cdot \frac{|H|}{|K|} = |G:H| \cdot |H:K|.$$

The full proof for an arbitrary, possibly infinite, group is omitted. It appears on Problem Sheet I.

Orders of elements and cyclic groups

Definition 1.11 If G is a group and x is an element of G, we define the *order* of x to be the smallest positive integer n such that $x^n = 1$ (if such exists) and otherwise say that x has *infinite order*. We write $o(x)$ for the order of the element x.

If $x^i = x^j$ for $i < j$, then $x^{j-i} = 1$ and x has finite order and $o(x) \leqslant j - i$. In particular, the powers of x are always distinct if x has infinite order. If the element x has finite order n and $k \in \mathbb{Z}$, write $k = nq + r$ where $0 \leqslant r < n$. Then

$$x^k = x^{nq+r} = (x^n)^q x^r = x^r \tag{1.1}$$

(since $x^n = 1$). Furthermore $1, x, x^2, \dots, x^{n-1}$ are distinct (by the first line of the previous paragraph). Hence:

Proposition 1.12 (i) *If x is an element of a group of infinite order, then the powers x^i (for $i \in \mathbb{Z}$) are distinct.*

(ii) *If x is an element of a group of order n, then x has precisely n distinct powers, namely $1, x, x^2, \dots, x^{n-1}$.* □

Corollary 1.13 *Let G be a group and $x \in G$. Then*

$$o(x) = |\langle x \rangle|.$$

If G is a finite group, then $o(x)$ divides $|G|$. □

Equation (1.1) yields a further observation, namely:

$$x^k = 1 \qquad \text{if and only if} \qquad o(x) \mid k.$$

In the case that a single element generates the whole group, we give it a special name:

Definition 1.14 A group G is called *cyclic* (with *generator* x) if $G = \langle x \rangle$.

The following fact about the subgroups of finite cyclic groups was established in *MT4003*. The proof basically depends upon the ideas just described. The details are omitted here and found instead, together with the analogous result for the infinite cyclic groups, on Problem Sheet I.

Theorem 1.15 *Let G be a finite cyclic group of order n. Then G has precisely one subgroup of order d for every divisor d of n.*

Normal subgroups and quotient groups

Definition 1.16 A subgroup N of a group G is called a *normal subgroup* of G if $g^{-1}xg \in N$ for all $x \in N$ and all $g \in G$. We write $N \trianglelefteqslant G$ to indicate that N is a normal subgroup of G.

The element $g^{-1}xg$ is called the *conjugate* of x by g and is often denoted by x^g. We shall discuss this operation a little towards the end of this chapter and in more detail in Chapter 2.

If $N \trianglelefteqslant G$, then we write G/N for the set of cosets of N in G:

$$G/N = \{\, Nx \mid x \in G \,\}.$$

Theorem 1.17 *Let G be a group and N be a normal subgroup of G. Then*

$$G/N = \{\, Nx \mid x \in G \,\},$$

the set of cosets of N in G, is a group when we define the multiplication by

$$Nx \cdot Ny = Nxy$$

for $x, y \in G$.

Proof (Omitted in lectures): The part of this proof requiring the most work is to show that this product is actually well-defined. Suppose that $Nx = Nx'$ and $Ny = Ny'$ for some elements $x, x', y, y' \in G$. Then $x = ax'$ and $y = by'$ for some $a, b \in N$. Then

$$xy = (ax')(by') = ax'b(x')^{-1}x'y' = ab^{(x')^{-1}}x'y'.$$

Since $N \trianglelefteqslant G$, it follows that $b^{(x')^{-1}} \in N$. Hence $(xy)(x'y')^{-1} = ab^{(x')^{-1}} \in N$ and we deduce $Nxy = Nx'y'$. This shows that the above multiplication of cosets is indeed well-defined.

It remains to show that the set of cosets forms a group under this multiplication. If $x, y, z \in G$, then

$$(Nx \cdot Ny) \cdot Nz = Nxy \cdot Nz = N(xy)z = Nx(yz) = Nx \cdot Nyz = Nx \cdot (Ny \cdot Nz).$$

Thus the multiplication is associative. We calculate

$$Nx \cdot N1 = Nx1 = Nx = N1x = N1 \cdot Nx$$

for all cosets Nx, so $N1$ is the identity element in G/N, while

$$Nx \cdot Nx^{-1} = Nxx^{-1} = N1 = Nx^{-1}x = Nx^{-1} \cdot Nx,$$

so Nx^{-1} is the inverse of Nx in G/N.

Thus G/N is a group. □

Definition 1.18 If G is a group and N is a normal subgroup of G, we call G/N (with the above multiplication) the *quotient group* of G by N.

We shall consider quotient groups in more detail later in this section. They are best discussed, however, in the context of homomorphisms, so we shall move onto these in a short while. We just mention some results (one part of which is proved here, the rest appear on Problem Sheet I) which will be needed later.

Lemma 1.19 *Let G be a group and let H and K be subgroups of G. Define $HK = \{\, hk \mid h \in H,\ k \in K \,\}$. Then*

(i) *HK is a subgroup of G if and only if $HK = KH$;*

(ii) *if K is a normal subgroup of G then HK is a subgroup of G (and consequently $HK = KH$);*

(iii) *if H and K are normal subgroups of G, then $H \cap K$ and HK are normal subgroups of G;*

(iv) *$|HK| \cdot |H \cap K| = |H| \cdot |K|$.*

When H and K are finite, then we can rearrange the last formula to give

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}.$$

This formula holds even when HK is not a subgroup.

PROOF: (iv) Define a map $\alpha \colon H \times K \to HK$ by

$$(h, k) \mapsto hk.$$

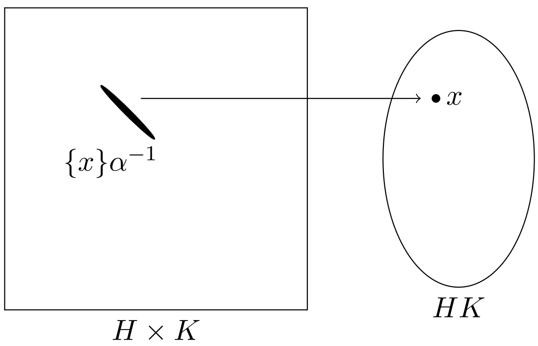
By definition, α is surjective. Fix a point $x \in HK$, say $x = h_0k_0$ for some fixed $h_0 \in H$ and $k_0 \in K$. Then for $(h, k) \in H \times K$,

$$\begin{aligned} (h,k)\alpha = x \qquad & \text{if and only if} \qquad hk = h_0k_0 \\ & \text{if and only if} \qquad h_0^{-1}h = k_0k^{-1} \in H \cap K \\ & \text{if and only if} \qquad h = h_0a,\ k = a^{-1}k_0 \quad \text{where } a \in H \cap K. \end{aligned}$$

Thus for each $x \in HK$, the set of elements mapped by α to x is

$$\{x\}\alpha^{-1} = \{\, (h,k) \in H \times K \mid (h,k)\alpha = x \,\} = \{\, (h_0a, a^{-1}k_0) \mid a \in H \cap K \,\}.$$

This set is in one-one correspondence with the set $H \cap K$ via the bijection $a \mapsto (h_0a, a^{-1}k_0)$. Hence we may partition $H \times K$ into $|HK|$ subsets, each corresponding to one point in HK and of size $|H \cap K|$.



This proves

$$|H \times K| = |HK| \cdot |H \cap K|;$$

that is,

$$|H| \cdot |K| = |HK| \cdot |H \cap K|. \qquad \square$$

Homomorphisms

Definition 1.20 Let G and H be groups. A *homomorphism* from G to H is a map $\phi \colon G \to H$ such that $(xy)\phi = (x\phi)(y\phi)$ for all $x, y \in G$.

Thus a homomorphism between two groups is a map which, in the sense of the above formula, "preserves" their multiplications. Note that I am writing my maps on the right, as is conventional in much of algebra. This has several advantages: the first is that when we compose a number of maps we can read from left to right, rather than from right to left. It will also be consistent with the notation that we use for group actions in Chapter 2 and it will make certain proofs more notationally convenient.

The following definition presents important subsets (actually subgroups) that are related to homomorphisms.

Definition 1.21 Let $\phi \colon G \to H$ be a homomorphism between two groups. Then the *kernel* of ϕ is

$$\ker \phi = \{ x \in G \mid x\phi = 1 \},$$

while the *image* of ϕ is

$$\operatorname{im} \phi = G\phi = \{ x\phi \mid x \in G \}.$$

Note here that $\ker \phi \subseteq G$ while $\operatorname{im} \phi \subseteq H$.

Lemma 1.22 *Let $\phi \colon G \to H$ be a homomorphism between two groups G and H. Then*

(i) $1\phi = 1$;

(ii) $(x^{-1})\phi = (x\phi)^{-1}$ *for all* $x \in G$;

(iii) *the kernel of ϕ is a normal subgroup of G;*

(iv) *the image of ϕ is a subgroup of H.*

Proof (Omitted in lectures): (i) $1\phi = (1 \cdot 1)\phi = (1\phi)(1\phi)$ and multiplying by the inverse of 1ϕ yields $1 = 1\phi$.

(ii) $(x\phi)(x^{-1}\phi) = (xx^{-1})\phi = 1\phi = 1$ and multiplying on the left by the inverse of $x\phi$ yields $(x^{-1})\phi = (x\phi)^{-1}$.

(iii) By (i), $1 \in \ker \phi$. If $x, y \in \ker \phi$, then $(xy)\phi = (x\phi)(y\phi) = 1 \cdot 1 = 1$ and $(x^{-1})\phi = (x\phi)^{-1} = 1^{-1} = 1$, so we deduce $xy \in \ker \phi$ and $x^{-1} \in \ker \phi$. Therefore $\ker \phi$ is a subgroup of G. Now if $x \in \ker \phi$ and $g \in G$, then $(g^{-1}xg)\phi = (g^{-1}\phi)(x\phi)(g\phi) = (g\phi)^{-1}1(g\phi) = 1$, so $g^{-1}xg \in \ker \phi$. Hence $\ker \phi$ is a normal subgroup of G.

(iv) Let $g, h \in \operatorname{im} \phi$. Then $g = x\phi$ and $h = y\phi$ for some $x, y \in G$. Then $gh = (x\phi)(y\phi) = (xy)\phi \in \operatorname{im} \phi$ and $g^{-1} = (x\phi)^{-1} = (x^{-1})\phi \in \operatorname{im} \phi$. Thus $\operatorname{im} \phi$ is a subgroup of G. □

The kernel is also useful for determining when a homomorphism is injective.

Lemma 1.23 *Let $\phi \colon G \to H$ be a homomorphism between two groups G and H. Then ϕ is injective if and only if* $\ker \phi = \mathbf{1}$.

Proof: Suppose ϕ is injective. If $x \in \ker \phi$, then $x\phi = 1 = 1\phi$, so $x = 1$ by injectivity. Hence $\ker \phi = \mathbf{1}$.

Conversely suppose that $\ker \phi = \mathbf{1}$. If $x\phi = y\phi$, then $(xy^{-1})\phi = (x\phi)(y\phi)^{-1} = 1$, so $xy^{-1} \in \ker \phi$. Hence $xy^{-1} = 1$ and, upon multiplying on the right by y, we deduce $x = y$. Hence ϕ is injective. □

Example 1.24 Let G be a group and N be a normal subgroup of G. Define a map $\pi \colon G \to G/N$ by

$$\pi \colon x \mapsto Nx.$$

The definition of the multiplication in the quotient group G/N ensures that π is a homomorphism. It is called the *natural map* (or *canonical homomorphism*). We see

$$\ker \pi = \{\, x \in G \mid Nx = N1 \,\};$$

that is, $\ker \pi = N$. By definition, $\operatorname{im} \pi = G/N$; that is, π is surjective.

Thus it is not just that every kernel is a normal subgroup, but also that every normal subgroup is the kernel of some homomorphism. Indeed, one can interpret this example together with the First Isomorphism Theorem (below) as saying that normal subgroups and homomorphisms in some sense correspond.

Isomorphism Theorems

We shall finish this section by discussing the four important theorems that relate quotient groups and homomorphisms. We shall need the concept of isomorphism, so we recall that first.

Definition 1.25 An *isomorphism* between two groups G and H is a homomorphism $\phi \colon G \to H$ which is a bijection. We write $G \cong H$ to indicate that there is an isomorphism between G and H, and we say that G and H are *isomorphic*.

What this means is that if G and H are isomorphic groups, then the elements of the two groups are in one-one correspondence in such a way that the group multiplications produce precisely corresponding elements. Thus essentially the groups concerned are identical: we may have given the groups different names and labelled the elements differently, but we are looking at identical objects in terms of their structure.

Theorem 1.26 (First Isomorphism Theorem) *Let G and H be groups and $\phi \colon G \to H$ be a homomorphism. Then $\ker \phi$ is a normal subgroup of G, $\operatorname{im} \phi$ is a subgroup of H and*

$$G/\ker \phi \cong \operatorname{im} \phi.$$

Proof (Sketch): We already know that $\ker \phi \trianglelefteq G$, so we can form $G/\ker \phi$. The isomorphism is the map

$$(\ker \phi)x \mapsto x\phi \qquad (\text{for } x \in G). \qquad \square$$

Omitted details: Let $K = \ker \phi$ and define $\theta \colon G/K \to \operatorname{im} \phi$ by $Kx \mapsto x\phi$ for $x \in G$. We note

$$\begin{aligned} Kx = Ky \quad &\text{if and only if} \quad xy^{-1} \in K \\ &\text{if and only if} \quad (xy^{-1})\phi = 1 \\ &\text{if and only if} \quad (x\phi)(y\phi)^{-1} = 1 \\ &\text{if and only if} \quad x\phi = y\phi. \end{aligned}$$

This shows that θ is well-defined and also that it is injective. By definition of the image, θ is surjective. Finally

$$\big((Kx)(Ky)\big)\theta = (Kxy)\theta = (xy)\phi = (x\phi)(y\phi) = (Kx)\theta \cdot (Ky)\theta$$

for all $x, y \in G$, so θ is a homomorphism. Hence θ is the required isomorphism. (All other parts of the theorem are found in Lemma 1.22.) $\square$

Rather than move straight on to the Second and Third Isomorphism Theorems, I shall deal with the Correspondence Theorem next so that I can use it when talking about the other Isomorphism Theorems. The Correspondence Theorem essentially tells us how to handle quotient groups, at least in terms of their subgroups, which is to some extent the principal way of handling them anyway.

Theorem 1.27 (Correspondence Theorem) *Let G be a group and let N be a normal subgroup of G.*

(i) *There is a one-one inclusion-preserving correspondence between subgroups of G containing N and subgroups of G/N given by*

$$H \mapsto H/N \qquad \textit{whenever } N \leqslant H \leqslant G.$$

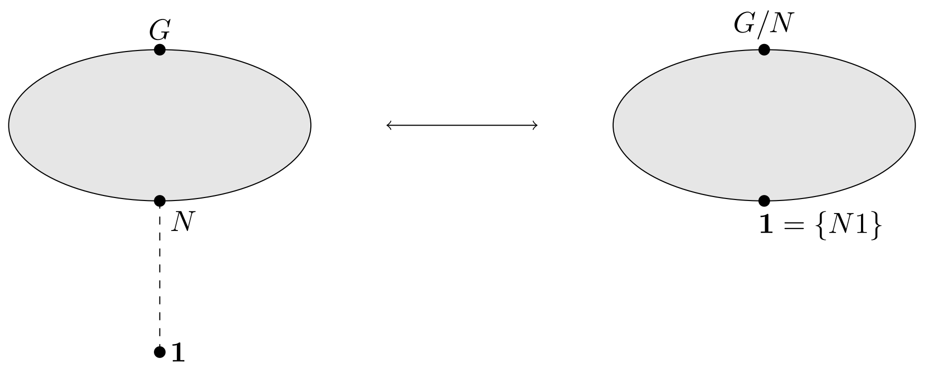
(ii) *Under the above correspondence, normal subgroups of G which contain N correspond to normal subgroups of G/N.*

The primary content of Part (i) of the Correspondence Theorem is that every subgroup of G/N can be uniquely expressed in the form H/N for some subgroup H of G that contains N. We shall frequently use the existence of this expression when trying to work with subgroups of the quotient group. Moreover, Part (ii) then tells us that this subgroup H/N is normal in G/N precisely when the corresponding subgroup H is normal in G; that is,

$$H \trianglelefteqslant G \qquad \text{if and only if} \qquad H/N \trianglelefteqslant G/N$$

for $N \leqslant H \leqslant G$.

If we view it that the 'structure' of a group is somehow the shape of the diagram of subgroups (with those 'special' subgroups which are normal indicated), then the Correspondence Theorem tells us how the structures of a group and a quotient are related. The diagram of subgroups of the quotient group G/N is simply that part of the diagram of subgroups sandwiched between G and N.



Proof (Omitted in lectures): Let $\mathscr{S}$ denote the set of subgroups of G that contain N (that is, $\mathscr{S} = \{ H \mid N \leqslant H \leqslant G \}$) and let $\mathscr{T}$ denote the set of subgroups of G/N. Let $\pi \colon G \to G/N$ denote the natural map $x \mapsto Nx$.

First note that if $H \in \mathscr{S}$, then N is certainly also a normal subgroup of H and we can form the quotient group H/N. This consists of some of the elements of G/N and forms a group, so is a subgroup of G/N. Thus we do indeed have a map $\Phi \colon \mathscr{S} \to \mathscr{T}$ given by $H \mapsto H/N$. Also note that if $H_1, H_2 \in \mathscr{S}$ with $H_1 \leqslant H_2$, then we immediately obtain $H_1/N \leqslant H_2/N$, so Φ preserves inclusions.

Suppose $H_1, H_2 \in \mathscr{S}$ and that $H_1/N = H_2/N$. Let $x \in H_1$. Then $Nx \in H_1/N = H_2/N$, so $Nx = Ny$ for some $y \in H_2$. Then $xy^{-1} \in N$, say $xy^{-1} = n$ for some $n \in N$. Since $N \leqslant H_2$, we then deduce $x = ny \in H_2$. This shows $H_1 \leqslant H_2$ and a symmetrical argument shows $H_2 \leqslant H_1$. Hence if $H_1\Phi = H_2\Phi$ then necessarily $H_1 = H_2$, so Φ is injective.

Finally let $J \in \mathscr{T}$. Let H be the inverse image of J under the natural map π; that is,

$$H = \{ x \in G \mid x\pi \in J \} = \{ x \in G \mid Nx \in J \}.$$

If $x \in N$, then $Nx = N1 \in J$, since $N1$ is the identity element in the quotient group. Therefore $N \leqslant H$. If $x, y \in H$, then $Nx, Ny \in J$ and so $Nxy = (Nx)(Ny) \in J$ and $Nx^{-1} = (Nx)^{-1} \in J$. Hence $xy, x^{-1} \in H$, so we deduce that H is a subgroup which contains N. Thus $H \in \mathscr{S}$. We now consider the image of this subgroup H under the map Φ. If $x \in H$, then $Nx \in J$, so $H/N \leqslant J$. On the other hand, an arbitrary element of J has the form Nx for some element x in G and, by definition, this element x belongs to H. Hence every element of J has the form Nx for some $x \in H$ and we conclude $J = H/N = H\Phi$. Thus Φ is surjective. This completes the proof of Part (i).

(ii) We retain the notation of Part (i). Suppose $H \in \mathscr{S}$ and that $H \trianglelefteqslant G$. Consider a coset Nx in H/N (with $x \in H$) and an arbitrary coset Ng in G/N. Now $g^{-1}xg \in H$ since $H \trianglelefteqslant G$, so $(Ng)^{-1}(Nx)(Ng) = Ng^{-1}xg \in H/N$. Thus $H/N \trianglelefteqslant G/N$.

Conversely suppose $J \trianglelefteqslant G/N$. Let $H = \{ x \in G \mid Nx \in J \}$, so that $J = H/N$ (as in the last part of the proof of (i)). Let $x \in H$ and $g \in G$. Then $Nx \in J$, so $Ng^{-1}xg = (Ng)^{-1}(Nx)(Ng) \in J$ by normality of J. Thus $g^{-1}xg \in H$, by definition of H, and we deduce that $H \trianglelefteqslant G$. Hence normality is preserved by the bijection Φ. □

Theorem 1.28 (Second Isomorphism Theorem) *Let G be a group, let H be a subgroup of G and let N be a normal subgroup of G. Then $H \cap N$ is a normal subgroup of H, NH is a subgroup of G, and*

$$H/(H \cap N) \cong NH/N.$$

Proof: The natural map $\pi \colon x \mapsto Nx$ is a homomorphism $G \to G/N$. Let ϕ be the restriction to H; i.e., $\phi \colon H \to G/N$ given by $x \mapsto Nx$ for all $x \in H$. Then ϕ is once again a homomorphism,

$$\ker \phi = H \cap \ker \pi = H \cap N$$

and

$$\operatorname{im} \phi = \{ Nx \mid x \in H \} = \{ Nnx \mid x \in H,\ n \in N \} = NH/N.$$

By the First Isomorphism Theorem, $H \cap N \trianglelefteqslant H$, $NH/N \leqslant G/N$ and

$$H/(H \cap N) \cong NH/N.$$

Finally NH is a subgroup of G by the Correspondence Theorem. □

Note: Since we shall need it later, we record that the isomorphism $H/(H \cap N) \to NH/N$ is given by

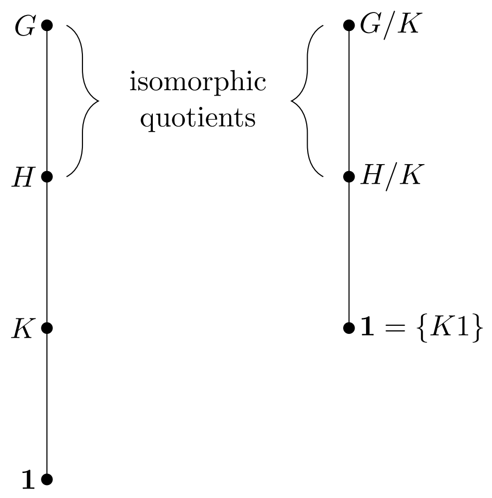
$$(H \cap N)x \mapsto Nx \qquad \text{for } x \in H$$

(as this is the isomorphism that the proof of the First Isomorphism Theorem provides for us).

Theorem 1.29 (Third Isomorphism Theorem) *Let G be a group and let H and K be normal subgroups of G such that $K \leqslant H \leqslant G$. Then H/K is a normal subgroup of G/K and*

$$\frac{G/K}{H/K} \cong G/H.$$

This theorem then tells us about the behaviour of normal subgroups of quotient groups and their associated quotients. Specifically, via the Correspondence Theorem, we know that a normal subgroup of the quotient group G/K has the form H/K where $K \leqslant H \trianglelefteq G$. Now we would like to know what the quotient group by this normal subgroup is, and the Third Isomorphism Theorem tells us that it is the same as the quotient in the original group. In terms of our diagrams of subgroups, the following occurs:



Proof: Define $\theta \colon G/K \to G/H$ by $Kx \mapsto Hx$ for $x \in G$. This is a well-defined map [if $Kx = Ky$, then $xy^{-1} \in K \leqslant H$, so $Hx = Hy$] which is easily seen to be a homomorphism [$\big((Kx)(Ky)\big)\theta = (Kxy)\theta = Hxy = (Hx)(Hy) = (Kx)\theta \cdot (Ky)\theta$ for all $x, y \in G$] and clearly $\operatorname{im}\theta = G/H$. The kernel is

$$\ker\theta = \{\, Kx \mid x \in H \,\} = H/K.$$

Hence, by the First Isomorphism Theorem, $H/K \trianglelefteq G/K$ and

$$\frac{G/K}{H/K} \cong G/H. \qquad \square$$

Sylow's Theorem

The final topic that we cover in this review section is the most important result from *MT4003*, indeed probably the most important theorem that is concerned with finite groups.

Definition 1.30 Let p be a prime number and G be a finite group.

(i) We say G is a *p-group* if its order is a power of p.

(ii) If G is any finite group, a *p-subgroup* of G is a subgroup whose order is a power of p.

(iii) Suppose that $|G| = p^n m$ where p does not divide m. A *Sylow p-subgroup* of G is a subgroup of order p^n.

Thus a Sylow p-subgroup of G is a p-subgroup of the largest possible order permitted by Lagrange's Theorem. Note that if $|G| = p^n m$ where $p \nmid m$ and P is a Sylow p-subgroup of G, then $|G : P| = m$ and this is coprime to p.

Theorem 1.31 (Sylow's Theorem) *Let p be a prime number, G be a finite group and write $|G| = p^n m$ where p does not divide m. Then*

(i) *G has a Sylow p-subgroup;*

(ii) *any two Sylow p-subgroups are conjugate in G;*

(iii) *the number of Sylow p-subgroups of G is congruent to $1 \pmod p$ and divides m;*

(iv) *any p-subgroup of G is contained in a Sylow p-subgroup.*

Before discussing initially the significance of this theorem, we will present some information about conjugation since this arises in part (ii) of the theorem.

Let G be a group and fix an element $x \in G$. Write τ_x for the map which is conjugation by x:

$$\begin{aligned} \tau_x \colon G &\to G \\ g &\mapsto g^x = x^{-1}gx. \end{aligned}$$

Observations:

(i)

$$(gh)\tau_x = x^{-1}ghx = x^{-1}gx \cdot x^{-1}hx = (g\tau_x)(h\tau_x)$$

for all $g, h \in G$; that is, τ_x is a homomorphism.

(ii)

$$g\tau_x\tau_{x^{-1}} = x(x^{-1}gx)x^{-1} = g$$

so that $\tau_x\tau_{x^{-1}} = \mathrm{id}_G$, and similarly $\tau_{x^{-1}}\tau_x = \mathrm{id}_G$. Hence τ_x is an invertible map (it has $\tau_{x^{-1}}$ as its inverse).

Invertible homomorphisms are, of course, called isomorphisms, but in the special case where the isomorphism is from a particular group to itself, we give it a special name:

Definition 1.32 Let G be a group. An *automorphism* of G is a map $G \to G$ which is an isomorphism.

We have shown that τ_x (conjugation by x) is an automorphism of our group. We use the term *inner automorphism* to refer to the specific automorphisms that arise in this form; that is, the inner automorphisms of G are all the maps $G \to G$ of the form $\tau_x \colon g \mapsto g^x$ for $x \in G$.

Now if H is a subgroup of G, its image under this automorphism must still be a subgroup. Hence the conjugate

$$H^x = \{ x^{-1}hx \mid h \in H \}$$

is also a subgroup of G. The inner automorphism τ_x restricts to an isomorphism $H \to H^x$ and hence this conjugate is isomorphic to the original subgroup H. In particular, the conjugates of H all have the same order as the original subgroup H.

Let us now return to Sylow's Theorem. What this significant theorem tells us is:

(i) Certain types of subgroups always exist in a finite group. In general, the converse of Lagrange's Theorem is false: there exist finite groups G with a divisor m of $|G|$ where there is no subgroup of order m. The first part of Sylow's Theorem essentially tells us that for the case of divisors of $|G|$ that are prime-powers, the converse *does* hold. This means that we can actually find subgroups to work with, as opposed to using Lagrange's Theorem which mostly functions as a tool to say certain subgroups do not exist.

(ii) If P is a Sylow p-subgroup, then certainly every conjugate P^x (for $x \in G$) is also a Sylow p-subgroup as it has the same order. The theorem tells us that all the Sylow p-subgroups arise in this way and so, in particular, they all have the same structure (that is, they are isomorphic).

(iii) We have strong numeric information about the Sylow subgroups and we shall exploit this at many points during the course.

(iv) The final part of the theorem tells us something stronger than Lagrange's Theorem about the p-subgroups of a group G. The earlier theorem tells us that Sylow p-subgroups are the largest p-subgroups of G in terms of their order. Part (iv) of Sylow's Theorem says they are also *maximal* in terms of *containment*. To be specific, if we were to draw a diagram of the subgroups of G as suggested earlier, then the p-subgroups of G all occur as the collections of the nodes below the Sylow p-subgroups within the diagram.

Chapter 2

Group Actions: Methods and Applications

The purpose of this chapter is to review and extend the material on group actions from *MT4003*. We shall also recall various examples of groups and of actions in the process. Group actions are significant because:

(i) Group actions are the main way that group theory applies to other branches of mathematics as well as to computer science and the physical sciences.

(ii) The important branch of group theory called *geometric group theory* is primarily concerned with the study of groups acting upon geometric structures.

(iii) Group actions give us a useful set of terminology and technology for referring to the behaviour of a group. For example, if we can say that a finite group G acts on its set of Sylow p-subgroups, then all the methods and results described in this section can be applied to deduce information about the original group G. This is the main reason we shall need this technology in this course.

We first recall the definition:

Definition 2.1 Let G be a group and Ω be a set. A *group action* of G on Ω is a map

$$\begin{aligned} \mu\colon \Omega \times G &\to \Omega \\ (\omega, x) &\mapsto \omega^x \end{aligned}$$

such that

(i) $(\omega^x)^y = \omega^{xy}$ for all $\omega \in \Omega$ and $x, y \in G$,

(ii) $\omega^1 = \omega$ for all $\omega \in \Omega$.

We then say that G *acts* on Ω.

We shall think of an action as a method of applying the element x of the group G to points in the set Ω. It is for this reason that we write ω^x to denote the image of the point ω under the element x. We shall frequently say "Let G act on the set Ω" and make no reference to the function μ appearing in the definition and instead just specify the images ω^x for each $\omega \in \Omega$ and $x \in G$. We shall adjust the chosen notation ω^x for the image of ω under the group element x when it is appropriate to do so.

An alternative way to view groups actions is via permutation representations. Recall that the *symmetric group* $\mathrm{Sym}(\Omega)$ on a set Ω consists of all bijections $\Omega \to \Omega$ (the *permutations* of Ω) and the binary operation is composition of maps. Any homomorphism $\rho\colon G \to \mathrm{Sym}(\Omega)$ is called a *permutation representation* of the group G. A permutation representation $\rho\colon G \to \mathrm{Sym}(\Omega)$ determines an action of G on the set Ω by defining ω^x to equal the image of $\omega \in \Omega$ under the permutation $x\rho \in \mathrm{Sym}(\Omega)$. (See Problem Sheet II for verification.) The following theorem from *MT4003* establishes the important converse that every group action determines a permutation representation.

Theorem 2.2 *Let G be a group that acts on the set Ω. For each $x \in G$, the map*

$$\rho_x \colon \omega \mapsto \omega^x \qquad \text{(for } \omega \in \Omega\text{)}$$

is a permutation of Ω. Furthermore, the map

$$\begin{aligned} \rho\colon G &\to \mathrm{Sym}(\Omega) \\ x &\mapsto \rho_x \end{aligned}$$

is a homomorphism.

Proof (Omitted in lectures): We compute

$$\omega\rho_x\rho_{x^{-1}} = (\omega^x)^{x^{-1}} = \omega^{xx^{-1}} = \omega^1 = \omega$$

and

$$\omega\rho_{x^{-1}}\rho_x = (\omega^{x^{-1}})^x = \omega^{x^{-1}x} = \omega^1 = \omega.$$

Hence $\rho_x\rho_{x^{-1}} = \rho_{x^{-1}}\rho_x = 1$ (the identity map $\Omega \to \Omega$), so ρ_x is a bijection and therefore $\rho_x \in \mathrm{Sym}(\Omega)$ for all $x \in G$.

We may therefore construct the map $\rho\colon G \to \mathrm{Sym}(\Omega)$ by setting $x\rho = \rho_x$ for each $x \in G$. Now

$$\omega\rho_x\rho_y = (\omega^x)^y = \omega^{xy} = \omega\rho_{xy}$$

for all $\omega \in \Omega$ and $x, y \in G$, so

$$\rho_x\rho_y = \rho_{xy} \qquad \text{for all } x, y \in G.$$

Thus

$$(x\rho)(y\rho) = (xy)\rho \qquad \text{for all } x, y \in G;$$

that is, ρ is a homomorphism. $\square$

In light of this theorem, we make the following definition:

Definition 2.3 Let G be a group that acts on the set Ω and let $\rho\colon G \to \mathrm{Sym}(\Omega)$ be the associated permutation representation. The kernel of ρ is called the *kernel of the action*. If $\ker\rho = \mathbf{1}$, then we say that the action is *faithful*.

If the action of G on the set Ω is faithful, then this tells us that the associated permutation representation $\rho\colon G \to \mathrm{Sym}(\Omega)$ is injective. Hence G is isomorphic to the image of ρ, which is a subgroup of a symmetric group. We use the term *permutation group* to refer to any subgroup of a symmetric group.

In general, the kernel of the permutation representation $\rho\colon G \to \mathrm{Sym}(\Omega)$ consists of those elements x of G such that

$$\omega^x = \omega \qquad \text{for all } \omega \in \Omega;$$

i.e., the elements of G which fix *all* points in Ω. Consequently, $x \in \ker\rho$ if and only if x belongs to every stabilizer G_ω (for all $\omega \in \Omega$). Thus:

Lemma 2.4 *Let G be a group that acts on a set Ω. If $\rho\colon G \to \mathrm{Sym}(\Omega)$ is the permutation representation associated to this action, then*

$$\ker \rho = \bigcap_{\omega \in \Omega} G_\omega$$

(the intersection of all stabilizers). □

Before we give examples of groups together with their natural actions, we recall some information from *MT4003* so that we can refer to it within our examples.

Definition 2.5 Let G be a group, Ω be a set and let G act on Ω. Let $\omega \in \Omega$.

(i) The *orbit* containing ω is defined to be the set

$$\omega^G = \{\, \omega^x \mid x \in G \,\}.$$

(ii) The *stabilizer* of ω in G is defined to be the following subset of G:

$$G_\omega = \{\, x \in G \mid \omega^x = \omega \,\};$$

that is, an element of G lies in this stabilizer if and only if it fixes the point ω.

Theorem 2.6 *Let G be a group, Ω be a set and let G act on Ω. Then Ω is the disjoint union of the orbits.*

We say that G acts *transitively* on Ω if it has precisely one orbit for its action. Observe that G acts transitively on Ω if and only if for all $\omega, \omega' \in \Omega$ there exists $x \in G$ such that $\omega' = \omega^x$.

Proof (Omitted in lectures): In the lecture notes for *MT4003*, this is established by showing that there is an equivalence relation on Ω for which the equivalence classes are the orbits. The proof presented here is similar. We shall establish that if $\omega, \omega' \in \Omega$ then

(i) $\omega \in \omega^G$;

(ii) either $\omega^G = (\omega')^G$ or $\omega^G \cap (\omega')^G = \varnothing$.

For the first part, $\omega = \omega^1$, so $\omega \in \omega^G$. For the second part, suppose that $\omega^G \cap (\omega')^G$ is non-empty; say $\alpha \in \omega^G \cap (\omega')^G$. Hence there exist $x, y \in G$ such that $\alpha = \omega^x = (\omega')^y$. Apply y^{-1}:

$$\omega^{xy^{-1}} = (\omega^x)^{y^{-1}} = ((\omega')^y)^{y^{-1}} = (\omega')^{yy^{-1}} = (\omega')^1 = \omega'.$$

Now, for $g \in G$,

$$(\omega')^g = (\omega^{xy^{-1}})^g = \omega^{xy^{-1}g} \in \omega^G$$

and we deduce $(\omega')^G \subseteq \omega^G$.

Similarly from $(\omega')^y = \omega^x$, we deduce $(\omega')^{yx^{-1}} = \omega$ and hence

$$\omega^g = (\omega')^{yx^{-1}g} \qquad \text{for all } g \in G.$$

This shows $\omega^G \subseteq (\omega')^G$. Hence if $\omega^G \cap (\omega')^G \neq \varnothing$, then $\omega^G = (\omega')^G$.

The above two observations show that every point in Ω lies in an orbit and that any two orbits are either disjoint or equal. Therefore Ω is indeed the disjoint union of its orbits. □

If G is a group acting on a set Ω, then *the stabilizer G_ω of a point ω is a subgroup of G.*

PROOF (OMITTED IN LECTURES): We check the conditions to be a subgroup. First $\omega^1 = \omega$, since we have an action, so $1 \in G_\omega$. In particular, the stabilizer G_ω is non-empty. Suppose $x, y \in G_\omega$. Then

$$\omega^{xy} = (\omega^x)^y = \omega^y = \omega$$

so $xy \in G_\omega$, while

$$\omega^{x^{-1}} = (\omega^x)^{x^{-1}} = \omega^{xx^{-1}} = \omega^1 = \omega$$

so $x^{-1} \in G_\omega$. Hence G_ω is a subgroup of G. □

The crucial way that stabilizers help us is that the *Orbit-Stabilizer Theorem* can be used to link orbits and stabilizers. If G is a group acting on a set Ω and $\omega \in \Omega$, then

$$|\omega^G| = |G : G_\omega|;$$

that is, the *length* (or size) of an orbit is equal to the index of the corresponding stabilizer. We shall not re-prove the Orbit-Stabilizer Theorem here, but later in this chapter we shall establish an improvement to it that will be useful when we attempt to describe permutation groups in more detail. We will make the following important observation now as we shall need this in a number of places.

Proposition 2.7 *Let G be a group that acts on a set Ω. Then two points in the same orbit of G on Ω have conjugate stabilizers. Indeed*

$$G_{\omega^x} = x^{-1} G_\omega x. \tag{2.1}$$

This observation only appears as an exercise in *MT4003*, so the proof will be covered in the lectures to emphasize its importance.

PROOF: If ω is an element of Ω, then any point in the same orbit has the form ω^x for some $x \in G$. So to establish the result, we need to prove the formula in Equation (2.1). Observe that

$$\begin{aligned} g \in G_{\omega^x} \qquad & \text{if and only if} \qquad \omega^{xg} = \omega^x \\ & \text{if and only if} \qquad \omega^{xgx^{-1}} = \omega \\ & \text{if and only if} \qquad xgx^{-1} \in G_\omega \\ & \text{if and only if} \qquad g \in x^{-1} G_\omega x. \end{aligned}$$

Hence Equation (2.1) holds. □

Examples of groups and actions

The following describes various (hopefully familiar!) groups and some of their natural actions.

Example 2.8 Let $\Omega = \{1, 2, \dots, n\}$. The *symmetric group of degree n* is denoted by S_n and consists of all bijections $\sigma \colon \Omega \to \Omega$. Such a bijection is called a *permutation* of Ω and we multiply permutations in the group S_n by composing them as maps.

It is an immediate consequence of this definition that the symmetric group S_n acts on the set Ω by

$$(\omega, \sigma) \mapsto \omega^\sigma$$

for $\omega \in \Omega$ and $\sigma \in S_n$. The notation ω^σ on the right-hand side denotes the result of applying the permutation σ to the point ω. The fact that multiplication in S_n is composition means that

$$\omega^{\sigma\tau} = (\omega^\sigma)^\tau \qquad \text{for } \omega \in \Omega \text{ and } \sigma, \tau \in S_n.$$

This establishes the second condition for an action. The first condition holds since the identity element of S_n is the identity map.

If $\omega, \omega' \in \Omega$, then we can construct a permutation $\sigma \in S_n$ that maps ω to ω' (just define the effect of σ on the remaining $n-1$ points of Ω arbitrarily subject to mapping distinct points to distinct points). Thus the orbit ω^{S_n} contains all points of Ω; that is, the action of S_n on $\Omega = \{1, 2, \dots, n\}$ is transitive.

If $\omega \in \Omega$, then an element of G_ω fixes the point ω and can permute the remaining $n-1$ points of Ω in any way that one chooses. Hence the stabilizer G_ω is a copy of the symmetric group S_{n-1} of degree $n-1$ (and when $\omega = n$, it is the natural copy of S_{n-1} inside S_n). Note that the index of S_{n-1} in S_n is equal to n and this consists with the size of the orbit ($|\Omega| = n$). These observations are therefore all consistent with the Orbit-Stabilizer Theorem (as they must be!).

Example 2.9 Let F be a field and n be a positive integer. Write $V = F^n$ for the vector space of all row vectors of length n. The *general linear group* $\mathrm{GL}_n(F)$ consists of all invertible $n \times n$ matrices over F with matrix multiplication as the binary operation.

The general linear group acts on the vector space V by right multiplication:

$$(v, A) \mapsto vA \qquad \text{for } v \in V \text{ and } A \in \mathrm{GL}_n(F).$$

The definition of matrix multiplication ensures that this is an action:

$$(vA)B = v(AB)$$

for all $v \in V$ and $A, B \in \mathrm{GL}_n(F)$.

More generally, if V is any vector space then the set of all invertible linear maps $g \colon V \to V$ forms a group, called the *general linear group* $\mathrm{GL}(V)$. This group acts on the vector space V by applying each linear map g to the vectors in V. If $\dim V = n$ and we fix a basis $\mathscr{B}$ for V, then every linear map $g \colon V \to V$ corresponds to a $n \times n$ matrix over the underlying field F. In this way, $\mathrm{GL}(V)$ is isomorphic to the general linear group $\mathrm{GL}_n(F)$.

Note that the zero vector is fixed in the action: $\mathbf{0}A = \mathbf{0}$ for all matrices A. Hence $\{\mathbf{0}\}$ is an orbit for the action of $\mathrm{GL}_n(F)$ on $V = F^n$. If v and w are non-zero vectors in V, we may extend them to bases for F^n: $\mathscr{B}_1 = \{v_1, v_2, \dots, v_n\}$ and $\mathscr{B}_2 = \{w_1, w_2, \dots, w_n\}$ where $v_1 = v$ and $w_1 = w$. Then $v_i \mapsto w_i$ for $i = 1, 2, \dots, n$ determines an invertible linear map $V \to V$ that maps v to w. Hence v and w lie in the same orbit. We conclude that $\mathrm{GL}_n(F)$ has precisely two orbits in its action on V:

$$\{\mathbf{0}\} \qquad \text{and} \qquad V \setminus \{\mathbf{0}\}.$$

Notational comment: Since we are placing our functions *on the right*, we use row vectors in the above example. In *MT2501* and *MT3501*, maps were written on the left, so lecture notes for those modules notes use column vectors $\boldsymbol{v}$ and each $A \in \mathrm{GL}_n(F)$ determines a map $\boldsymbol{v} \mapsto A\boldsymbol{v}$.

Example 2.10 Again let F be a field and V be a vector space of dimension n over F. Take Ω to be the set of all subspaces of V. Now define an action of the general linear group $\mathrm{GL}(V)$ on Ω by letting each linear map $g \in \mathrm{GL}(V)$ map a subspace U to its image U^g under g. Since $(v^g)^h = v^{gh}$ for each vector $v \in V$ and all $g, h \in \mathrm{GL}(V)$, it follows that

$$(U^g)^h = U^{gh}$$

when U is a subspace of V (since these sets are just the collections of images of vectors $u \in U$ under the corresponding linear maps). Furthermore, as is well-known from basic linear algebra, the image of a space under a linear map is again a subspace, so each $g \in \mathrm{GL}(V)$ does map $U \in \Omega$ to a subspace $U^g \in \Omega$. Hence $\mathrm{GL}(V)$ acts on Ω.

Furthermore, since each $g \in \mathrm{GL}(V)$ is invertible (so $\ker g = \mathbf{0}$), the Rank-Nullity Theorem tells us $\dim U^g = \dim U$ for all subspaces $U \in \Omega$. Conversely, if U and W are subspaces of V of the same dimension, we pick bases $\{u_1, u_2, \dots, u_k\}$ and $\{w_1, w_2, \dots, w_k\}$, respectively, for them, extend to bases $\{u_1, u_2, \dots, u_n\}$ and $\{w_1, w_2, \dots, w_n\}$ to V, and then the linear map g given by $u_i \mapsto w_i$ for each i is an invertible linear map such that $U^g = W$. Hence two subspaces in Ω lie in the same $\mathrm{GL}(V)$-orbit if and only if they have the same dimension.

Lemma 2.11 *Suppose V is a vector space of dimension n. Then in the action of the general linear group $\mathrm{GL}(V)$ on the collection Ω of all subspaces of V, there are $n+1$ orbits each consisting of the subspaces of V of dimension k for $0 \leqslant k \leqslant n$.* □

Example 2.12 Let n be an integer with $n \geqslant 3$. Recall that the *dihedral group* D_{2n} of order $2n$ consists of all isometries that preserve the regular n-sided polygon X. (Some authors use the notation D_n instead of D_{2n} for this group. The notation is not consistent either way within the literature.) Let Ω be the set of vertices of the polygon X. Since the multiplication in D_{2n} is composition of isometries, it follows that

$$(\omega^g)^h = \omega^{gh}$$

for all vertices $\omega \in \Omega$ and all isometries $g, h \in D_{2n}$. From this we see that the dihedral group D_{2n} acts on the set Ω of vertices of the polygon X. This action is transitive, since we can simply rotate the polygon to move any one vertex to any other.

Suppose that a group G acts on a set Ω and that Δ is an orbit for this action. By definition, $\delta^g \in \Delta$ for all $\delta \in \Delta$ and $g \in G$; that is, Δ is closed under the action of G. Both conditions of Definition 2.1 are satisfied when we apply them to elements of Δ and of G, so we deduce that there is also an induced action of G on the orbit Δ. By a similar argument, one can also see that there is an induced action of any subgroup of G on the set Ω. Thus an action of a group G on a set Ω induces further actions of any subgroup of G and further actions on any union of orbits in Ω.

Actions arising in group theory

We now recall some ways of the ways that actions arise when we study groups.

Example 2.13 (Conjugation Action) Let G be a group and define an action of G on itself by

$$\begin{aligned} G \times G &\to G \\ (g, x) &\mapsto x^{-1}gx = g^x, \end{aligned}$$

the *conjugate* of g by x. We check the conditions for a group action:

(i) $(g^x)^y = y^{-1}(x^{-1}gx)y = y^{-1}x^{-1}gxy = (xy)^{-1}g(xy) = g^{xy}$ for all $g, x, y \in G$;

(ii) $g^1 = 1^{-1}g1 = 1g1 = g$ for all $g \in G$.

Thus we have a genuine action of G on itself. If $g \in G$, the orbit of G containing g (for this conjugation action) is the *conjugacy class* of g in G:

$$g^G = \{\, g^x \mid x \in G \,\} = \{\, x^{-1}gx \mid x \in G \,\},$$

the set of all conjugates of g. The stabilizer of g under this action is the *centralizer* of g in G:

$$\begin{aligned} \mathrm{C}_G(g) = G_g &= \{\, x \in G \mid g^x = g \,\} \\ &= \{\, x \in G \mid x^{-1}gx = g \,\} \\ &= \{\, x \in G \mid gx = xg \,\}; \end{aligned}$$

that is, the set of elements of G that commute with g.

The standard facts about group actions immediately yield the following facts about conjugacy classes and centralizers. The fourth one is simply the interpretation of Equation (2.1) in this example.

Proposition 2.14 *Let G be a group. Then*

(i) *G is the disjoint union of its conjugacy classes;*

(ii) *the centralizer of an element g is a subgroup of G;*

(iii) *the number of conjugates of an element g equals the index of its centralizer;*

(iv) *if $g, x \in G$ then*
$$\mathrm{C}_G(g^x) = \mathrm{C}_G(g)^x.$$

Example 2.15 (Conjugation action on subsets and subgroups)
Let G be a group and let $\mathscr{P}(G)$ denote the set of all subsets of G (the *power set* of G). We define an action of G on $\mathscr{P}(G)$ by

$$\begin{aligned} &\mathscr{P}(G) \times G \to \mathscr{P}(G) \\ &\qquad (A, x) \mapsto A^x = x^{-1}Ax = \{\, x^{-1}ax \mid a \in A \,\}. \end{aligned}$$

A similar argument to Example 2.13 checks that this is indeed an action (this only relies on associativity of the group multiplication and the formula for the inverse of a product of two elements). The orbit containing the subset A is the set of all conjugates of A and the stabilizer is the *normalizer* of A in G:

$$\mathrm{N}_G(A) = \{\, x \in G \mid A^x = A \,\}.$$

Since this is a stabilizer, it is always a subgroup of G. We shall be most interested in the case when we conjugate subgroups and here we make use of what we know about conjugation from Chapter 1. We know that the conjugation map $\tau_x \colon g \mapsto g^x = x^{-1}gx$ is an automorphism of the group G. (Recall that we use the term *inner automorphism* for this map τ_x.) As a consequence, it maps subgroups to subgroups and hence the conjugate H^x of a subgroup H is another subgroup. This tells us that the orbit of a subgroup H under the conjugation action consists of some subgroups of G that all are isomorphic to the original subgroup H. Therefore:

Proposition 2.16 *Let G be a group and H be a subgroup of G.*

(i) *The normalizer $\mathrm{N}_G(H)$ of H in G is a subgroup of G.*

(ii) *The conjugates of H are subgroups of G that are isomorphic to H.*

(iii) *The number of conjugates of H in G is equal to the index $|G : \mathrm{N}_G(H)|$ of the normalizer in G.* □

In particular, if we consider conjugation of the Sylow subgroups of a finite group G, some of what Sylow's Theorem tells us is that, for a given prime p, the Sylow p-subgroups form a single orbit under the conjugation action of G on its subgroups.

Our final example is extremely important. We shall observe that if we can find a subgroup H of a group G then we can define an action on the cosets of H and hence produce the associated permutation representation.

Example 2.17 (Action on Cosets) Let G be a group and H be a subgroup of G. Let $\Omega = \{ Hg \mid g \in G \}$, the set of cosets of H in G. We shall define an action of G on Ω as follows:

$$\begin{aligned} \Omega \times G &\to \Omega \\ (Hg, x) &\mapsto Hgx. \end{aligned}$$

The first thing to do is check that this is well-defined, that is, the image of a coset Hg when we apply an element x of G does not depend upon the choice of representative g for the coset. Let $g, h, x \in G$ and suppose that $Hg = Hh$. Then $gh^{-1} \in H$. Now

$$(gx)(hx)^{-1} = gxx^{-1}h^{-1} = gh^{-1} \in H$$

and therefore $Hgx = Hhx$. Hence the action is well-defined. We shall write $Hg \cdot x$ to denote this action in what follows.

It is now straightforward to check that this is a group action:

$$(Hg \cdot x) \cdot y = Hgx \cdot y = Hgxy = Hg \cdot xy$$

and

$$Hg \cdot 1 = Hg1 = Hg$$

for all $g, x, y \in G$.

We shall now establish the main properties of this action on cosets:

Theorem 2.18 *Let H be a subgroup of a group G, let $\Omega = \{ Hg \mid g \in G \}$ be the set of cosets of H, and let G act on Ω by right multiplication. Let $\rho \colon G \to \mathrm{Sym}(\Omega)$ be the permutation representation associated to the action. Then:*

(i) *The action of G on Ω is transitive.*

(ii) *If H is a proper subgroup of G (that is, if $H \neq G$), then $G\rho$ is a non-trivial subgroup of $\mathrm{Sym}(\Omega)$.*

(iii) *The stabilizer of each coset Hg in G is equal to the conjugate H^g of the original subgroup H. In particular, the stabilizer of $H1$ is H.*

(iv) *The kernel of* ρ *is the intersection of the conjugates of* H*:*

$$\ker \rho = \bigcap_{g \in G} H^g.$$

(v) *The kernel of* ρ *is the largest normal subgroup of* G *contained in* H*.*

Definition 2.19 We call this intersection $\bigcap_{g \in G} H^g$ (that occurs as the kernel of the above permutation representation) the *core* of H in G and shall denote it by $\mathrm{Core}_G(H)$.

PROOF: (i) Consider two cosets Hg and Hh in Ω. Take $x = g^{-1}h$. Then

$$Hg \cdot x = Hgx = Hgg^{-1}h = Hh.$$

This shows that Hg and Hh lie in the same orbit. Hence G acts transitively on Ω.

(ii) Suppose $H \neq G$. Then Ω contains more than one coset: $|\Omega| > 1$. Since G acts transitively, it must induce at least one non-trivial permutation on Ω and so $G\rho \neq \mathbf{1}$.

(iii) We compute the stabilizer of each coset Hg:

$$\begin{aligned} G_{Hg} &= \{ x \in G \mid Hg \cdot x = Hg \} \\ &= \{ x \in G \mid Hgx = Hg \} \\ &= \{ x \in G \mid gxg^{-1} \in H \} \\ &= \{ x \in G \mid x \in g^{-1}Hg \} = H^g. \end{aligned}$$

(iv) The kernel of ρ is determined by Lemma 2.4:

$$\ker \rho = \bigcap_{g \in G} G_{Hg} = \bigcap_{g \in G} H^g.$$

(v) Certainly $\ker \rho$ is a normal subgroup of G (since kernels are always normal subgroups). By the previous part,

$$\ker \rho = \bigcap_{g \in G} H^g \leqslant H^1 = H.$$

On the other hand, if K is any normal subgroup of G that is contained in H then $K = K^g \leqslant H^g$ for all $g \in G$. Hence

$$K \leqslant \bigcap_{g \in G} H^g = \ker \rho.$$

$\square$

If H is any subgroup of G with trivial core, then the permutation representation $\rho \colon G \to \mathrm{Sym}(\Omega)$ is injective and $G \cong \mathrm{im}\, \rho$. In particular this happens when we take $H = \mathbf{1}$ and the action of G is one on the set of singletons $\{g\}$ (for $g \in G$). This is equivalent to the *right regular action* of G on itself by right multiplication:

$$(g, x) \mapsto gx \qquad \text{for } x, g \in G.$$

(In this case, the effect of applying x to an element g is to multiply on the right by x.) We therefore deduce the following (noting that the action is transitive by Theorem 2.18(i)):

Theorem 2.20 (Cayley's Theorem) *Every group is isomorphic to a transitive permutation group.* $\square$

Cayley's Theorem tells us that *every* group is essentially the same as a permutation group. One might therefore wonder how much one could say about permutation groups. In the final chapter of the notes we shall pull together various threads within the notes to describe the structure of a permutation group in more detail. The primary idea is to use the action of the permutation group to establish ways in which the group can be decomposed and to reduce to various types of standard "building blocks." The smallest form of building block (the *primitive* permutation groups) turn out to have far more restricted structure than arbitary (transitive) permutation groups.

In preparation for this work in Chapter 7, we shall establish the promised extension to Orbit-Stabilizer Theorem.

Definition 2.21 (i) Let G be a group that has actions on two sets Ω and Ω'. We say that these actions are *equivalent* if there is a bijection $\phi\colon \Omega \to \Omega'$ such that

$$(\omega^x)\phi = (\omega\phi)^x \qquad \text{for all } \omega \in \Omega \text{ and } x \in G.$$

(ii) Let G be a permutation group on a set Ω and H be a permutation group on a set Ω'. We say that G and H are *permutation isomorphic* if there is a bijection $\phi\colon \Omega \to \Omega'$ and an isomorphism $\theta\colon G \to H$ such that

$$(\omega^x)\phi = (\omega\phi)^{x\theta} \qquad \text{for all } \omega \in \Omega \text{ and } x \in G.$$

We can interpret both parts of the definition as saying the actions are essentially the same. The two concepts are closely linked with each other. Indeed, if the actions of G on two sets Ω and Ω' are equivalent, then the two permutations groups obtained via the associated permutations representations turn out to be permutation isomorphic. (See Problem Sheet II for more details.)

Theorem 2.22 (Orbit-Stabilizer Theorem, Improved) *Let G be a group that acts transitively on a set Ω and let $\omega \in \Omega$. Let*

$$\Omega' = \{\, G_\omega g \mid g \in G \,\}$$

be the set of cosets of the stabilizer G_ω and let G act on Ω' by right multiplication. Then these actions of G are equivalent.

PROOF: We define a map

$$\begin{aligned} \phi\colon \Omega' &\to \Omega \\ G_\omega g &\mapsto \omega^g. \end{aligned}$$

The proof of the Orbit-Stabilizer Theorem from *MT4003* shows that ϕ is a well-defined bijection as follows:

$$\begin{aligned} G_\omega g = G_\omega h \qquad &\text{if and only if} \qquad gh^{-1} \in G_\omega \\ &\text{if and only if} \qquad \omega^{gh^{-1}} = \omega \\ &\text{if and only if} \qquad \omega^g = \omega^h \end{aligned}$$

This calculation shows ϕ is well-defined ($\Rightarrow$) and injective ($\Leftarrow$). The map is surjective since G acts transitively on Ω.

In fact, ϕ is an equivalence between the actions of G on Ω' and Ω as the following calculation shows: If $g, x \in G$, then

$$(G_\omega g \cdot x)\phi = (G_\omega g x)\phi = \omega^{gx} = (\omega^g)^x = \big((G_\omega g)\phi\big)^x.$$

Hence the actions of G on Ω' and Ω are equivalent. $\square$

This theorem then tells us that any transitive action of a group G is essentially the same as its action on the cosets of some subgroup. Conversely, we observed in Theorem 2.18 that the action of a group G on the cosets of a subgroup H is always transitive. As a consequence, we can observe something about the significance of the right regular action described earlier.

Definition 2.23 Let G be a group that acts on a set Ω. We say that this action is *regular* if it is transitive and the stabilizer of any point is trivial.

Note that, by transitivity, Proposition 2.7 tells us that if one stabilizer is trivial then all stabilizers are trivial.

Corollary 2.24 *Let G be a group. Then any regular action of G is equivalent to the right regular action of G on itself.*

PROOF: Taking $G_\omega = \mathbf{1}$ in Theorem 2.22 tells us that any regular action of G is equivalent to that on the cosets of the trivial group. Hence it is equivalent to the right regular action. □

Applications: The search for simple groups

To illustrate the significance of the action on the cosets, we shall give some examples of how this can be used to establish information about finite simple groups. We start by recalling the term that was introduced in *MT4003*:

Definition 2.25 A non-trivial group G is *simple* if the only normal subgroups it has are $\mathbf{1}$ and G.

The idea here is that if a non-trivial group G is *not* simple then it has a non-trivial proper normal subgroup N and we can break it down into two smaller groups N and G/N which are hopefully easier to handle than G. On the other hand, when G is simple this process yields nothing new: one of these groups is trivial and the other is just a copy of G. As a consequence, simple groups are the minimal building blocks (analogous, in some sense, to primes in number theory) from which all other (finite) groups are built. One of the things we shall discuss in the next chapter is how we can describe the factorization of a (finite) group into simple factors and to what extent this factorization is unique. Chapter 4 will give some descriptions of ways in which the factorization may be reversed (i.e., how the simple factors might be put back together), but we shall only start to describe what is a somewhat complicated topic.

To start we shall give one example of the use of the methods of *MT4003* (i.e., the following does not use group actions):

Example 2.26 *Show that there is no simple group of order* 858.

SOLUTION: Let G be a group of order $858 = 2 \cdot 3 \cdot 11 \cdot 13$. Let n_{11} and n_{13} denote the number of Sylow 11- and Sylow 13-subgroups of G, respectively. By Sylow's Theorem,

$$\begin{aligned} n_{11} &\equiv 1 \pmod{11} &\quad \text{and} \quad & n_{11} \mid 78 \\ n_{13} &\equiv 1 \pmod{13} &\quad \text{and} \quad & n_{13} \mid 66. \end{aligned}$$

If $n_{11} = 1$, then the unique Sylow 11-subgroup of G would be normal in G and hence G would not be simple. Similarly if $n_{13} = 1$.

Suppose that $n_{11} = 78$. Consider two distinct Sylow 11-subgroups E_1 and E_2 of G. Then $|E_1| = |E_2| = 11$. The intersection $E_1 \cap E_2$ is a proper subgroup of E_1, so $|E_1 \cap E_2|$ divides 11, by Lagrange's Theorem. Hence $E_1 \cap E_2 = \mathbf{1}$. It follows that each Sylow 11-subgroup of G contains 10 non-identity elements (all of order 11) and these are contained in no other Sylow 11-subgroup of G. Hence the Sylow 11-subgroups account for

$$78 \times 10 = 780 \text{ elements of order } 11.$$

By exactly the same argument, if $n_{13} = 66$ then the Sylow 13-subgroups contain $66 \times 12 = 792$ elements of order 13.

Since $780 + 792 > |G|$, it must be the case that either $n_{11} = 1$ or $n_{13} = 1$. So G either has a normal subgroup of order 13 or a normal subgroup of order 11. Hence G is not simple. □

Let us now see how the use of group actions can help with this type of problem.

Example 2.27 *Show that there is no simple group of order* 36.

Solution: Let H be a Sylow 3-subgroup of G. Then $|G : H| = 4$. Let G act on the set of right cosets of H by right multiplication,

$$(Hx, g) \mapsto Hxg,$$

and let $\rho \colon G \to S_4$ be the associated permutation representation. Since $|G| = 36 \geqslant |S_4| = 24$, it is certainly the case that $\ker \rho \neq \mathbf{1}$. On the other hand, Theorem 2.18(ii) tells us that $\ker \rho \neq G$ (because $G\rho \neq \mathbf{1}$). It follows that $\ker \rho$ is a non-trivial proper normal subgroup of G and hence G is not simple. □

Proposition 2.28 *Let p and q be distinct primes and let G be a finite group of order p^2q. Then one of the following holds:*

(i) *$p > q$ and G has a normal Sylow p-subgroup;*

(ii) *$q > p$ and G has a normal Sylow q-subgroup;*

(iii) *$p = 2$, $q = 3$, $G \cong A_4$ and G has a normal Sylow 2-subgroup.*

Proof: (i) Suppose $p > q$. Let n_p denote the number of Sylow p-subgroups of G. Then

$$n_p \equiv 1 \pmod{p} \qquad \text{and} \qquad n_p \mid q.$$

The latter forces $n_p = 1$ or q. But $1 < q < p + 1$, so $q \not\equiv 1 \pmod{p}$. Hence $n_p = 1$, so G has a unique Sylow p-subgroup which must be normal.

(ii) and (iii): Suppose $q > p$. Let n_q be the number of Sylow q-subgroups of G. If $n_q = 1$ then the unique Sylow q-subgroup of G would be normal in G (and Case (ii) would hold). So suppose that $n_q \neq 1$ (and we shall endeavour to show Case (iii) holds). Now

$$n_q \equiv 1 \pmod{q} \qquad \text{and} \qquad n_q \mid p^2.$$

So $n_q = p$ or p^2. But $1 < p < q + 1$, so $p \not\equiv 1 \pmod{q}$. Hence $n_q = p^2$, so

$$p^2 \equiv 1 \pmod{q};$$

that is,

$$q \text{ divides } p^2 - 1 = (p+1)(p-1).$$

But q is prime, so either q divides $p-1$ or it divides $p+1$. However, $1 \leqslant p-1 < q$, so the only one of these possibilities is that q divides $p+1$. However $p < q$, so $p+1 \leqslant q$ so we are forced into the situation where $p+1 = q$. Hence

$$p = 2, \qquad q = 3$$

and

$$|G| = 2^2 \cdot 3 = 12.$$

Let Q be a Sylow 3-subgroup of G $(q = 3)$ and let G act on the set of right cosets of Q by right multiplication. This gives rise to the permutation representation

$$\rho\colon G \to S_4$$

(as $|G : Q| = 4$, so we are acting on four points). Theorem 2.18 tells us that the kernel of ρ is contained in Q, while it must be a proper subgroup of Q as $Q \ntrianglelefteq G$. Since $|Q| = 3$, this forces $\ker\rho = \mathbf{1}$, so ρ is injective. Hence

$$G \cong \operatorname{im}\rho.$$

Now $\operatorname{im}\rho$ is a subgroup of S_4 of order 12 and therefore index 2. We deduce that $\operatorname{im}\rho = A_4$ and so $G \cong A_4$.

Finally

$$V_4 = \{1, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$$

is a subgroup of A_4 that is isomorphic to the Klein 4-group (hence our choice of notation V_4 for it). This is a Sylow 2-subgroup of A_4 and $V_4 \trianglelefteq S_4$ (and hence $V_4 \trianglelefteq A_4$). Therefore G has a normal Sylow 2-subgroup in this case. □

It is worth pointing out that at this point in time much is actually now known about finite simple groups. A mammoth effort by a large collection of mathematicians from the 1950s to the 1980s succeeded in establishing a full classification. The complete proof runs to tens of thousands of pages of extremely complicated mathematics and there has been doubt as to what extent this is truly complete. Since then there have been many developments where people have established detailed information about the simple groups in this list (for example, what their *maximal subgroups* are and what we can say about some of their actions and (permutation) representations). More work is still currently being done so as to check, clarify and simplify the proof. Nevertheless it is generally accepted that this Classification is correct, though typically when relying upon it a mathematician would normally state that they are doing so.

Theorem 2.29 (Classification of Finite Simple Groups) *Let G be a finite simple group. Then G is one of the following:*

(i) *a cyclic group of prime order;*

(ii) *an alternating group A_n where $n \geqslant 5$;*

(iii) *one of sixteen (infinite) families of groups of Lie type;*

(iv) *one of twenty-six sporadic simple groups.*

The fact that cyclic groups of prime order are simple is already known: it follows immediately from Lagrange's Theorem. The proof that the alternating groups of degree at least 5 are simple is longer and was one of the highlights of *MT4003*.

The groups of Lie type are essentially 'matrix-like' groups which preserve geometric structures on vector spaces over finite fields. For example, the first (and most easily described) family is the collection of groups $\mathrm{A}_n(q)$ where n is a positive integer and q is a prime-power (and where we require $q \geqslant 4$ if $n = 1$). The definition of this family is as follows: Start with the group $\mathrm{GL}_{n+1}(q)$ of invertible $(n+1) \times (n+1)$ matrices with entries from the field $\mathbb{F}_q$ of q elements; then take those of determinant 1

$$\mathrm{SL}_{n+1}(q) = \{ A \in \mathrm{GL}_{n+1}(q) \mid \det A = 1 \}$$

(the *special linear group*); then factor out the centre (which happens to consist of all scalar matrices in $\mathrm{SL}_{n+1}(q)$)

$$\mathrm{Z}(\mathrm{SL}_{n+1}(q)) = \{ \lambda I \mid \lambda^{n+1} = 1 \text{ in } \mathbb{F}_q \},$$

to form

$$\mathrm{A}_n(q) = \mathrm{PSL}_{n+1}(q) = \mathrm{SL}_{n+1}(q)/\mathrm{Z}(\mathrm{SL}_{n+1}(q))$$

and we have constructed a simple group (provided either $n \geqslant 2$, or $n = 1$ and $q \geqslant 4$).

The twenty-six sporadic groups are as listed in Table 2.1. The order of the Monster is too large to appear in the table, it is

$$|\mathrm{M}| = 808\,017\,424\,794\,512\,875\,886\,459\,904\,961\,710\,757\,005\,754\,368\,000\,000\,000$$

It will not surprise anyone that the proof of Theorem 2.29 is not included in these notes!

Mathieu	M_{11}	7 920
Mathieu	M_{12}	95 040
Janko	J_1	175 560
Mathieu	M_{22}	443 520
Janko	J_2	604 800
Mathieu	M_{23}	10 200 960
Higman–Sims	HS	44 352 000
Janko	J_3	50 232 960
Mathieu	M_{24}	244 823 040
McLaughlin	McL	898 128 000
Held	He	4 030 387 200
Rudvalis	Ru	145 926 144 000
Suzuki	Suz	448 345 497 600
O'Nan	O'N	460 815 505 920
Conway	Co_3	495 766 656 000
Conway	Co_2	42 305 421 312 000
Fischer	Fi_{22}	64 561 751 654 400
Harada–Norton	HN	273 030 912 000 000
Lyons	Ly	51 765 179 004 000 000
Thompson	Th	90 745 943 887 872 000
Fischer	Fi_{23}	4 089 470 473 293 004 800
Conway	Co_1	4 157 776 806 543 360 000
Janko	J_4	86 775 571 046 077 562 880
Fischer	Fi'_{24}	1 255 205 709 190 661 721 292 800
Baby Monster	B	4 154 781 481 226 426 191 177 580 544 000 000
Monster	M	see text

Table 2.1: The sporadic simple groups

Chapter 3

Composition Series, Chief Series and the Jordan–Hölder Theorem

The purpose of this chapter is to describe some ways in which one might decompose a group into simple factors and to establish the uniqueness of this factorization. We begin this chapter with the following general definition.

Definition 3.1 Let G be a group. A *series* for G is a finite chain of subgroups

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1}$$

such that G_{i+1} is a normal subgroup of G_i for $i = 0, 1, \ldots, n-1$. The collection of quotient groups

$$G_0/G_1, \quad G_1/G_2, \quad \ldots, \quad G_{n-1}/G_n$$

are called the *factors* of the series and we call the number n the *length* of the series.

Some authors refer to this as a *subnormal series* (coming from the fact that a subgroup that is normal in a normal subgroup of ... of a normal subgroup is often called *subnormal*). In particular, we do not require each subgroup in the series to be normal in the whole group G, only that it is normal in the previous subgroup in the chain. However, the following term is used for a stronger situation:

Definition 3.2 Let G be a group. A *normal series* for G is a series

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1}$$

such that G_i is a normal subgroup of G for every $i = 0, 1, \ldots, n$.

Observe that in both cases the length n is the number of factors that occur in the series.

The first definition is rather general and does not give much information on its own. Indeed, one example of a (normal) series of a non-trivial group G is simply of length 1, namely

$$G > \mathbf{1},$$

with factor isomorphic to G. To actually obtain useful and interesting information about the group, we need stronger conditions on the nature of the series we are considering. There will be several examples of series occurring in this course and this chapter will consider two specific types of series. The first case is the following where the factors are all required to be simple groups.

Composition series

Definition 3.3 A *composition series* for a group G is a finite chain of subgroups

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1}$$

such that, for $i = 1, 2, \ldots, n$, G_i is a normal subgroup of G_{i-1} and the quotient group G_{i-1}/G_i is simple. In this case, the factors

$$G_0/G_1, \quad G_1/G_2, \quad \ldots, \quad G_{n-1}/G_n$$

are called the *composition factors* of G.

The idea here is that if G possesses a composition series, then the composition factors are a collection of simple groups that arise as some sort of factorization of G. One of the main goals of this chapter is to establish the following theorem that says that these factors are uniquely determined.

Theorem 3.4 (Jordan–Hölder Theorem) *Let G be a group and let*

$$G = G_0 > G_1 > G_2 > \cdots > G_m = \mathbf{1}$$

and

$$G = H_0 > H_1 > H_2 > \cdots > H_n = \mathbf{1}$$

be composition series for G. Then $m = n$ and there is a one-one correspondence between the two collections of composition factors

$$G_0/G_1, \quad G_1/G_2, \quad \ldots, \quad G_{m-1}/G_m$$

and

$$H_0/H_1, \quad H_1/H_2, \quad \ldots, \quad H_{n-1}/H_n$$

such that the corresponding factors are isomorphic.

We shall delay the proof of the theorem until later in the chapter. At this stage, it is illustrative to consider some examples and to examine what the condition for a factor in a series to be simple actually means.

Example 3.5 (i) The trivial group $\mathbf{1}$ has composition series of length 0:

$$G_0 = \mathbf{1}$$

and so has no composition factors. This example is included to point out what a series of length zero looks like!

(ii) Let $G = S_4$, the symmetric group of degree 4. Then we can construct the following chain of subgroups:

$$S_4 > A_4 > V_4 > \langle (1\ 2)(3\ 4) \rangle > \mathbf{1}. \tag{3.1}$$

We know that $A_4 \trianglelefteq S_4$ with quotient $S_4/A_4 \cong C_2$ and that $V_4 \trianglelefteq A_4$ with quotient $A_4/V_4 \cong C_3$. Since V_4 is abelian, each of its subgroups is normal. Finally

$$V_4/\langle (1\ 2)(3\ 4) \rangle \cong C_2 \qquad \text{and} \qquad \langle (1\ 2)(3\ 4) \rangle \cong C_2.$$

This shows that the chain (3.1) is a composition series for S_4 and that the composition factors of S_4 (which are unique up to isomorphism by the Jordan–Hölder Theorem) are

$$C_2, \ C_3, \ C_2, \ C_2$$

(all of which are indeed simple groups as they are cyclic of prime order).

(iii) Let $n \geqslant 5$ and consider the symmetric group S_n of degree n. Then

$$S_n > A_n > \mathbf{1} \tag{3.2}$$

is a composition series for S_n with composition factors isomorphic to C_2 and A_n. (Here we use the fact that the alternating group A_n is simple for $n \geqslant 5$.) Furthermore, it can be shown that S_n has precisely three normal subgroups (namely those appearing in (3.2)) and hence this is the only composition series for S_n.

(iv) The following is a straightforward example of a group with more than one (actually it has precisely two) composition series. It is built using the direct product construction which was introduced in *MT4003* and which was reviewed on Problem Sheet I. Let $G = A_5 \times C_2$, the *direct product* of the alternating group A_5 of degree 5 and the cyclic group of order 2. It follows from the definition of the multiplication in a direct product that the projection maps

$$\begin{aligned} \pi_1 \colon G &\to A_5 \\ (x,y) &\mapsto x \end{aligned} \qquad\qquad \begin{aligned} \pi_2 \colon G &\to C_2 \\ (x,y) &\mapsto y \end{aligned}$$

are surjective homomorphisms.

Hence $M = \ker \pi_1 = \mathbf{1} \times C_2 \cong C_2$ and $N = \ker \pi_2 = A_5 \times \mathbf{1} \cong A_5$ are normal subgroups of G with simple quotients

$$G/M \cong A_5 \qquad \text{and} \qquad G/N \cong C_2.$$

Thus we obtain composition series

$$G = A_5 \times C_2 > \mathbf{1} \times C_2 > \mathbf{1} \qquad \text{and} \qquad G = A_5 \times C_2 > A_5 \times \mathbf{1} > \mathbf{1}$$

for G. The composition factors of the first are isomorphic to A_5 and C_2 (in that order), while those of the second are C_2 and A_5.

The third example above illustrates that, although the Jordan–Hölder Theorem says that the composition factors are uniquely determined, it does not mean that the composition series are unique, nor that the composition factors have to occur within the composition series in the same order. It is also easy to construct similar examples using abelian groups (for context, see also Example 3.9 below where we describe what the composition factors of a finite abelian group are).

Example 3.6 The infinite cyclic group has no composition series.

Proof: Let $G = \langle x \rangle$, where $o(x) = \infty$, and suppose that

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1}$$

is a composition series for G. Then G_{n-1} is a non-trivial subgroup of G, so $G_{n-1} = \langle x^k \rangle$ for some $k > 0$ and this is an infinite cyclic group, so is not simple. This contradicts the assumption that the above is a composition series for G. □

To understand the behaviour of composition series, consider some subgroups of a group G

$$G \geqslant M > N \geqslant \mathbf{1}$$

where N is a normal subgroup of M. (The context here is that we might be considering a particular series $G = G_0 > G_1 > \cdots > G_n = \mathbf{1}$ and that $M = G_{i-1}$ and $N = G_i$ for some i. We seek to investigate what property ensures that G_{i-1}/G_i is simple so that we understand when the series is a composition series.) The Correspondence Theorem tells us that subgroups of M/N correspond to subgroups of M which contain N. Furthermore under this correspondence, normal subgroups of M/N correspond to normal subgroups of M which contain N. We conclude that

> *M/N is simple if and only if the only normal subgroups of M containing N are M and N themselves.*

Accordingly, we can now describe what it means for a series of subgroups to be a composition series:

Proposition 3.7 *Let G be a group and*

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1} \tag{3.3}$$

be a series of subgroups of G (i.e., $G_{i+1} \trianglelefteqslant G_i$ for each i). Then this is a composition series for G if and only if it is maximal*, in the sense that one cannot create a longer series by inserting an additional subgroup.*

Proof: It is possible to insert such a subgroup H into (3.3) to form another series precisely when there is a value i where

$$G = G_0 > G_1 > \cdots > G_{i-1} > H > G_i > \cdots > G_n = \mathbf{1}$$

with $H \trianglelefteqslant G_{i-1}$. (The additional requirement that $G_i \trianglelefteqslant H$ follows immediately because $G_i \trianglelefteqslant G_{i-1}$ by assumption.) Thus (3.3) is maximal when, for every i, there is no normal subgroup H of G_{i-1} with $G_{i-1} > H > G_i$; that is, by the Correspondence Theorem when G_{i-1}/G_i is simple. This establishes the claim. $\square$

We shall use the term *refine* to refer to this process of inserting additional terms into a series.

Corollary 3.8 *Let G be a finite group. Then every series for G can be refined to a composition series. In particular, every finite group has a composition series.*

Proof: Start with any series for the finite group G. If it is not a composition series, then we can insert an additional subgroup by Proposition 3.7. We repeat this process until we reach a composition series. The process cannot continue forever, since G is finite so only has finitely many subgroups. To see every finite group G has a composition series, start with the series $G > \mathbf{1}$ and refine to a composition series. $\square$

Example 3.9 Let G be a finite abelian group of order n and write

$$n = p_1^{r_1} p_2^{r_2} \dots p_k^{r_k}$$

where $p_1, p_2, \dots, p_k$ are the distinct prime factors of n. If

$$G = G_0 > G_1 > G_2 > \cdots > G_m = \mathbf{1}$$

is a composition series, then the composition factors

$$G_0/G_1, \quad G_1/G_2, \quad \dots, \quad G_{m-1}/G_m$$

are *abelian* simple groups and so are cyclic of prime order. Now

$$|G| = |G_0/G_1| \cdot |G_1/G_2| \cdot \ldots \cdot |G_{m-1}/G_m|$$

must be the prime factorization of $|G| = n$ and hence the composition factors of G are

$$\underbrace{C_{p_1}, C_{p_1}, \ldots, C_{p_1}}_{r_1 \text{ times}}, \underbrace{C_{p_2}, C_{p_2}, \ldots, C_{p_2}}_{r_2 \text{ times}}, \ldots, \underbrace{C_{p_k}, C_{p_k}, \ldots, C_{p_k}}_{r_k \text{ times}},$$

arising in some order.

Chief series

Before we establish the Jordan–Hölder Theorem, we shall introduce another type of series. Recall that a *normal series* is a series for a group G where every term is a normal subgroup of G.

Definition 3.10 A *chief series* for a group G is a normal series

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1}$$

which is maximal; that is, one cannot create a longer normal series by inserting an additional normal subgroup of G. The quotient groups

$$G_0/G_1, \quad G_1/G_2, \quad \ldots, \quad G_{n-1}/G_n$$

are called the *chief factors* of G.

It is important to distinguish between the concept of a composition series and a chief series. Both are maximal series of a particular type, but in a composition series one requires each term G_i to be normal in the previous one but not necessarily in the whole group G. For a chief series, one requires each term G_i to be maximal in the group G and the series is maximal subject to this condition. Thus, in some sense, chief series are the analogue for *normal series* of what a composition series is. The same argument as used for Corollary 3.8 (i.e., inserting normal subgroups of G until we cannot insert any more) establishes:

Proposition 3.11 *Let G be a finite group. Then every normal series for G can be refined to a chief series. In particular, every finite group has a chief series.* □

Example 3.12 (i) If G is any simple group, then

$$G > \mathbf{1}$$

is its only chief series and it has a single chief factor, namely G.

(ii) Let $n \geqslant 5$. Then the composition series

$$S_n > A_n > \mathbf{1}$$

for the symmetric group S_n of degree n is also a chief series (as each term happens to be normal in S_n) and the chief factors of S_n coincide with the composition series.

(iii) Let $G = S_4$, the symmetric group of degree 4. Consider the following chain of normal subgroups of G:

$$S_4 > A_4 > V_4 > \mathbf{1} \tag{3.4}$$

We know that each is a normal subgroup of S_4. Since V_4 consists of the identity together with the conjugacy class of all the permutations of the form $(a\ b)(c\ d)$, there is no normal subgroup N that can be inserted between $\mathbf{1}$ and V_4. The quotients $S_4/A_4 \cong C_2$ and $A_4/V_4 \cong C_3$ are simple. Hence (3.4) is a chief series for S_4 and its chief factors are:

$$C_2,\ C_3,\ C_2 \times C_2.$$

This example illustrates the difference between chief series and composition series. The symmetric group S_4 has four composition factors, as we observed in Example 3.5(ii), but it has three chief factors. The difference arises because the last term of the composition series (3.1) is not normal in S_4 itself.

If it happens that every term in some composition series for G happens to be a normal subgroup of G (as in Example 3.12(ii)) then this will also be a chief series. In this case, its maximality amongst all series implies maximal amongst normal series. However, in general, a chief series does not need to be a composition series. If

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1} \tag{3.5}$$

is a chief series, then there may exist some subgroup H with $G_i < H < G_{i-1}$ and $H \trianglelefteq G_{i-1}$ (but H cannot be normal in G since (3.5) is a chief series). We could then insert H into (3.5) to create a longer series. This would, however, not be a normal series since $H \ntrianglelefteq G$.

In the description of chief series, we have referred to *the* chief factors of a group. Accordingly, we would like to know that they are uniquely determined by the group under consideration. The relevant theorem is:

Theorem 3.13 (Jordan–Hölder Theorem for Chief Series) *Let G be a group and let*

$$G = G_0 > G_1 > G_2 > \cdots > G_m = \mathbf{1}$$

and

$$G = H_0 > H_1 > H_2 > \cdots > H_n = \mathbf{1}$$

be chief series for G. Then $m = n$ and there is a one-one correspondence between the two collections of chief factors

$$G_0/G_1, \quad G_1/G_2, \quad \ldots, \quad G_{m-1}/G_m$$

and

$$H_0/H_1, \quad H_1/H_2, \quad \ldots, \quad H_{n-1}/H_n$$

such that the corresponding factors are isomorphic.

Comparing this to the Jordan–Hölder Theorem for composition series (Theorem 3.4), one can see that the two theorems basically tell us the same thing. They both say that for *maximal* series of a particular type (i.e., series or normal series, respectively), the factors occurring are essentially unique. In view of this, in these lecture notes we shall give a single proof that covers both cases simultaneously.

The Jordan–Hölder Theorem

The goal now is to present a unified argument that establishes the two versions of the Jordan–Hölder Theorem. Accordingly, fix a group G and a collection $\mathcal{S} = \mathcal{S}(G)$ of subgroups of G with the following properties:

($\mathcal{S}1$) the trivial group $\mathbf{1}$ and the group G are in $\mathcal{S}$;

($\mathcal{S}2$) if $H, K \in \mathcal{S}$, then $H \cap K \in \mathcal{S}$;

($\mathcal{S}3$) if $H, K, L \in \mathcal{S}$ with $H \leqslant L$ and $K \trianglelefteq L$, then $HK \in \mathcal{S}$.

One observes that if $\mathcal{S}$ is the collection of all subgroups of G or is the collection of all normal subgroups of G, then it satisfies these three properties.

If $M \in \mathcal{S}$, then define

$$\mathcal{S}(M) = \{ H \in \mathcal{S} \mid H \leqslant M \};$$

that is, the set of subgroups of M that belong to the collection $\mathcal{S}$. Observe that the collection $\mathcal{S}(M)$ satisfies the same three conditions ($\mathcal{S}1$)–($\mathcal{S}3$) when expressed to relate to subgroups of M (that is, $\mathbf{1}, M \in \mathcal{S}(M)$ and this collection of subgroups of M satisfies ($\mathcal{S}2$) and ($\mathcal{S}3$).) This will enable us to apply an induction argument in our proof.

Now consider two series for G whose terms belong to the collection $\mathcal{S}$:

$$G = G_0 > G_1 > G_2 > \cdots > G_m = \mathbf{1}$$

and

$$G = H_0 > H_1 > H_2 > \cdots > H_n = \mathbf{1}$$

(so we assume $G_i, H_j \in \mathcal{S},\ G_i \trianglelefteq G_{i-1}$ and $H_j \trianglelefteq H_{j-1}$ for all i and j). We shall say that these series are *isomorphic* if $m = n$ and there is an *isomorphism* between their factors; that is, a bijection between the collections

$$G_0/G_1,\ G_1/G_2,\ \dots,\ G_{m-1}/G_m \qquad \text{and} \qquad H_0/H_1,\ H_1/H_2,\ \dots,\ H_{n-1}/H_n$$

that maps each factor G_{i-1}/G_i to an isomorphic factor H_{j-1}/H_j.

We shall now prove the following theorem:

Theorem 3.14 (General Version of Jordan–Hölder Theorem) *Let G be a group with a collection of subgroups $\mathcal{S} = \mathcal{S}(G)$ satisfying Conditions ($\mathcal{S}1$)–($\mathcal{S}3$) and let*

$$G = G_0 > G_1 > G_2 > \cdots > G_m = \mathbf{1} \tag{3.6}$$

and

$$H = H_0 > H_1 > H_2 > \cdots > H_n = \mathbf{1} \tag{3.7}$$

be series for G consisting of subgroups from $\mathcal{S}$ that are maximal (in the sense that one cannot create a longer series by inserting an additional subgroup from $\mathcal{S}$). Then these two maximal series of subgroups from $\mathcal{S}$ are isomorphic.

The Jordan–Hölder Theorem (Theorem 3.4) and the version for chief series (Theorem 3.13) then follow as special cases when $\mathcal{S}$ is the set of all subgroups of G and when $\mathcal{S}$ is the set of all normal subgroups of G, respectively.

Proof: Assume without loss of generality that $m \leqslant n$. We shall proceed by induction on m. If $m = 0$, then $G = \mathbf{1}$ and hence necessarily $n = 0$ also. In this case there are no factors and the claimed isomorphism exists vacuously.

Suppose then that $m > 0$ and that the Jordan–Hölder Theorem holds when applied to maximal series of $\mathcal{S}$-subgroups of length $< m$ in any group with a suitable collection $\mathcal{S}$ of subgroups. To complete the induction step, we split into two cases.

Case 1: $G_1 = H_1$.

In this case, $G_0/G_1 = G/G_1 = H_0/H_1$. Furthermore,

$$G_1 > G_2 > \cdots > G_m = \mathbf{1}$$

and

$$G_1 = H_1 > H_2 > \cdots > H_n = \mathbf{1}$$

are maximal series for G_1 consisting of subgroups in $\mathcal{S}(G_1)$ of lengths $m-1$ and $n-1$, respectively. Hence, by induction, $m-1 = n-1$ (that is, $m = n$) and there is an isomorphism between the factors

$$G_1/G_2,\ \ldots,\ G_{m-1}/G_m \qquad \text{and} \qquad H_1/H_2,\ \ldots,\ H_{n-1}/H_n.$$

Since $G_0/G_1 = H_0/H_1$, this extends to the required isomorphism between the factors of the two series (3.6) and (3.7) for G.

Case 2: $G_1 \neq H_1$.

Conditions ($\mathcal{S}2$) and ($\mathcal{S}3$) ensure that $D = G_1 \cap H_1$ and G_1H_1 are subgroups in $\mathcal{S}$. Note that $G_1 \leqslant G_1H_1 \trianglelefteq G_0 = G$ and $H_1 \leqslant G_1H_1 \trianglelefteq H_0 = G$. Since the series (3.6) and (3.7) are maximal, we cannot insert G_1H_1 as a new term into them. Hence either $G_1H_1 = G$ or $G_1 = G_1H_1 = H_1$. We have assumed the latter does not occur and so $G_1H_1 = G$.

By the Second Isomorphism Theorem,

$$\frac{H_0}{H_1} = \frac{G}{H_1} = \frac{G_1H_1}{H_1} \cong \frac{G_1}{G_1 \cap H_1} = \frac{G_1}{D},$$

and this isomorphism $\theta\colon G_1/D \to G/H_1$ is given by $Dx \mapsto H_1x$ for $x \in G_1$. Similarly $G_0/G_1 = G/G_1 \cong H_1/D$. Now as D is a normal subgroup of G belonging to $\mathcal{S}$, it has a maximal series with terms in $\mathcal{S}$:

$$D = D_2 > D_3 > \cdots > D_k = \mathbf{1}$$

(For finite groups, this is established by the same argument as used for Corollary 3.8 and Proposition 3.11: just keep refining the series $D > \mathbf{1}$ by inserting terms from $\mathcal{S}$ until we cannot insert any more. When G is infinite, we are using a theorem, which appears on Problem Sheet III, that says if G has a maximal series with terms from $\mathcal{S}$ then so does any normal subgroup in $\mathcal{S}$.) We now have four series for G with terms from $\mathcal{S}$:

$$G = G_0 > G_1 > G_2 > \cdots > G_m = \mathbf{1} \tag{3.6}$$
$$G = G_0 > G_1 > D = D_2 > \cdots > D_k = \mathbf{1} \tag{3.8}$$
$$G = H_0 > H_1 > D = D_2 > \cdots > D_k = \mathbf{1} \tag{3.9}$$
$$G = H_0 > H_1 > H_2 > \cdots > H_n = \mathbf{1} \tag{3.7}$$

By assumption, (3.6) and (3.7) are maximal series with terms from $\mathcal{S}$. We shall show that the same is true for (3.8) and (3.9). We just need to ensure that one cannot insert a term from $\mathcal{S}$ between G_1 and D nor between H_1 and D.

Suppose $D \leqslant M \trianglelefteq G_1$ with $M \in \mathcal{S}$. Since $H_1 \trianglelefteq G$, use of Condition ($\mathcal{S}3$) shows that $MH_1 \in \mathcal{S}$ and this satisfies $H_1 \leqslant MH_1 \leqslant G$. Moreover the isomorphism $\theta\colon G_1/D \to G/H_1$, given by $Dx \mapsto H_1x$ for $x \in G_1$, maps the normal subgroup M/D to MH_1/H_1, so $MH_1 \trianglelefteq G$. Since the series (3.7) is maximal, we cannot insert MH_1 as a new term and therefore $MH_1 = H_1$ or G. If $MH_1 = H_1$, then M/D is mapped to the trivial subgroup of G/H_1 by the isomorphism θ and hence $M = D$. If $MH_1 = G$, then M/D is mapped to the whole quotient group G/H_1 and hence $M = G_1$. Therefore the series (3.8), and by symmetry the series (3.9) also, is maximal.

Now Case 1 applies to the series (3.6) and (3.8). We deduce that $m = k$ and there is an isomorphism between their factors. We have already observed that $G_0/G_1 \cong H_1/D$ and $H_0/H_1 \cong G_1/D$ and hence there is an isomorphism between the series (3.8) and (3.9) that simply interchanges the first pair of factors. Finally Case 1 applies to the series (3.9) and (3.7) (as we know the former has length $k = m$) and hence $m = k = n$ and there is an isomorphism between the factors of these series. In conclusion, $m = n$ and, upon composing the isomorphisms, we obtain the required isomorphism between the factors of (3.6) and (3.7).

This completes the induction step and establishes the Jordan–Hölder Theorem (in its general form, and therefore also the specific versions stated earlier). $\square$

Characteristic subgroups

We know that composition factors of a group are simple groups. One might ask what can be said about the chief factors. In the remainder of the chapter, we aim to describe these chief factors. To do so, we first introduce the following concept.

Definition 3.15 A subgroup H of a group G is said to be a *characteristic subgroup* of G if $x\phi \in H$ for all $x \in H$ and all automorphisms ϕ of G.

The definition requires that $H\phi \leqslant H$ for all automorphisms ϕ of G. It then follows that $H\phi^{-1} \leqslant H$ for any automorphism ϕ and applying ϕ then yields $H \leqslant H\phi$. Thus H is a characteristic subgroup if and only if $H\phi = H$ for all automorphisms ϕ of G.

The notation used for a characteristic subgroup is less consistent in the literature than for, say, being a normal subgroup. In these notes, we shall write

$$H \operatorname{char} G$$

to indicate that H is a characteristic subgroup of G.

Lemma 3.16 *Let G be a group.*

(i) *If $H \operatorname{char} G$, then $H \trianglelefteq G$.*

(ii) *If $K \operatorname{char} H$ and $H \operatorname{char} G$, then $K \operatorname{char} G$.*

(iii) *If $K \operatorname{char} H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.*

There are differences between characteristic subgroups and normal subgroups that should be noted. For example, note that in general:

- $K \trianglelefteq H \trianglelefteq G$ does not imply $K \trianglelefteq G$, in contrast to the assertion of part (ii) of the lemma.
- If $\phi \colon G \to K$ is a homomorphism and $H \operatorname{char} G$, then it does not follow necessarily that $H\phi \operatorname{char} G\phi$. (Consequently there is no version of the Correspondence Theorem that will work well with characteristic subgroups.)
- If $H \leqslant L \leqslant G$ and $H \operatorname{char} G$, then it does not necessarily follow that $H \operatorname{char} L$.

Problem Sheet III contains a question that addresses the existence of examples illustrating the above three points.

PROOF OF LEMMA 3.16: (i) If $x \in G$, then $\tau_x \colon g \mapsto g^x$ is an (inner) automorphism of G. Hence if $H \operatorname{char} G$, then

$$H^x = H\tau_x = H \qquad \text{for all } x \in G,$$

so $H \trianglelefteqslant G$.

(ii) Let ϕ be an automorphism of G. Then $H\phi = H$ (as $H \operatorname{char} G$). Hence the restriction $\phi|_H$ of ϕ to H is an automorphism of H and so we deduce

$$x\phi \in K \qquad \text{for all } x \in K$$

(since this is the effect that the restriction $\phi|_H$ has when applied to elements of K). Thus $K \operatorname{char} G$.

(iii) Let $x \in G$. Then $H^x = H$ (as $H \trianglelefteqslant G$) and therefore $\tau_x \colon g \mapsto g^x$ (for $g \in H$) is a bijective homomorphism $H \to H$; that is, τ_x induces an automorphism of H. Since $K \operatorname{char} H$, we deduce that $K^x = K\tau_x = K$. Thus $K \trianglelefteqslant G$. □

Minimal normal subgroups

Definition 3.17 Let G be a non-trivial group. A *minimal normal subgroup* of G is a non-trivial normal subgroup of G which has no non-trivial proper subgroup that is also normal in G.

Thus M is a minimal normal subgroup of G if

(i) $\mathbf{1} < M \trianglelefteqslant G$;

(ii) if $\mathbf{1} \leqslant N \leqslant M$ and $N \trianglelefteqslant G$, then either $N = \mathbf{1}$ or $N = M$.

An arbitrary group need not possess a minimal normal subgroup. For example, every non-trivial subgroup of the infinite cyclic group $G = \langle x \rangle$ is normal and infinite cyclic. Hence this G has no minimal normal subgroup. On the other hand, non-trivial finite groups always possess minimal normal subgroups: one simply takes a non-trivial normal subgroup of smallest order.

Lemma 3.18 *Let G be a non-trivial group and*

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1}$$

be a chief series for G. Then each chief factor G_{i-1}/G_i is a minimal normal subgroup of G/G_i.

In particular, the last term G_{n-1} of the chief series is a minimal normal subgroup of G.

PROOF: Since a chief series is a maximal normal series, there is no normal subgroup N of G satisfying $G_i < N < G_{i-1}$. Hence, by the Correspondence Theorem, there is no normal subgroup N/G_i of G/G_i satisfying $\mathbf{1} < N/G_i < G_{i-1}/G_i$. □

We shall prove the following description of minimal normal subgroups, which as a consequence of Lemma 3.18 therefore also describes all chief factors of a *finite* group.

Theorem 3.19 *A minimal normal subgroup of a finite group G is a direct product of isomorphic simple groups.*

We shall work towards the proof of this theorem in the remainder of the chapter. First we make the following definition.

Definition 3.20 A non-trivial group G is called *characteristically simple* if the only characteristic subgroups it has are $\mathbf{1}$ and G.

Lemma 3.21 *A minimal normal subgroup of a group is characteristically simple.*

PROOF: Let M be a minimal normal subgroup of the group G. Let K be a characteristic subgroup of M. Then

$$K \operatorname{char} M \trianglelefteq G,$$

so $K \trianglelefteq G$ by Lemma 3.16(iii). Thus minimality of M forces $K = \mathbf{1}$ or $K = M$. Hence M is indeed characteristically simple. □

Theorem 3.19 then follows immediately from the following result. (The advantage of proving Theorem 3.22 over a direct attempt on Theorem 3.19 is that we can concentrate only on the characteristically simple group rather than having to juggle both the minimal normal subgroup together with its embedding in the larger group.) The hypothesis that the group is finite is important for both theorems. We shall see that the proof uses the existence of certain minimal normal subgroups and this depends upon the finiteness hypothesis.

Theorem 3.22 *A characteristically simple finite group is a direct product of isomorphic simple groups.*

PROOF: Let G be a finite group which is characteristically simple. Let S be a minimal normal subgroup of G. (It is at this point that we use the fact that G is finite to guarantee the existence of S. Certainly $S \neq \mathbf{1}$ by definition. It is possible that $S = G$.) Consider the following set

$$\begin{aligned}\mathcal{D} = \{\, N \trianglelefteq G \mid N = S_1 \times S_2 \times \cdots \times S_k \text{ where each } S_i \text{ is a} \\ \text{minimal normal subgroup of } G \text{ isomorphic to } S \,\}.\end{aligned}$$

(Recall what is needed to be a direct product here: We need $N = S_1 S_2 \dots S_k$ and $S_i \cap S_1 \dots S_{i-1} S_{i+1} \dots S_k = \mathbf{1}$ for each i. The condition that each S_i is normal in N comes for free, since we required $S_i \trianglelefteq G$ in the definition of the set $\mathcal{D}$.)

Note that our original minimal normal subgroup S is a member of $\mathcal{D}$, so $\mathcal{D}$ certainly contains non-trivial members. Choose $N \in \mathcal{D}$ of largest possible order. (Here again we use finiteness of G to guarantee the existence of N.)

Claim: $N = G$.

Suppose our maximal member N of $\mathcal{D}$ is not equal to G. Since it belongs to our set $\mathcal{D}$,

$$N = S_1 \times S_2 \times \cdots \times S_k$$

where each S_i is a minimal normal subgroup of G isomorphic to S. As G is characteristically simple, N cannot be a characteristic subgroup of G. Hence there exists an automorphism ϕ of G such that

$$N\phi \not\leqslant N.$$

Therefore there exists i such that
$$S_i\phi \nleqslant N.$$

Now ϕ is an automorphism of G, so $S_i\phi$ is a minimal normal subgroup of G. Observe that $N \cap S_i\phi \trianglelefteq G$ and $N \cap S_i\phi$ is properly contained in $S_i\phi$ (as $S_i\phi \nleqslant N$). Therefore, by minimality of $S_i\phi$, we deduce $N \cap S_i\phi = \mathbf{1}$. It follows that

$$N \cdot S_i\phi = N \times S_i\phi = S_1 \times S_2 \times \cdots \times S_k \times S_i\phi$$

and
$$N \cdot S_i\phi \trianglelefteq G.$$

This shows that $N \cdot S_i\phi \in \mathscr{D}$. This contradicts N being a maximal member of $\mathscr{D}$.

This establishes the claim, so

$$G = N = S_1 \times S_2 \times \cdots \times S_k,$$

where each S_i is a minimal normal subgroup of G isomorphic to our original minimal normal subgroup S.

It remains to check that S is simple. If $J \trianglelefteq S_1$, then

$$J \trianglelefteq S_1 \times S_2 \times \cdots \times S_k = G.$$

Therefore, as S_1 is a minimal normal subgroup of G, we must have $J = \mathbf{1}$ or $J = S_1$. Hence S_1 (and, as it is isomorphic, S also) is simple.

We have consequently shown that, indeed, G is a direct product of isomorphic simple groups. □

Chapter 4

Semidirect Products

The purpose of this chapter is to discuss some ways in which groups can be built from smaller groups. In some sense, we are considering how to reverse the decomposition that composition series (or, indeed, chief series) and the Jordan–Hölder Theorem give us. One question you might ask is:

> If a finite group G has composition factors S_1, S_2, $\dots$, S_k, what are the possibilities for G?

This turns out to be an extremely difficult question to answer in general, but the content of this chapter will provide one of the more straightforward ways to build a group with a given normal subgroup N and a given quotient Q.

This construction arises naturally in the context of attempting to classify groups given specific information. We shall first illustrate the sort of question one might consider and answer it using the direct product construction. First recall that if a group G has normal subgroups M and N such that $G = MN$ and $M \cap N = \mathbf{1}$, then

$$G \cong M \times N = \{ (x, y) \mid x \in M,\ y \in N \}.$$

The multiplication in the direct product $M \times N$ is componentwise:

$$(x_1, y_1)(x_2, y_2) = (x_1x_2, y_1y_2)$$

for $x_1, x_2 \in M$ and $y_1, y_2 \in N$. Accordingly if we find such normal subgroups M and N and we fully understand their structure, then this implies that we have fully determined the possibility for G. (A verification of the above isomorphism, as revision of direct products, appeared on Problem Sheet I.)

Example 4.1 *Classify the finite groups of order* 45*.*

Solution: Let G be a group of order $45 = 3^2 \cdot 5$. The number of Sylow 3-subgroups is congruent to $1 \pmod 3$ and divides 5. Therefore G has a unique Sylow 3-subgroup T. The number of Sylow 5-subgroups is congruent to $1 \pmod 5$ and divides 9 and we conclude that G also has a unique Sylow 5-subgroup F. Since the conjugate of a Sylow p-subgroup is again a Sylow p-subgroup, it follows that $T \trianglelefteq G$ and $F \trianglelefteq G$. By Lagrange's Theorem, $T \cap F = \mathbf{1}$ (as these subgroups have coprime order) and hence

$$|TF| = \frac{|T| \cdot |F|}{|T \cap F|} = 3^2 \cdot 5 = 45.$$

We deduce that $G = TF$. This shows that G satisfies the criteria to be a direct product:

$$G \cong T \times F.$$

Now T is a group of order 9 and we know (from *MT4003*) that a group of order p^2 (for p prime) is abelian and is isomorphic to either C_{p^2} or $C_p \times C_p$. Hence $T \cong C_9$ or $C_3 \times C_3$. Equally, F is a group of order 5 (which is prime), so $F \cong C_5$. We conclude that there are (up to isomorphism) two possibilities for G:

$$G \cong C_9 \times C_5 \qquad \text{or} \qquad G \cong C_3 \times C_3 \times C_5$$

The Classification of Finite Abelian Groups (again from *MT4003*) tells us that these groups are not isomorphic. Hence there are precisely two groups (up to isomorphism) of order 45. □

This method of using direct products works very well when the number theory happens to land nicely and one can find the required normal subgroups. However, consider the case of trying to classify groups of order $20 = 2^2 \cdot 5$. Application of Sylow's Theorem shows that there is a unique Sylow 5-subgroup, but all can conclude about the Sylow 2-subgroups is that the number of them is either 1 or 5. We would have one normal subgroup (the Sylow 5-subgroup) but not necessarily the two normal subgroups required for the direct product. We now introduce a new construction to cover this situation.

Semidirect products

Suppose that a group G can be expressed as $G = NH$ with $H \cap N = \mathbf{1}$ and only $N \trianglelefteq G$. (The direct product case is when $H \trianglelefteq G$ also holds.) An element in G is expressible as $g = nh$ where $h \in H$ and $n \in N$. If we attempt to multiply two elements of G, then we calculate

$$(n_1h_1)(n_2h_2) = n_1(h_1n_2h_1^{-1}) \cdot h_1h_2$$

Here $h_1h_2 \in H$, $h_1n_2h_1^{-1} \in N$ (as $N \trianglelefteq G$) and so $n_1(h_1n_2h_1^{-1}) \in N$. This formula shows us that to be able to work effectively in G, we need to be able to (i) multiply in H, (ii) multiply in N, and (iii) conjugate elements of N by elements of H. The semidirect product construction is designed to encode these three pieces of information.

We shall need the following object as part of the construction. Recall (from Definition 1.32) that an automorphism of a group G is a bijective homomorphism $G \to G$.

Definition 4.2 Let G be a group. The *automorphism group* of G is denoted by $\operatorname{Aut} G$ and consists of all automorphisms of G:

$$\operatorname{Aut} G = \{\, \phi \colon G \to G \mid \phi \text{ is an automorphism} \,\}.$$

The product of two automorphisms ϕ and ψ is the composite $\phi\psi$.

It was verified on Problem Sheet I that $\operatorname{Aut} G$ is a group. The verification is very similar to the proof that a symmetric group forms a group. Indeed one can observe $\operatorname{Aut} G$ is a subgroup of the symmetric group $\operatorname{Sym}(G)$.

Definition 4.3 Let H and N be groups and let $\phi \colon H \to \operatorname{Aut} N$ be a homomorphism. The (*external*) *semidirect product* of N by H via ϕ is denoted by $N \rtimes_\phi H$ and is the set

$$N \rtimes_\phi H = \{\, (n, h) \mid n \in N,\ h \in H \,\}$$

with multiplication given by

$$(n_1, h_1)(n_2, h_2) = (n_1 n_2^{(h_1\phi)^{-1}}, h_1 h_2).$$

If $h \in H$, then $h\phi$ is an automorphism of N and we are writing $n^{h\phi}$ for the image of an element $n \in N$ under the automorphism $h\phi$. (The reason for using exponential notation is twofold: firstly to make the notation easier to distinguish and secondly to be suggestive of conjugation in a way that we shall use later.) This means that the above multiplication in $N \rtimes_\phi H$ at least has meaning.

Aside: If $G = NH$ where $N \trianglelefteq G$ and $H \leqslant G$, then also $G = HN$ (see Lemma 1.19). As a consequence, it should not be surprising to learn that we can also define an external semidirect product as $H \ltimes_\phi N$ with respect to some homomorphism $\phi\colon H \to \operatorname{Aut} N$ with elements denoted by pair (h, n) where $h \in H$ and $n \in N$. This can indeed be done and the result is a group isomorphic to the one given in Definition 4.3. In some ways the formulae involved are slightly more pleasant with this alternative construction, but the majority of the literature on group theory seems to use the $N \rtimes H$ version with the normal subgroup N on the left. Accordingly these lecture notes follow the standard convention of writing $N \rtimes_\phi H$ for the semidirect product.

Proposition 4.4 *The semidirect product $N \rtimes_\phi H$ is a group.*

PROOF: We need to check the axioms of a group. First associativity (which is straightforward, but messy):

$$\begin{aligned}
\big((n_1, h_1)(n_2, h_2)\big)(n_3, h_3) &= \big(n_1 n_2^{(h_1\phi)^{-1}}, h_1 h_2\big)(n_3, h_3) \\
&= \big(n_1 n_2^{(h_1\phi)^{-1}} n_3^{((h_1 h_2)\phi)^{-1}}, h_1 h_2 h_3\big) \\
&= \big(n_1 n_2^{(h_1\phi)^{-1}} n_3^{(h_2\phi)^{-1}(h_1\phi)^{-1}}, h_1 h_2 h_3\big),
\end{aligned}$$

using the fact that ϕ is a homomorphism, so that $\big((h_1 h_2)\phi\big)^{-1} = \big((h_1\phi)(h_2\phi)\big)^{-1} = (h_2\phi)^{-1}(h_1\phi)^{-1}$. On the other hand

$$\begin{aligned}
(n_1, h_1)\big((n_2, h_2)(n_3, h_3)\big) &= (n_1, h_1)\big(n_2 n_3^{(h_2\phi)^{-1}}, h_2 h_3\big) \\
&= \big(n_1 (n_2 n_3^{(h_2\phi)^{-1}})^{(h_1\phi)^{-1}}, h_1 h_2 h_3\big) \\
&= \big(n_1 n_2^{(h_1\phi)^{-1}} n_3^{(h_2\phi)^{-1}(h_1\phi)^{-1}}, h_1 h_2 h_3\big)
\end{aligned}$$

using the fact that the inverse of $h_1\phi$ is an automorphism, in particular a homomorphism, and so maps products to products. Comparing these formulae we deduce that the binary operation on the semidirect product is associative.

Identity:

$$(1, 1)(n, h) = (1n^{(1\phi)^{-1}}, 1h) = (n^{\mathrm{id}}, h) = (n, h)$$

(since the automorphism 1ϕ must be the identity) and

$$(n, h)(1, 1) = (n1^{(h\phi)^{-1}}, h1) = (n1, h) = (n, h)$$

Hence $(1, 1)$ is the identity element in $N \rtimes_\phi H$.

Inverses:

$$\begin{aligned}
(n,h)\big((n^{h\phi})^{-1},h^{-1}\big) &= \big(n((n^{h\phi})^{-1})^{(h\phi)^{-1}}, hh^{-1}\big) \\
&= \big(n(n^{(h\phi)(h\phi)^{-1}})^{-1}, 1\big) && (*) \\
&= (n(n^{\mathrm{id}})^{-1}, 1) \\
&= (nn^{-1}, 1) \\
&= (1,1),
\end{aligned}$$

since $(*)$ $(h\phi)^{-1}$ is a homomorphism so maps inverses to inverses, and

$$\begin{aligned}
\big((n^{h\phi})^{-1},h^{-1}\big)(n,h) &= \big((n^{h\phi})^{-1}n^{((h^{-1})\phi)^{-1}}, h^{-1}h\big) \\
&= \big((n^{h\phi})^{-1}n^{h\phi}, 1\big) && (\dagger) \\
&= (1,1)
\end{aligned}$$

since $(\dagger)$ ϕ is a homomorphism so maps inverses to inverses. Thus $\big((n^{h\phi})^{-1},h^{-1}\big)$ is the inverse of (n,h) in $N \rtimes_\phi H$.

This completes the proof that $N \rtimes_\phi H$ is a group. $\square$

We have now successful constructed a group G from two groups N and H and the specified homomorphism $\phi\colon H \to \operatorname{Aut} N$. The following summarizes the properties of this group, though we leave the proofs to Problem Sheet IV since they are mostly an exercise in computation (and therefore a good way to practice working with this construction).

Theorem 4.5 *Let H and N be groups, $\phi\colon H \to \operatorname{Aut} N$ be a homomorphism and $G = N \rtimes_\phi H$ be the semidirect product of N by H via ϕ. Then*

(i) $\bar{N} = \{\, (n,1) \mid n \in N \,\}$ *is a normal subgroup of G that is isomorphic to N;*

(ii) $\bar{H} = \{\, (1,h) \mid h \in H \,\}$ *is a subgroup of G that is isomorphic to H;*

(iii) $\bar{H} \cap \bar{N} = \mathbf{1}$ *and* $G = \bar{N}\bar{H}$;

(iv) *the quotient group $G/\bar{N}$ is isomorphic to H;*

(v) *conjugation of an element of $\bar{N}$ by an element of $\bar{H}$ is determined by the homomorphism ϕ:*

$$(1,h)^{-1}(n,1)(1,h) = (n^{h\phi}, 1) \qquad \text{for all } n \in N \text{ and } h \in H.$$

In the case of a direct product, sometimes one refers to the *external* direct product by which one means the group defined as ordered pairs

$$G = G_1 \times G_2 = \{\, (x,y) \mid x \in G_1,\ y \in G_2 \,\}$$

with componentwise multiplication and the *internal* direct product by which one means a group G with two normal subgroups G_1 and G_2 satisfying $G = G_1G_2$ and $G_1 \cap G_2 = \mathbf{1}$. What we has observed previously (see, for example, Problem Sheet I for one direction) is that an internal direct product is isomorphic to the corresponding external direct product and that the external direct product has normal subgroups that satisfy the properties of an internal direct product. As a consequence, we tend to move fluidly between the two viewpoints and rarely distinguish between them. (These groups are isomorphic; i.e., essentially the same.)

In the case of semidirect products, a similar viewpoint is taken. We have just described an external version, namely the construction appearing in Definition 4.3, and we have just recorded its basic structural properties in Theorem 4.5. We shall take some of these properties as being the *internal* description for a group to be a semidirect product and show that indeed a group satisfying these properties is isomorphic to a group built using the semidirect product construction.

We shall therefore assume that G is a group with two subgroups H and N such that

(i) N is a normal subgroup of G,

(ii) $G = NH$, and

(iii) $H \cap N = \mathbf{1}$.

When these conditions hold, we shall sometimes say that H is a *complement* to N and write $G = N \rtimes H$. Our goal is to show that it is isomorphic to some external semidirect product of N by H. There are various stages to proceed through, but the most significant is to work out what the required homomorphism $\phi\colon H \to \operatorname{Aut} N$ should be.

We shall define a map $\theta\colon N \times H \to G$ by

$$(n, h) \mapsto nh.$$

(At this point, we do not assume any group theoretical structure on the *set* $N \times H$. It will eventually become a semidirect product.) Now $G = NH$, so every element of G can be written in the form nh where $n \in N$ and $h \in H$. Therefore θ is surjective. Suppose $nh = n'h'$ where $n, n' \in H$ and $h, h' \in N$. Then

$$h(h')^{-1} = n^{-1}n' \in H \cap N = \mathbf{1}.$$

This forces $h = h'$ and $n = n'$. Therefore this expression for an element of G as a product is unique and we deduce that θ is injective.

We now know that θ is a bijection, but we seek to endow the domain of θ with the structure of an (external) semidirect product and consequently need to specify a homomorphism $\phi\colon H \to \operatorname{Aut} N$ to use when constructing this group. Let $h \in H$. Then $N^h = N$ in the group G since $N \trianglelefteq G$. Hence conjugation by h determines a map

$$\begin{aligned} \phi_h\colon N &\to N \\ n &\mapsto n^h. \end{aligned}$$

Its inverse is $\phi_{h^{-1}}\colon n \mapsto n^{h^{-1}}$, so ϕ_h is a bijection. Also

$$(mn)\phi_h = h^{-1}(mn)h = h^{-1}mh \cdot h^{-1}nh = (m\phi_h)(n\phi_h).$$

Hence $\phi_h \in \operatorname{Aut} N$. Finally

$$n\phi_{hk} = (hk)^{-1}n(hk) = k^{-1}(h^{-1}nh)k = n\phi_h\phi_k$$

for $n \in N$, so

$$\phi_{hk} = \phi_h\phi_k \qquad \text{for all } h, k \in H.$$

We deduce that $\phi\colon h \mapsto \phi_h$ is a homomorphism $H \to \operatorname{Aut} N$. We use this map ϕ to construct the semidirect product $N \rtimes_\phi H$.

In view of what we have already shown, we can now view the bijection θ as a map

$$\begin{aligned}\theta\colon N \rtimes_\phi H &\to G\\ (n,h) &\mapsto nh.\end{aligned}$$

Let $(n_1,h_1),(n_2,h_2) \in N \rtimes_\phi H$. Then

$$\begin{aligned}\big((n_1,h_1)(n_2,h_2)\big)\theta &= (n_1 n_2^{(h_1\phi)^{-1}}, h_1h_2)\theta\\ &= (n_1 n_2^{h_1^{-1}}, h_1h_2)\theta\\ &= n_1 n_2^{h_1^{-1}} h_1h_2\\ &= n_1 \cdot h_1 n_2 h_1^{-1} \cdot h_1 h_2\\ &= n_1h_1n_2h_2\\ &= (n_1,h_1)\theta \cdot (n_2,h_2)\theta.\end{aligned}$$

Hence θ is a homomorphism and consequently is an isomorphism. We have established the following theorem:

Theorem 4.6 *Let G be a group with a normal subgroup N and a subgroup H such that $G = NH$ and $H \cap N = \mathbf{1}$. Then G is isomorphic to the semidirect product $N \rtimes_\phi H$ where $\phi\colon H \to \operatorname{Aut} N$ is the homomorphism given by*

$$h\phi\colon n \mapsto n^h$$

for $n \in N$ and $h \in H$. □

In view of this theorem, we tend to view groups that satisfy the hypotheses as being "the same" (since that is what isomorphic means) as a semidirect product as constructed by Definition 4.3. We use the term *semidirect product* to refer to both situations and we try to move between both viewpoints quite smoothly.

Indeed, for notational simplicity, from now on we shall drop the brackets in the semidirect product construction. To be precise, let N and H be groups and $\phi\colon H \to \operatorname{Aut} N$ be a homomorphism from H to the automorphism group of N. We shall write $G = N \rtimes_\phi H$ (and sometimes simply $N \rtimes H$ if ϕ is sometimes understood) to be the set of products

$$G = \{\, nh \mid n \in N,\ h \in H \,\}$$

with multiplication given by

$$n_1h_1 \cdot n_2h_2 = n_1 n_2^{(h_1\phi)^{-1}} h_1h_2.$$

We identify N with the set of elements of the form $n1$ for $n \in N$ and H with the set of elements of the form $1h$ for $h \in H$. Then, according to Theorem 4.5, the conjugation in G of an element of N by an element of H is determined by the homomorphism ϕ:

$$h^{-1}nh = n^{h\phi} \qquad \text{for } h \in H \text{ and } n \in N.$$

It is for this reason that we sometimes omit reference to the homomorphism ϕ: a semidirect product is determined by the two groups N and H from which it is built together with a description of the result of conjugating an element of N by an element of H.

Warnings:

(i) Note, however, that to construct a semidirect product $G = N \rtimes H$ you cannot simply specify $h^{-1}nh$ for each $h \in H$ and $n \in N$. If there isn't some homomorphism $\phi\colon H \to \operatorname{Aut} N$ involved (as we have described above) then the result will not be a group.

(ii) If G is a group which is not simple, it does not necessarily decompose as a semidirect product. There will exist a non-trivial proper normal subgroup N of G, but there is no guarantee that there will be a complement H to N in G. For example, if N is *any* non-trivial proper normal subgroup of the quaternion group Q_8, there does not exist a subgroup H of Q_8 with $N \cap H = \mathbf{1}$ and $NH = Q_8$. Consequently Q_8 cannot be decomposed in a proper way as a semidirect product.

Applications of semidirect products

We shall now give a few examples of semidirect products. In the first example below, we shall classify the groups of order 20. The method will be to show that if $|G| = 20$ then $G \cong N \rtimes_\phi H$ for some N, H and ϕ using Theorem 4.6. We shall describe all the options for the three ingredients N, H and ϕ in the semidirect product construction and hence will have determined all the groups of order 20.

Example 4.7 *Determine how many groups of order* 20 *there are (up to isomorphism).*

Solution: Let G be a group of order $20 = 2^2 \cdot 5$. By Sylow's Theorem, the number of Sylow 5-subgroups is congruent to 1 (mod 5) and divides 4. Hence G has a unique Sylow 5-subgroup F. Then we know $F \trianglelefteq G$ and $|F| = 5$.

Let T be a Sylow 2-subgroup of G, so $|T| = 4$. Then $T \cap F = \mathbf{1}$ by Lagrange's Theorem, while Lemma 1.19 tells us that

$$|FT| = \frac{|F| \cdot |T|}{|F \cap T|} = \frac{5 \cdot 4}{1} = 20.$$

Hence $G = FT$, $F \trianglelefteq G$ and $T \cap F = \mathbf{1}$. Thus, by Theorem 4.6, $G \cong F \rtimes_\phi T$, the semidirect product of F by T with respect to a suitable homomorphism $\phi\colon T \to \operatorname{Aut} F$. In order to describe all possible such groups G, we need to determine the possibilities for F, T and the homomorphism ϕ.

We know that $|F| = 5$, so $F \cong C_5$. Fix a generator x for F, so $o(x) = 5$. Since $|T| = 4$, there are two possibilities for T: either $T \cong C_4$ or $C_2 \times C_2$. In order to determine the possibilities for ϕ, we first need to be able to describe the automorphism group of F.

An automorphism α of F of is determined by its effect on the generator x, since if we know what $x\alpha$ is then $(x^i)\alpha = (x\alpha)^i$ is determined by α. Furthermore, since the powers of x are mapped to powers of $x\alpha$, in order that α be surjective, necessarily $x\alpha$ must be a generator of F. Thus $x\alpha = x$, x^2, x^3 or x^4. (Note that all the non-identity elements of F have order 5 and therefore generate F.) Finally, if g is a generator of F, then $x^i \mapsto g^i$ does define an automorphism of F. Thus all four choices do indeed give automorphisms of F and so

$$|\operatorname{Aut} F| = 4.$$

In fact, we can determine which group of order 4 this automorphism group is. Consider the automorphism β given by $x \mapsto x^2$ and compute its powers:

$$\begin{aligned}
x\beta^2 &= (x\beta)\beta = (x^2)\beta = (x\beta)^2 = (x^2)^2 = x^4 \\
x\beta^3 &= (x\beta^2)\beta = (x^4)\beta = (x\beta)^4 = (x^2)^4 = x^8 = x^3 \\
x\beta^4 &= (x\beta^3)\beta = (x^3)\beta = (x\beta)^3 = (x^2)^3 = x^6 = x.
\end{aligned}$$

So $\beta^4 = \mathrm{id}_F$ and $o(\beta) = 4$. Thus $\mathrm{Aut}\,F = \langle \beta \rangle$ is cyclic of order 4.

We are now able to describe all the groups of order 20. We shall consider each possibility for T in turn and determine what the options are for the homomorphism $\phi \colon T \to \mathrm{Aut}\,F$.

Case 1: $T \cong C_4$.

If $\phi \colon T \to \mathrm{Aut}\,F$ is a homomorphism, then the image of ϕ must be a subgroup of $\mathrm{Aut}\,F = \langle \beta \rangle$ that could occur as an image of the cyclic group T. Since $|T| = |\mathrm{Aut}\,F| = 4$, there are three possibilities: $T\phi = \mathbf{1}$, $\langle \beta^2 \rangle$ or $\langle \beta \rangle$ (these three being the unique subgroups of $\mathrm{Aut}\,F$ of order 1, 2 and 4, respectively).

(i) If $T\phi = \mathbf{1}$, then every element of T commutes with all elements of F (since if $g\phi$ is the automorphism that $g \in T$ induces when it acts by conjugation on F). Therefore in this case, G is a direct product:

$$G \cong F \times T = C_5 \times C_4 \cong C_{20}$$

(using the standard fact, from *MT4003*, that $C_m \times C_n \cong C_{mn}$ when $\gcd(m, n) = 1$).

(ii) If $T\phi = \langle \beta^2 \rangle$, then $|\ker \phi| = 2$; that is $\ker \phi$ is the unique subgroup of order 2 in $T \cong C_4$. If y is a generator for T, then $y \notin \ker \phi$ (as y has order 4) and therefore $y\phi = \beta^2$. Hence in this case

$$G \cong F \rtimes_\phi T$$

where a generator y of T induces the automorphism $\beta^2 \colon x \mapsto x^4$ when it acts by conjugation: $y^{-1}xy = x^4$.

(iii) If $T\phi = \langle \beta \rangle$, then ϕ is actually an isomorphism from T to $\mathrm{Aut}\,F$ (both are cyclic of order 4). Hence one of the generators of T will be mapped to our chosen generator β of $\mathrm{Aut}\,F$: there exists a generator y for T satisfying $y\phi = \beta$. Hence in this case

$$G \cong F \rtimes_\phi T$$

where some generator y of T induces the automorphism β when it acts by conjugation: $y^{-1}xy = x^2$.

We have therefore constructed three semidirect products $F \rtimes_\phi T$ of $F = C_5$ by $T = C_4$ using three homomorphisms $\phi \colon T \to \mathrm{Aut}\,F$. Furthermore, we have shown that every group of order 20 with cyclic Sylow 2-subgroup has one of these forms. It remains to observe that none of these examples are isomorphic to each other.

First note that the group in (i) is abelian, but the other two semidirect products constructed are non-abelian (as the two elements x and y used do not commute). Let $G = F \rtimes_\phi T$ be the group occurring in construction (iii), so $\phi \colon T \to \mathrm{Aut}\,F$ is an isomorphism, $F = \langle x \rangle$, $T = \langle y \rangle$ and $y\phi = \beta$. Suppose that $g \in \mathrm{Z}(G)$. Since $G = FT$, we can write $g = x^i y^j$ where $0 \leqslant i \leqslant 4$ and $0 \leqslant j \leqslant 3$. Then

$$x = x^g = x^{x^i y^j} = x^{y^j} = x\beta^j.$$

Since an automorphism of F is determined by its effect on x, we deduce $\beta^j = \mathrm{id}_F$ and so $j = 0$. If $i \neq 0$, then some power of $g = x^i$ equals x and we would deduce $x \in \mathrm{Z}(G)$. However, this is not true since $x^y = x^2 \neq x$. This shows that $g = 1$ and we have shown that the group constructed in (iii) has trivial centre.

In the case of the group $G = F \rtimes_\phi T$ constructed in (ii), the homomorphism $\phi\colon T \to \operatorname{Aut} F$ has kernel of order 2. We chose y to be a generator for $T \cong C_4$ and so $y^2 \in \ker \phi$. This means that y^2 commutes with x and all its powers. On the other hand, clearly y^2 commutes with all powers of y. Therefore y^2 commutes with all elements of $G = FT$ and so y^2 is a non-identity element of $\mathrm{Z}(G)$ in this case.

We conclude that the two non-abelian groups of order 20 constructed above are not isomorphic to each other as they have different centres. Therefore there are precisely *three* distinct groups of order 20 with cyclic Sylow 2-subgroups.

Case 2: $T \cong C_2 \times C_2$.

If $\phi\colon T \to \operatorname{Aut} F$ in this case, then the image of ϕ must be cyclic as it is a subgroup of $\operatorname{Aut} F = \langle \beta \rangle$ but also every element in $T\phi$ has order dividing 2 (as this is true for all elements of T). Therefore there are just two possibilities: $T\phi = \mathbf{1}$ or $\langle \beta^2 \rangle$.

(i) If $T\phi = \mathbf{1}$, then every element of T commutes with every element of F and so we obtain a direct product:

$$\begin{aligned} G \cong F \times T &= C_5 \times C_2 \times C_2 \\ &\cong C_2 \times C_{10}. \end{aligned}$$

(ii) If $T = \langle \beta^2 \rangle$, then $|\ker \phi| = 2$. Choose an element y that generates $\ker \phi$ and $z \in T \setminus \ker \phi$. Then $y\phi = 1$ and $z\phi = \beta^2$. Note that these two elements y and z generate T and so from these two images we can construct the image of every element of T under ϕ. Thus we have completely determined the semidirect product:

$$G \cong F \rtimes_\phi T$$

where two generators y and z for T are such that y commutes with all elements in F and $z^{-1}xz = x\beta^2 = x^4$.

As in Case 1, both these constructions define different groups: The second group is non-abelian and so is not isomorphic to $C_2 \times C_{10}$. (In fact, the group in construction (ii) is isomorphic to the dihedral group D_{20} of order 20, as is observed on Problem Sheet IV.) Finally, none of them are isomorphic to any of the three groups in Case 1 since the Sylow 2-subgroups are different.

Conclusion: There are, up to isomorphism, precisely *five* groups of order 20. □

The following example makes use of ideas from linear algebra to describe the conjugation action in the semidirect product.

Example 4.8 In this example, we shall construct a non-abelian group G of order $147 = 3 \cdot 7^2$ with non-cyclic Sylow 7-subgroup. The number of Sylow 7-subgroups in a group of order 147 divides 3 and is congruent to 1 (mod 7). Hence there is a unique Sylow 7-subgroup P. By assumption, $P \cong C_7 \times C_7$.

Now (temporarily) write the group operation in the abelian group P additively, so $P = \mathbb{F}_7 \oplus \mathbb{F}_7$, where $\mathbb{F}_7 = \mathbb{Z}/7\mathbb{Z}$ is the field containing 7 elements. Thus we can view P as a vector space of dimension 2 over the field $\mathbb{F}_7$. A homomorphism $P \to P$ then corresponds to a linear transformation, so automorphisms correspond to invertible linear transformations:

$$\operatorname{Aut} P \cong \mathrm{GL}_2(\mathbb{F}_7) = \{ A \mid A \text{ is a } 2 \times 2 \text{ matrix over } \mathbb{F}_7 \text{ with } \det A \neq 0 \}.$$

If z is a generator for the Sylow 3-subgroup of G, then z induces an automorphism of P via conjugation; that is, z induces an invertible linear transformation T of P such that $T^3 = I$. Hence the *minimal polynomial* $m_T(X)$ of T divides

$$X^3 - 1 = (X-1)(X-2)(X-4) \qquad (\text{over } \mathbb{F}_7)$$

and must be of degree at most 2. In particular, $m_T(X)$ is a product of linear factors, so T is diagonalisable. Hence we may choose a basis $\{x, y\}$ for P such that the matrix of T with respect to this basis is

$$\begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix},$$

where $\lambda, \mu \in \{1, 2, 4\}$. For example, one such group occurs when we select $\lambda = 2$ and $\mu = 4$. Returning to multiplicative notation, with this choice of λ and μ, we will have constructed a semidirect product $(C_7 \times C_7) \rtimes C_3$ with generators x and y for the Sylow 7-subgroup and a generator z for the Sylow 3-subgroup that acts by conjugation in the following way:

$$z^{-1}xz = x^2 \qquad \text{and} \qquad z^{-1}yz = y^4.$$

Wreath products

We finish this chapter by giving another example of the use of the semidirect product construction.

Let G and H be any groups and suppose that G acts on the set $\Omega = \{1, 2, \dots, n\}$. (In fact, we can use any finite set Ω — and with some minor adjustment even an infinite set — but we choose to use this specific set to make the notation more convenient.) Take B to be the direct product of n copies H_i of the group H:

$$B = H_1 \times H_2 \times \cdots \times H_n$$

Elements of B are written as $b = (h_1, h_2, \dots, h_n)$, a sequence of elements of H indexed by Ω; that is,

$$B = \{ (h_1, h_2, \dots, h_n) \mid h_1, h_2, \dots, h_n \in H \}.$$

Now define an action of G on B by permuting the entries of these elements:

$$(h_1, h_2, \dots, h_n)^g = (h_{1g^{-1}}, h_{2g^{-1}}, \dots, h_{ng^{-1}}) \tag{4.1}$$

This formula indicates that applying g move the contents of entry i of the element $b \in B$ into the entry ig in b^g. We verified on Problem Sheet II that this is an action of G.

Recall that the subgroup H_i is identified with the following set of elements of B:

$$H_i = \{ (1, \dots, 1, h, 1, \dots, 1) \mid h \in H \}$$

(where the h is in the ith position). If $g \in G$ has the effect of moving i to j (that is, $ig = j$) and $b = (1, \dots, 1, h, 1, \dots, 1) \in H_i$, then the jth entry of b^g is the entry in position $jg^{-1} = i$ of b. Hence, for this g and b, we compute

$$b^g = (1, \dots, 1, h, 1, \dots, 1) \qquad \text{with } h \text{ in the } j\text{th position.}$$

This shows that

$$H_i^g = H_j = H_{ig}$$

and, in conclusion, we deduce that G permutes the factors of B in the same way as it permutes the elements of Ω.

Write $\phi\colon G \to \mathrm{Sym}(B)$ for the associated permutation representation and let $\phi_g = g\phi$ for $g \in G$; that is,

$$\phi_g\colon (h_1, h_2, \dots, h_n) \mapsto (h_{1g^{-1}}, h_{2g^{-1}}, \dots, h_{ng^{-1}}).$$

In fact, ϕ_g is an automorphism of B because

$$\begin{aligned}\big((h_1, h_2, \dots, h_n)(k_1, k_2, \dots, k_n)\big)^{\phi_g} &= (h_1k_1, h_2k_2, \dots, h_nk_n)^g \\ &= (h_{1g^{-1}}k_{1g^{-1}}, h_{2g^{-1}}k_{2g^{-1}}, \dots, h_{ng^{-1}}k_{ng^{-1}}) \\ &= (h_{1g^{-1}}, h_{2g^{-1}}, \dots, h_{ng^{-1}})(k_{1g^{-1}}, k_{2g^{-1}}, \dots, k_{ng^{-1}}) \\ &= (h_1, h_2, \dots, h_n)^{\phi_g} \cdot (k_1, k_2, \dots, k_n)^{\phi_g}.\end{aligned}$$

This shows that ϕ_g is a homomorphism, while we already know that it is a permutation and consequently bijective. Hence $\phi\colon G \to \mathrm{Aut}\, B$. We may therefore construct the semidirect product

$$W = B \rtimes_\phi G$$

with respect to this action.

Definition 4.9 Let G and H be groups and let G act on the set $\Omega = \{1, 2, \dots, n\}$. The semidirect product W constructed above is called the *wreath product* of H by G with respect to the action of G on Ω. We shall use the notation $W = H \mathbin{\mathrm{wr}_\Omega} G$ to denote this group. The normal subgroup B is usually called the *base group* of the wreath product.

In summary, a wreath product is a special type of semidirect product that is built with normal subgroup B that is a direct product of copies of one of our starting ingredients H and the complement G to B acts to permute the direct factors of B in the same way that it permutes the elements of the set Ω. (The formula (4.1) specifies explicitly how one computes the conjugation of an element of B by an element of H. It is this that we have to depend upon when computing in the wreath product.) We shall use wreath products in our discussion of subgroups of symmetric groups in Chapter 7.

Let us now consider an example of the wreath product construction:

Example 4.10 Let $G = S_2 \cong C_2$ acting on the set $\Omega = \{1, 2\}$ in the natural way. Take $H = \langle a \rangle \cong C_2$. We then construct the wreath product $W = C_2 \mathbin{\mathrm{wr}_\Omega} S_2 = C_2 \mathbin{\mathrm{wr}_\Omega} C_2$ with respect to this action. The base group of W is

$$B = C_2 \times C_2 = \{(1,1), (a,1), (1,a), (a,a)\} \cong V_4.$$

To simplify notation, and in line with what we have done earlier, we shall write elements of W as products $b\sigma$ where $b \in B$ and $\sigma \in S_2$.

Observe that $W = B \rtimes S_2$ is a group of order 8. It is non-abelian because

$$(a,1)^{(1\ 2)} = (1,a) \neq (a,1).$$

Hence W is isomorphic to one of the two non-abelian groups of order 8. Observe that W contains many elements of order 2, for example, $(a,1)$, $(1,a)$, (a,a) and $(1\ 2)$. Hence $W \cong D_8$, the dihedral group of order 8.

As an example of computing within this wreath product, observe that

$$\begin{aligned}\big((a,1)\,(1\;2)\big)^2 &= (a,1)\,(1\;2)\,(a,1)\,(1\;2)\\ &= (a,1)\,(1,a)\\ &= (a,a).\end{aligned}$$

We conclude that the element $g = (a,1)\,(1\;2)$ satisfies $g^4 = 1$ but $g^2 \neq 1$. Hence this g is one of the two elements of order 4 in the wreath product W.

Chapter 5

Soluble Groups

We have already met the concept of a composition series for a group. In this chapter we shall consider groups whose composition factors are all abelian. We can think of this as the class of groups that can be built only using abelian groups. To give a general description of these groups we use the following concept.

Definition 5.1 Let G be a group and $x, y \in G$. The *commutator* of x and y is the element

$$[x, y] = x^{-1}y^{-1}xy.$$

Note that the following equations hold immediately:

$$[x, y] = x^{-1}x^{y}$$
$$[x, y] = (y^{-1})^{x}y$$

and

$$xy = yx\,[x, y]. \tag{5.1}$$

The latter tells us that the commutator can be viewed as a measure of by how much x and y fail to commute.

Lemma 5.2 *Let G and H be groups, let $\phi\colon G \to H$ be a homomorphism and let $x, y, z \in G$. Then*

(i) $[x, y]^{-1} = [y, x]$;

(ii) $[x, y]\phi = [x\phi, y\phi]$;

(iii) $[x, yz] = [x, z]\,[x, y]^{z}$;

(iv) $[xy, z] = [x, z]^{y}\,[y, z]$.

Proof: (i) $[x, y]^{-1} = (x^{-1}y^{-1}xy)^{-1} = y^{-1}x^{-1}yx = [y, x]$.

(ii) $[x, y]\phi = (x^{-1}y^{-1}xy)\phi = (x\phi)^{-1}(y\phi)^{-1}(x\phi)(y\phi) = [x\phi, y\phi]$.

(iii) For this and part (iv), we shall rely on Equation (5.1) which we view as telling us how to exchange group elements at the expense of introducing commutators. (This process is known as 'collection'.) So

$$xyz = yzx\,[x, yz]$$

but if we collect one term at a time we obtain

$$\begin{aligned} xyz &= yx\,[x,y]\,z \\ &= yxz\,[x,y]^z \\ &= yzx\,[x,z]\,[x,y]^z. \end{aligned}$$

Hence

$$yzx\,[x,yz] = yzx\,[x,z]\,[x,y]^z,$$

so

$$[x,yz] = [x,z]\,[x,y]^z.$$

(iv)

$$xyz = zxy\,[xy,z]$$

and

$$\begin{aligned} xyz &= xzy\,[y,z] \\ &= zx\,[x,z]\,y\,[y,z] \\ &= zxy\,[x,z]^y\,[y,z]. \end{aligned}$$

Comparing we deduce

$$[xy,z] = [x,z]^y\,[y,z].$$

□

Both parts (iii) and (iv) can be proved by a more simple-minded expansion of the terms on both sides, but more insight can be obtained with use of the above collection process.

Definition 5.3 Let G be a group. The *derived subgroup* (or *commutator subgroup*) G' of G is the subgroup generated by all commutators of elements from G:

$$G' = \langle\, [x,y] \mid x, y \in G \,\rangle.$$

Part (i) of Lemma 5.2 tells us that the inverse of a commutator is again a commutator, but we have no information about products of commutators. Consequently, a typical element of G' has the form

$$[x_1,y_1]\,[x_2,y_2]\ldots[x_n,y_n]$$

where $x_i, y_i \in G$ for each i. In general, a product of commutators need not itself be a commutator. There do exist examples of groups where a product of two commutators is not itself a commutator, but they are not so easy to construct and the smallest finite example has order 96. Nevertheless do note that the elements of the derived subgroup are *not* necessarily themselves commutators.

Observe that if $x, y \in G$ are two elements that commute, then $[x,y] = x^{-1}y^{-1}xy = x^{-1}y^{-1}yx = 1$. On the other hand, it follows immediately from Equation (5.1) that if $[x,y] = 1$, then x and y commute. In particular:

Lemma 5.4 *Let G be a group. Then $G' = \mathbf{1}$ if and only if G is an abelian group.* □

We can strengthen this observation in the following key properties that characterize the derived subgroup. The second part of the following lemma was established in *MT4003*.

Lemma 5.5 *Let G be a group and N be a normal subgroup of G. Then:*

(i) *the derived subgroup G' is a characteristic subgroup of G and hence normal in G;*

(ii) *G/N is abelian if and only if $G' \leqslant N$.*

In particular, G/G' is an abelian group and it is the largest quotient group of G which is abelian. The quotient G/G' is often called the *abelianization* of G.

PROOF: (i) By Lemma 5.2(ii), if $x, y \in G$ and ϕ is an automorphism of G, then $[x, y]\phi = [x\phi, y\phi] \in G'$. Since ϕ is a homomorphism, it follows that any product of commutators in G is mapped into G' for ϕ. Thus $G'\phi \leqslant G'$ for all automorphisms ϕ of G and so G' is a characteristic subgroup of G. Therefore $G' \trianglelefteqslant G$ by Lemma 3.16(i).

(ii) Suppose G/N is abelian. Then

$$Nx \cdot Ny = Ny \cdot Nx \qquad \text{for all } x, y \in G,$$

so

$$N[x, y] = (Nx)^{-1}(Ny)^{-1}(Nx)(Ny) = N1 \qquad \text{for all } x, y \in G.$$

Thus $[x, y] \in N$ for all $x, y \in G$ and we deduce $G' \leqslant N$.

Conversely if $G' \leqslant N$, then $[x, y] \in N$ for all $x, y \in G$ and reversing the above steps shows that G/N is abelian. □

Iterating the construction of the derived subgroup yields the derived series:

Definition 5.6 The *derived series* $(G^{(i)})$ (for $i \geqslant 0$) is the chain of subgroups of the group G defined by

$$G^{(0)} = G$$

and

$$G^{(i+1)} = (G^{(i)})' \qquad \text{for } i \geqslant 0.$$

So $G^{(1)} = G'$, $G^{(2)} = (G')' = G''$, etc. This produces a chain of subgroups

$$G = G^{(0)} \geqslant G^{(1)} \geqslant G^{(2)} \geqslant \cdots .$$

We shall see later that, when G is soluble, this is indeed a series in the sense of Definition 3.1 (in that each term is normal in the previous). In fact, it is a normal series (each term is normal in G) and more: each term is a characteristic subgroup of G.

Definition 5.7 A group G is called *soluble* (*solvable* in the U.S.) if $G^{(d)} = \mathbf{1}$ for some d. The least such d is called the *derived length* of G.

When forming the derived series, we take the derived subgroup at each stage of the previous term. Consequently, once we obtain a repetition then the series is constant from that point. Thus if G is a soluble group of derived length d, its derived series has the form

$$G = G^{(0)} > G^{(1)} > G^{(2)} > \cdots > G^{(d)} = \mathbf{1}$$

with strict inclusions: $G^{(i)}$ is a proper subgroup of $G^{(i-1)}$ for $i = 1, 2, \ldots, d$.

Example 5.8 We shall compute the derived series of some familiar groups. The process involves *some* direct computation of commutators combined with the use of Lemma 5.5 that characterizes the derived subgroup.

(i) Let $G = S_4$, the symmetric group of degree 4. Observe

$$[(1\ 2), (1\ 3)] = (1\ 2)\,(1\ 3)\,(1\ 2)\,(1\ 3) = (1\ 3\ 2)$$

and similarly every 3-cycle is a commutator. Since the 3-cycles generate the alternating group, we deduce that $A_4 \leqslant G'$. On the other hand, $S_4/A_4 \cong C_2$, which is abelian, so $G' \leqslant A_4$ by Lemma 5.5. Therefore

$$G' = A_4.$$

We now repeat this process:

$$[(1\ 2\ 3), (1\ 2\ 4)] = (1\ 3\ 2)\,(1\ 4\ 2)\,(1\ 2\ 3)\,(1\ 2\ 4) = (1\ 2)\,(3\ 4)$$

and similarly every product of two disjoint transpositions is a commutator. Hence $V_4 \leqslant G'' = A_4'$. On the other hand, $A_4/V_4 \cong C_3$, which is abelian, so $A_4' \leqslant V_4$ by Lemma 5.5. Therefore

$$G'' = A_4' = V_4.$$

Finally V_4 is an abelian group, so $G''' = V_4' = \mathbf{1}$. Thus the symmetric group S_4 is soluble of derived length 3.

(ii) Let $n \geqslant 5$ and let $G = A_n$, the alternating group of degree n. Then G is a non-abelian simple group. Therefore $G' \neq \mathbf{1}$ but, since G' is a normal subgroup of G, this forces $G' = G$. Hence $G' = A_n$. If we repeat the process, then we will always produce the same subgroup and hence $G^{(i)} = G = A_n$ for all $i \geqslant 0$. We conclude that the alternating group A_n, for $n \geqslant 5$, is not soluble.

To move beyond such computations and to fully understand the properties of a soluble group, we shall produce some equivalent formulations. This will enable us to describe examples of soluble groups more easily. We begin by establishing basic properties of the derived subgroup and the derived series.

Lemma 5.9 (i) *If H is a subgroup of G, then $H' \leqslant G'$.*

(ii) *If $\phi\colon G \to K$ is a homomorphism, then $G'\phi \leqslant K'$.*

(iii) *If $\phi\colon G \to K$ is a surjective homomorphism, then $G'\phi = K'$.*

Proof: (i) If $x, y \in H$, then $[x, y]$ is in particular a commutator of elements from G so belongs to the derived subgroup of G:

$$[x, y] \in G' \qquad \text{for all } x, y \in H.$$

Therefore

$$\langle\, [x, y] \mid x, y \in H \,\rangle \leqslant G',$$

that is, $H' \leqslant G'$.

(ii) This is similar to Lemma 5.5(i). If $x, y \in G$, then $[x, y]\phi = [x\phi, y\phi] \in K'$. Since K' is closed under products, it follows that any product of commutators in G is mapped into K' by the homomorphism ϕ. Thus $G'\phi \leqslant K'$.

(iii) Let $a, b \in K$. Since ϕ is surjective, there exists $x, y \in G$ such that $a = x\phi$ and $b = y\phi$. Thus

$$[a, b] = [x\phi, y\phi] = [x, y]\phi \in G'\phi.$$

Thus

$$[a,b] \in G'\phi \qquad \text{for all } a,b \in K.$$

The image of a subgroup under the homomorphism ϕ is itself a subgroup of the codomain, so we see that $G'\phi$ is a subgroup of K that contains all commutators $[a,b]$ for $a,b \in K$. The definition of the derived subgroup now forces $K' \leqslant G'\phi$. Using (ii) gives $K' = G'\phi$, as required. □

Corollary 5.10 (i) *If H is a subgroup of G, then $H^{(i)} \leqslant G^{(i)}$ for all i.*

(ii) *If $\phi\colon G \to K$ is a surjective homomorphism, then $G^{(i)}\phi = K^{(i)}$ for all i.*

PROOF: We prove both parts by induction on i using the previous lemma as the tool to establish the induction step.

(i) When $i = 0$, the claimed inclusion is $H \leqslant G$ which holds by assumption. Suppose then that $H^{(i)} \leqslant G^{(i)}$ for some $i \geqslant 0$. Apply Lemma 5.9(i) to give

$$(H^{(i)})' \leqslant (G^{(i)})';$$

that is,

$$H^{(i+1)} \leqslant G^{(i+1)}.$$

This completes the induction.

(ii) The assumption that ϕ is a surjective homomorphism tells us that $G\phi = K$. This is the claimed formula when $i = 0$. Now suppose that $G^{(i)}\phi = K^{(i)}$ for some $i \geqslant 0$. Hence, upon restricting the $G^{(i)}$, ϕ induces a surjective homomorphism $G^{(i)} \to K^{(i)}$. Apply Lemma 5.9(iii) to this homomorphism to give

$$(G^{(i)})'\phi = (K^{(i)})';$$

that is,

$$G^{(i+1)}\phi = K^{(i+1)}.$$

This completes the induction. □

We can now deduce the following observation about soluble groups:

Theorem 5.11 *Subgroups and homomorphic images of soluble groups are themselves soluble.*

PROOF: Let G be a soluble group and H be a subgroup of G. The assumption tells us that $G^{(d)} = \mathbf{1}$ for some d. Therefore, using Corollary 5.10(i), since $H^{(d)} \leqslant G^{(d)}$, we obtain $H^{(d)} = \mathbf{1}$ and so we deduce that H is soluble.

Now let K be a homomorphic image of G; that is, there exists a surjective homomorphism $\phi\colon G \to K$. We have assumed $G^{(d)} = \mathbf{1}$ for some d and thus, by Corollary 5.10(ii),

$$K^{(d)} = G^{(d)}\phi = \mathbf{1}\phi = \mathbf{1}.$$

Hence K is soluble. □

It follows that quotient groups (which are the same as homomorphic images) of soluble groups are themselves soluble. There is a rather strong converse to the above lemma as well.

Proposition 5.12 *Let G be a group and N be a normal subgroup of G such that both G/N and N are soluble. Then G is soluble.*

PROOF: Let $\pi\colon G \to G/N$ be the natural map. By assumption $(G/N)^{(d)} = \mathbf{1}$ and $N^{(e)} = \mathbf{1}$ for some d and e. Now, by Corollary 5.10(ii),

$$G^{(d)}\pi = (G/N)^{(d)} = \mathbf{1}.$$

Hence

$$G^{(d)} \leqslant \ker \pi = N.$$

Therefore, by Corollary 5.10(i),

$$(G^{(d)})^{(e)} \leqslant N^{(e)} = \mathbf{1};$$

that is,

$$G^{(d+e)} = \mathbf{1}.$$

Thus G is soluble. □

We have already observed that the derived subgroup G' is a characteristics subgroup of G; that is,

$$G' \operatorname{char} G.$$

Since the terms of the derived series are defined recursively using the derived subgroup (that is, $G^{(i+1)} = (G^{(i)})'$ for all $i \geqslant 0$), it follows that each term of the derived series is characteristic in the previous one:

$$G^{(i)} \operatorname{char} G^{(i-1)} \operatorname{char} G^{(i-2)} \operatorname{char} \cdots \operatorname{char} G^{(1)} \operatorname{char} G^{(0)} = G$$

for all $i \geqslant 0$. Now apply Lemma 3.16(ii) to conclude that each term $G^{(i)}$ of the derived series is a characteristic subgroup (and hence a normal subgroup) of G.

Proposition 5.13 *The derived series*

$$G = G^{(0)} \geqslant G^{(1)} \geqslant G^{(2)} \geqslant \cdots$$

is a chain of subgroups each of which is a characteristic subgroup of G and hence each of which is a normal subgroup of G. □

Consequently, if G is a soluble group of derived length d then its derived series has the form

$$G = G^{(0)} > G^{(1)} > \cdots > G^{(d)} = \mathbf{1}$$

and this is a *normal series* (that is, each term is normal in G). Moreover, each factor has the form

$$G^{(i)}/G^{(i+1)} = G^{(i)}/(G^{(i)})'$$

and this is an abelian group by Lemma 5.5. Thus if G is soluble then it has a normal series with abelian factors. This property fully describes what it means for a group to be soluble:

Theorem 5.14 *Let G be a group. The following conditions are equivalent:*

(i) *G is soluble;*

(ii) *G has a series with abelian factors.*

PROOF: (i) $\Rightarrow$ (ii): The derived series is such a series (indeed, it is a *normal* series with abelian factors).

(ii) $\Rightarrow$ (i): Suppose that

$$G = G_0 \geqslant G_1 \geqslant G_2 \geqslant \ldots \geqslant G_n = \mathbf{1}$$

is a series where G_{i-1}/G_i is abelian for all i.

Claim: $G^{(i)} \leqslant G_i$ for all i.

We prove the claim by induction on i. Since $G^{(0)} = G = G_0$, the claim holds for $i = 0$.

Suppose $G^{(i)} \leqslant G_i$ for some $i \geqslant 0$. Now $G_{i+1} \trianglelefteq G_i$ and G_i/G_{i+1} is abelian. Hence $(G_i)' \leqslant G_{i+1}$ by Lemma 5.5. Further, by Lemma 5.9(i), $(G^{(i)})' \leqslant (G_i)'$ and consequently

$$G^{(i+1)} = (G^{(i)})' \leqslant (G_i)' \leqslant G_{i+1}.$$

Therefore, by induction, $G^{(n)} \leqslant G_n = \mathbf{1}$, so $G^{(n)} = \mathbf{1}$ and G is soluble. $\square$

The above theorem provides a general characterization of soluble groups. We shall also describe a characterization of soluble groups in terms of composition series (as introduced in Definition 3.3). Note, however, that the infinite cyclic group is abelian, so soluble, but it does not have a composition series (see Example 3.6). Consequently, we cannot hope for composition series to give us complete information about soluble groups. However, as long as we avoid *infinite* soluble groups, the composition factors do determine whether or not a group is soluble.

Theorem 5.15 *Let G be a group. Then the following conditions are equivalent:*

(i) *G is a finite soluble group;*

(ii) *G has a composition series with all composition factors cyclic of prime order.*

Recall that the abelian simple groups are precisely the cyclic groups of (various) prime orders. Thus condition (ii) describes the groups with abelian composition factors.

PROOF: (ii) $\Rightarrow$ (i): Let

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1}$$

be a composition series for G and suppose that all the factors are cyclic. This is, in particular, a series for G with abelian factors and so G is soluble by Theorem 5.14. Furthermore,

$$|G| = |G_0/G_1| \cdot |G_1/G_2| \cdot \ldots \cdot |G_{n-1}/G_n|,$$

a product of finitely many primes, so G is finite.

(i) $\Rightarrow$ (ii): Let G be a finite soluble group. Then by Theorem 5.14, G possesses a series

$$G = G_0 > G_1 > G_2 > \cdots > G_n = \mathbf{1} \tag{5.2}$$

with abelian factors. Now use Corollary 3.8 to refine this to a composition series for G (that is, repeatedly insert additional terms into the series until we cannot insert any more). Observe that if $G_{i+1} < N < G_i$, then N/G_{i+1} is a subgroup of G_i/G_{i+1} and so is abelian, while by the Third Isomorphism Theorem

$$G_i/N \cong \frac{G_i/G_{i+1}}{N/G_{i+1}},$$

which is a quotient of an abelian group and so is abelian. Hence the refinement process preserves the property that the factors are abelian. The result is a composition series for G with abelian factors. Since the only abelian simple groups are cyclic of prime order, we deduce that all the composition factors of G are cyclic of prime order (for various primes). □

This characterization of a finite soluble group in terms of its composition factors gives further insight into our collection of examples given in Example 5.8. For example, we observed that S_4 is soluble but we also know (from Example 3.5(ii)) that its composition factors are C_2 (three times) and C_3. Equally, if G is a non-abelian simple group (for example, $G = A_n$ for $n \geqslant 5$) than G is its only composition factor and so G is not soluble.

The following examples, and non-examples, of soluble groups can also be deduced using the theory that we have developed.

Example 5.16 (i) We have already observed that the symmetric group S_4 of degree 4 is soluble. We know, from Theorem 5.11, that a subgroup of a soluble group is soluble. Hence, for example, the alternating group A_4 of degree 4 and the dihedral group D_8 of order 8 are also soluble.

(ii) Since A_n is not soluble for $n \geqslant 5$, it follows that the symmetric group S_n, for $n \geqslant 5$, is not soluble. (Indeed, the composition factors of S_n are C_2 and A_n.)

(iii) Let $n \geqslant 3$. We shall show that the dihedral group D_{2n} of order $2n$ is soluble. First D_{2n} contains an element α of order n, so $\langle \alpha \rangle$ has index 2 and so is normal. Thus

$$D_{2n} > \langle \alpha \rangle > \mathbf{1}$$

is a series for D_{2n} with both factors cyclic. Hence D_{2n} is soluble by Theorem 5.14.

Careful analysis of the examples at the end of Chapter 2 shows that the groups we considered that were not simple in 2.26–2.28 are also soluble groups.

Hall's Theorem

For the remainder of this chapter we shall work only with *finite* soluble groups. Our goal is to prove Hall's Theorem concerning the existence of Hall subgroups in a finite soluble groups. We shall prove this by induction on the group order and one key step will be to take the quotient by a minimal normal subgroup.

Recall from Definition 3.17 that a minimal normal subgroup of a finite group G is a non-trivial normal subgroup M such that if $1 \leqslant N \leqslant M$ with $N \trianglelefteq G$ then either $N = \mathbf{1}$ or $N = M$. Theorem 3.19 tells us that a minimal normal subgroup of a finite group is a direct product of isomorphic simple groups. However, if G is a finite *soluble* group then non-abelian simple groups cannot occur by Theorem 5.15. The only possible simple groups that can occur are cyclic groups of prime order. Thus in a finite soluble group, a minimal normal subgroup is a direct product of cyclic groups of order p (for some prime p). We give a special name to these groups:

Definition 5.17 Suppose that p is a prime number. An *elementary abelian p-group* G is an abelian group such that

$$x^p = 1 \qquad \text{for all } x \in G.$$

Recall from the Fundamental Theorem of Finite Abelian Groups (see *MT4003*) that a finite abelian group is a direct product of cyclic groups. It follows that a finite group is an elementary abelian p-group if and only if

$$G \cong \underbrace{C_p \times C_p \times \cdots \times C_p}_{d \text{ times}}$$

for some d. We have therefore observed that combining Theorem 5.15 with Theorem 3.19 gives:

Theorem 5.18 *A minimal normal subgroup of a finite soluble group is an elementary abelian p-group for some prime number p.* □

This result will be used in the induction step of our proof of Hall's Theorem. Before stating that theorem, we describe the type of subgroup with which this theorem is concerned.

Hall subgroups

Definition 5.19 Let π be a set of prime numbers and let G be a finite group. A *Hall π-subgroup* of G is a subgroup H of G such that $|H|$ is a product involving only the primes in π and $|G : H|$ is a product involving only primes not in π.

If p is a prime number, then this definition means that a Hall $\{p\}$-subgroup (that is, when $\pi = \{p\}$ contains just one prime) is precisely the same thing as a Sylow p-subgroup.

Example 5.20 Consider the alternating group A_5 of degree 5. Here

$$|A_5| = 60 = 2^2 \cdot 3 \cdot 5,$$

so a Hall $\{2,3\}$-subgroup of A_5 has order 12. We already know of a subgroup with this order: namely, A_4 is a Hall $\{2,3\}$-subgroup of A_5.

A Hall $\{2,5\}$-subgroup of A_5 would have order 20 and index 3, while a Hall $\{3,5\}$-subgroup of A_5 would have order 15 and index 4. Suppose that one of these subgroups exists, say H with $|A_5 : H| = r = 3$ or 4. Let A_5 act on the cosets of H to determine a non-trivial homomorphism $\rho\colon A_5 \to S_r$. However A_5 is simple, so necessarily ρ is then injective, but this is impossible as $|A_5| > |S_r|$.

Hence A_5 does not have any Hall π-subgroups for $\pi = \{2,5\}$ or $\pi = \{3,5\}$.

So in insoluble groups, some Hall π-subgroups might exist, while others might not (in fact, it is a theorem that some definitely do not!). This is in stark contrast to soluble groups where we shall observe that Hall π-subgroups always do exist:

Theorem 5.21 (P. Hall, 1928) *Let G be a finite soluble group and let π be a set of prime numbers. Then G has a Hall π-subgroup.*

Hall subgroups and this theorem are named after Philip Hall (1904–1982), a British mathematician who did groundbreaking research into the theory of finite and infinite groups in the early and mid-parts of the twentieth century. In fact, he established more properties of Hall subgroups that are analogous to the behaviour of Sylow subgroups. He also showed that, in a finite soluble group G,

(i) any two Hall π-subgroups of G are conjugate, and

(ii) any π-subgroup (that is, a subgroup whose order is a product involving only primes in π) of G is contained in a Hall π-subgroup.

The proofs of these two facts will not be established in this module, but similar methods are used to what we do when proving Theorem 5.21.

A number of tools are needed in the course of this theorem. The one remaining fact that has not already been established is the following result. It first appeared in the study of nilpotent groups (which are discussed in the next chapter). We shall use this lemma in the most complicated step in the proof of Hall's Theorem.

Lemma 5.22 (Frattini Argument) *Let G be a finite group, N be a normal subgroup of G and P be a Sylow p-subgroup of N. Then*

$$G = \mathrm{N}_G(P)\,N.$$

The name of the lemma suggests (correctly) that it is the method of proof that is actually most important here. The idea can be adapted to many situations and turns out to be very useful.

Proof: Let $x \in G$. Since $N \trianglelefteq G$, we have

$$P^x \leqslant N^x = N,$$

so P^x is a Sylow p-subgroup of N. Sylow's Theorem then tells us that P^x and P are conjugate in N:

$$P^x = P^n \qquad \text{for some } n \in N.$$

Therefore

$$P^{xn^{-1}} = P,$$

so $y = xn^{-1} \in \mathrm{N}_G(P)$. Hence $x = yn \in \mathrm{N}_G(P)\,N$. The reverse inclusion is obvious, so

$$G = \mathrm{N}_G(P)\,N. \qquad \square$$

Proof of Theorem 5.21: Let G be a finite soluble group and write $|G| = mn$ where m is a product involving primes in π and n is a product involving primes not in π. We shall show, by induction on the order of G, that G has a subgroup of order m. If $m = 1$, then the trivial subgroup is the subgroup of order m and, in particular, this establishes the base case. We can therefore assume that $m > 1$ and that the result holds for all soluble groups of order less than $|G|$.

Since we are now considering a non-trivial soluble group G, it has a minimal normal subgroup and, by Theorem 5.18, this is elementary abelian. We consider two cases according to the prime dividing the order of a minimal normal subgroup.

Case 1: G possesses a minimal normal subgroup M that is an elementary abelian p-group where $p \in \pi$. Write $|M| = p^\alpha$.

Then

$$|G/M| = mn/p^\alpha = m_1 n,$$

where $m = m_1 p^\alpha$. By induction, G/M has a Hall π-subgroup; that is, a subgroup of order m_1. The Correspondence Theorem tells us this has the form H/M for some subgroup H of G containing M. Then

$$|H/M| = m_1$$

so
$$|H| = m_1|M| = m_1 p^\alpha = m.$$
Hence H is a Hall π-subgroup of G.

Case 2: No minimal normal subgroup of G is an elementary abelian p-group with $p \in \pi$.

Let M be a minimal normal subgroup of G, so M is an elementary abelian q-group for some prime $q \notin \pi$. Write $|M| = q^\beta$ so

$$|G/M| = mn/q^\beta = mn_1$$

where $n = n_1 q^\beta$. We now further subdivide according to whether or not $n_1 = 1$. (The case when $n_1 = 1$ is the most complicated.)

Subcase 2A: $n_1 \neq 1$.

By induction, G/M has a Hall π-subgroup, which has the form K/M where K is a subgroup of G containing M and
$$|K/M| = m.$$
Then
$$|K| = m|M| = mq^\beta = mn/n_1 < mn.$$
We shall further apply induction to K. This has smaller order than G and hence possesses a Hall π-subgroup. Let H be a Hall π-subgroup of K. Then $|H| = m$, so H is also a Hall π-subgroup of G.

Subcase 2B: $n_1 = 1$, so $|G| = mq^\beta$.

Note also that the general assumption of Case 2 still applies: G has a minimal normal subgroup M with $|M| = q^\beta$ and it has no minimal normal subgroup whose order is a power of a prime in π. The previous steps essentially just depended on careful application of induction. In this case, we need to do far more group theory and we shall establish that our Hall π-subgroup arises in a specific manner.

Now $|G/M| = m > 1$. Let N/M be a minimal normal subgroup of G/M. Then N/M is an elementary abelian p-group for some $p \in \pi$ (since m is a product involving only primes in π), say $|N/M| = p^\alpha$. Then $N \trianglelefteq G$, by the Correspondence Theorem, and

$$|N| = p^\alpha q^\beta.$$

Let P be a Sylow p-subgroup of N. Let us now apply the Frattini Argument (Lemma 5.22):

$$G = \mathrm{N}_G(P)\, N.$$

But we know $|P| = p^\alpha$ and $|M| = q^\beta$, so $N = PM$. Hence

$$G = \mathrm{N}_G(P)\, PM = \mathrm{N}_G(P)\, M$$

(as $P \leqslant \mathrm{N}_G(P)$).

Now consider $J = \mathrm{N}_G(P) \cap M$. Since M is abelian, $J \trianglelefteq M$. Also since $M \trianglelefteq G$, $J = \mathrm{N}_G(P) \cap M \trianglelefteq \mathrm{N}_G(P)$. Hence

$$J \trianglelefteq \mathrm{N}_G(P)\, M = G.$$

But M is a minimal normal subgroup of G, so $J = \mathbf{1}$ or $J = M$.

If $J = \mathrm{N}_G(P) \cap M = M$, then $M \leqslant \mathrm{N}_G(P)$, so $G = \mathrm{N}_G(P)$. Hence P is a normal p-subgroup of G and some subgroup of P is a minimal normal subgroup of G and this must be an elementary abelian p-group with $p \in \pi$. This is contrary to the general assumption made for Case 2.

Thus $J = \mathbf{1}$, so $\mathrm{N}_G(P) \cap M = \mathbf{1}$. Now

$$mq^\beta = |G| = |\mathrm{N}_G(P)\, M| = |\mathrm{N}_G(P)| \cdot |M|,$$

by Lemma 1.19(iv). Therefore $|\mathrm{N}_G(P)| = m$. Hence $H = \mathrm{N}_G(P)$ is a Hall π-subgroup, which is what we were trying to find.

This completes the induction and establishes the existence of Hall subgroups in a finite soluble group. □

Chapter 6

Nilpotent Groups

In Chapter 5, we defined a collection of subgroups of a group called the derived series using the concept of the commutator

$$[x,y] = x^{-1}y^{-1}xy.$$

In this chapter, we shall define another collection of subgroups and hence define a class of groups called the nilpotent groups. Recall that if $x, y \in G$, then $[x,y] = 1$ if and only if x and y commute.

Definition 6.1 Let A and B be subgroups of a group G. Define the *commutator subgroup* $[A,B]$ by

$$[A,B] = \langle\, [a,b] \mid a \in A,\ b \in B \,\rangle,$$

the subgroup generated by all commutators $[a,b]$ with $a \in A$ and $b \in B$.

In this notation, the derived series is then given recursively by setting $G^{(0)} = G$ and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$ for all integers $i \geqslant 0$. We now make a new definition which is similar but slightly less symmetrical in its form.

Definition 6.2 The *lower central series* $\big(\gamma_i(G)\big)$ (for integers $i \geqslant 1$) is the collection of subgroups of the group G defined by $\gamma_1(G) = G$ and

$$\gamma_{i+1}(G) = [\gamma_i(G), G] \qquad \text{for } i \geqslant 1.$$

Definition 6.3 A group G is called *nilpotent* if $\gamma_{c+1}(G) = \mathbf{1}$ for some c. The least such c is called the *nilpotency class* of G.

It is easy to see that $G^{(i)} \leqslant \gamma_{i+1}(G)$ for all i (by induction on i). Thus if G is nilpotent, then it is soluble. Note also that $\gamma_2(G) = G'$. Consequently, we observe:

Lemma 6.4 *Let G be a group. Then $\gamma_2(G) = \mathbf{1}$ if and only if G is abelian. In particular, the nilpotent groups of class $\leqslant 1$ are precisely the abelian groups.* □

In order that we can study nilpotent groups, we shall need some basic properties of the lower central series. The first two parts are proved by a very similar argument to that used for the derived series in Lemma 5.9 and Corollary 5.10.

Lemma 6.5 (i) *If H is a subgroup of G, then $\gamma_i(H) \leqslant \gamma_i(G)$ for all i.*

(ii) *If $\phi\colon G \to K$ is a surjective homomorphism, then $\gamma_i(G)\phi = \gamma_i(K)$ for all i.*

(iii) $\gamma_i(G)$ *is a characteristic subgroup of* G *for all* i.

(iv) *The lower central series of* G *is a chain of subgroups*

$$G = \gamma_1(G) \geqslant \gamma_2(G) \geqslant \gamma_3(G) \geqslant \cdots .$$

In particular, if G is a nilpotent group, then the lower central series

$$G = \gamma_1(G) > \gamma_2(G) > \cdots > \gamma_{c+1}(G) = \mathbf{1}$$

is a normal series for G.

Proof: (i) Proceed by induction on i. Note that $\gamma_1(H) = H \leqslant G = \gamma_1(G)$. If we assume that $\gamma_i(H) \leqslant \gamma_i(G)$, then this together with $H \leqslant G$ gives $[\gamma_i(H), H] \leqslant [\gamma_i(G), G]$, so $\gamma_{i+1}(H) \leqslant \gamma_{i+1}(G)$.

(ii) Again proceed by induction on i. Note that $\gamma_1(G)\phi = G\phi = K = \gamma_1(K)$. Suppose that $\gamma_i(G)\phi = \gamma_i(K)$ for some i. If $x \in \gamma_i(G)$ and $y \in G$, then

$$[x, y]\phi = [x\phi, y\phi] \in [\gamma_i(G)\phi, G\phi] = [\gamma_i(K), K] = \gamma_{i+1}(K),$$

so $\gamma_{i+1}(G)\phi = [\gamma_i(G), G]\phi \leqslant \gamma_{i+1}(K)$.

On the other hand, if $a \in \gamma_i(K)$ and $b \in K$, then $a = x\phi$ and $b = y\phi$ for some $x \in \gamma_i(G)$ and $y \in G$. So

$$[a, b] = [x\phi, y\phi] = [x, y]\phi \in [\gamma_i(G), G]\phi = \gamma_{i+1}(G)\phi.$$

Thus $\gamma_{i+1}(K) = [\gamma_i(K), K] \leqslant \gamma_{i+1}(G)\phi$.

Putting the two inclusions together, We deduce that $\gamma_{i+1}(G)\phi = \gamma_{i+1}(K)$ to complete the inductive step.

(iii) If ϕ is an automorphism of G, then $\phi\colon G \to G$ is, in particular, a surjective homomorphism, so from (ii) we conclude that

$$\gamma_i(G)\phi = \gamma_i(G).$$

Thus $\gamma_i(G) \operatorname{char} G$.

(iv) From (iii), $\gamma_i(G) \trianglelefteq G$. Hence if $x \in \gamma_i(G)$ and $y \in G$, then

$$[x, y] = x^{-1}x^y \in \gamma_i(G).$$

Hence

$$\gamma_{i+1}(G) = [\gamma_i(G), G] \leqslant \gamma_i(G) \qquad \text{for all } i.$$

□

We deduce two consequences immediately:

Lemma 6.6 *Subgroups and homomorphic images of nilpotent groups are themselves nilpotent.*

Proof: Suppose that $\gamma_{c+1}(G) = \mathbf{1}$ and $H \leqslant G$. Then by Lemma 6.5(i), $\gamma_{c+1}(H) \leqslant \gamma_{c+1}(G) = \mathbf{1}$, so $\gamma_{c+1}(H) = \mathbf{1}$ and H is nilpotent.

If K is a homomorphic image of G, say $\phi\colon G \to K$ is a surjective homomorphism, then Lemma 6.5(ii) tells us that $\gamma_{c+1}(K) = \gamma_{c+1}(G)\phi = \mathbf{1}\phi = \mathbf{1}$, so K is nilpotent. □

In contrast to soluble groups, if N is a normal subgroup of a group G such that G/N and N are both nilpotent, one cannot deduce that G is nilpotent.

Example 6.7 First observe that if G is a nilpotent group of class c then its lower central series has the form

$$G = \gamma_1(G) > \gamma_2(G) > \cdots > \gamma_c(G) > \gamma_{c+1}(G) = \mathbf{1}$$

and by definition $[\gamma_c(G), G] = \gamma_{c+1}(G) = \mathbf{1}$. Hence every element of the last non-trivial term $\gamma_c(G)$ of the lower central series commutes with every element of G; that is $\gamma_c(G) \leqslant \mathrm{Z}(G)$. Thus if G is a non-trivial nilpotent group, then its centre is non-trivial.

We know that $\mathrm{Z}(S_3) = \mathbf{1}$ (as can be shown by a direct computation), so S_3 is *not* nilpotent. However A_3 and $S_3/A_3 \cong C_2$ are both abelian and hence nilpotent.

This last observation tells us that being nilpotent has something to do with non-trivial centre. This idea will enable us to show that every finite p-group is nilpotent. We shall start with the following observation, which was proved in *MT4003* and which we have used regularly on the problem sheets already. However, for completeness we shall give a proof based on group actions.

Lemma 6.8 *Let G be a non-trivial finite p-group. Then the centre $\mathrm{Z}(G)$ of G is non-trivial.*

PROOF: Let G be a finite group of order $|G| = p^n$ for some prime p and some integer $n \geqslant 1$. Let G act on itself by conjugation. It is then the disjoint union of the orbits, which are the conjugacy classes of G:

$$G = \mathscr{C}_1 \cup \mathscr{C}_2 \cup \cdots \cup \mathscr{C}_k$$

The size of each conjugacy class equals the index of the centralizer of an element within it:

$$|\mathscr{C}_i| = |G : \mathrm{C}_G(x_i)|$$

where $x_i \in \mathscr{C}_i$. Hence $|\mathscr{C}_i| = 1$ if and only if $\mathscr{C}_i = \{x_i\} \subseteq \mathrm{Z}(G)$. If $x_i \notin \mathrm{Z}(G)$, then $|G : \mathrm{C}_G(x_i)|$ is a non-trivial power of p. Consequently,

$$p^n = |\mathscr{C}_1| + |\mathscr{C}_2| + \cdots + |\mathscr{C}_k| \equiv |\mathrm{Z}(G)| \pmod{p}.$$

Hence $|\mathrm{Z}(G)| \neq 1$ and so $\mathrm{Z}(G) \neq \mathbf{1}$. □

Proposition 6.9 *A finite p-group is nilpotent.*

PROOF: Let G be a finite p-group, say $|G| = p^n$. We proceed by induction on $|G|$. If $|G| = 1$, then immediately G is nilpotent (as $\gamma_1(G) = G = \mathbf{1}$).

Now suppose $|G| > 1$. Apply Lemma 6.8: $\mathrm{Z}(G) \neq \mathbf{1}$. Consider the quotient group $G/\mathrm{Z}(G)$. This is a p-group of order smaller than G, so by induction it is nilpotent, say

$$\gamma_{c+1}(G/\mathrm{Z}(G)) = \mathbf{1}$$

for some c. Let $\pi \colon G \to G/\mathrm{Z}(G)$ be the natural homomorphism. Then by Lemma 6.5(ii),

$$\gamma_{c+1}(G)\pi = \gamma_{c+1}(G/\mathrm{Z}(G)) = \mathbf{1},$$

so $\gamma_{c+1}(G) \leqslant \ker \pi = \mathrm{Z}(G)$. Thus

$$\gamma_{c+2}(G) = [\gamma_{c+1}(G), G] \leqslant [\mathrm{Z}(G), G] = \mathbf{1},$$

so G is nilpotent. □

The goal in the remainder of this chapter is to demonstrate that essentially all finite groups are built as direct products of p-groups. Consequently, the examples from Proposition 6.9 are in some sense archetypal.

Proposition 6.10 *Let G be a nilpotent group. Then every proper subgroup of G is properly contained in its normalizer:*

$$H < \mathrm{N}_G(H) \qquad \text{whenever } H < G.$$

PROOF: Let

$$G = \gamma_1(G) \geqslant \gamma_2(G) \geqslant \cdots \geqslant \gamma_{c+1}(G) = \mathbf{1}$$

be the lower central series of G and let H be a proper subgroup of G. Then $\gamma_{c+1}(G) \leqslant H$ but $\gamma_1(G) \nleqslant H$. Choose i as small as possible so that $\gamma_i(G) \leqslant H$. Then $\gamma_{i-1}(G) \nleqslant H$. Now

$$[\gamma_{i-1}(G), H] \leqslant [\gamma_{i-1}(G), G] = \gamma_i(G) \leqslant H,$$

so

$$x^{-1}hxh^{-1} = [x, h^{-1}] \in H \qquad \text{for all } x \in \gamma_{i-1}(G) \text{ and } h \in H.$$

Therefore

$$H^x = \{\, x^{-1}hx \mid h \in H \,\} \leqslant H \qquad \text{for all } x \in \gamma_{i-1}(G).$$

Note that also that $H^{x^{-1}} \leqslant H$ for all $x \in \gamma_{i-1}(G)$, so $H^x = H$ for all $x \in \gamma_{i-1}(G)$. Hence $\gamma_{i-1}(G) \leqslant \mathrm{N}_G(H)$. Therefore, since $\gamma_{i-1}(G) \nleqslant H$, we deduce $\mathrm{N}_G(H) > H$. □

The final lemma we need to analyze the behaviour of nilpotent finite groups is the following observation about Sylow subgroups:

Lemma 6.11 *Let G be a finite group and let P be a Sylow p-subgroup of G for some prime p. Then*

$$\mathrm{N}_G(\mathrm{N}_G(P)) = \mathrm{N}_G(P).$$

PROOF: Let $H = \mathrm{N}_G(P)$. Then $P \trianglelefteqslant H$, so P is the unique Sylow p-subgroup of H. (Note that as it is a Sylow p-subgroup of G and since $P \leqslant H$, it is also a Sylow p-subgroup of H, as it must have the largest possible order for a p-subgroup of H.) Let $g \in \mathrm{N}_G(H)$. Then

$$P^g \leqslant H^g = H,$$

so P^g is also a Sylow p-subgroup of H and we deduce $P^g = P$; that is, $g \in \mathrm{N}_G(P) = H$. Thus $\mathrm{N}_G(H) \leqslant H$, so we deduce

$$\mathrm{N}_G(H) = H,$$

as required. □

We can now characterise finite nilpotent groups as being built from p-groups in the most simple way.

Theorem 6.12 *Let G be a finite group. Then G is nilpotent if and only if it is the direct product of its Sylow subgroups.*

Proof: First suppose that G is a nilpotent finite group. Let P be a Sylow p-subgroup of G (for some prime p) and $H = \mathrm{N}_G(P)$. By Lemma 6.11, $\mathrm{N}_G(H) = H$. Hence, by Proposition 6.10, it must be the case that $H = G$; that is, $\mathrm{N}_G(P) = G$ and so $P \trianglelefteqslant G$.

Now if $P_1, P_2, \dots, P_k$ are all the Sylow subgroups of G (one for each prime divisor of $|G|$), then we have just observed that each P_i is a normal subgroup of G. Then, with repeated use of Lemma 1.19,

$$P_1 \dots P_{i-1}P_{i+1} \dots P_k$$

is a normal subgroup of G of order $|P_1| \dots |P_{i-1}||P_{i+1}| \dots |P_k|$. This order is not divisible by the prime corresponding to P_i and hence, by Lagrange's Theorem,

$$P_1 \dots P_{i-1}P_{i+1} \dots P_k \cap P_i = \mathbf{1}$$

for each i. Also $G = P_1P_2 \dots P_k$ (again by the formula from Lemma 1.19) and we have verified the condition to be a direct product:

$$G = P_1 \times P_2 \times \dots \times P_k.$$

Conversely, if $G = P_1 \times P_2 \times \dots \times P_k$ is the direct product of its Sylow subgroups, then a straightforward calculation shows

$$\gamma_i(G) = \gamma_i(P_1) \times \gamma_i(P_2) \times \dots \times \gamma_i(P_k)$$

for each $i \geqslant 1$. Since each P_i is nilpotent by Proposition 6.9, there exists some c such that $\gamma_{c+1}(P_i) = \mathbf{1}$ for each i. Therefore $\gamma_{c+1}(G) = \mathbf{1}$ and G is nilpotent. $\square$

Chapter 7

Primitive Permutation Groups

In this chapter, we return to the general theme of understanding the action of a group and we focus on the important class of actions known as primitive actions. Cayley's Theorem (Theorem 2.20) tells us that every group is isomorphic to a transitive permutation group. In view of this, our primary focus will be on transitive permutation groups.

Blocks

We shall use the following concept to analyze the action of a transitive group.

Definition 7.1 Let G be a group that acts transitively on a set Ω. A *block* for G is a non-empty set Δ of Ω such that, for every $x \in G$, either $\Delta^x = \Delta$ or $\Delta^x \cap \Delta = \varnothing$.

For a transitive group, there is only one orbit. Blocks provide us with an alternative way to break down the action as we obtain a partition of the set Ω that interacts well with the action, as we shall now describe:

Lemma 7.2 *Let G be a group that acts transitively on a set Ω. If Δ is a block, then $\Sigma = \{\, \Delta^x \mid x \in G \,\}$ is a partition of Ω.*

PROOF: Fix some $\delta \in \Delta$. If $\omega \in \Omega$, then as G acts transitively, there exists $x \in G$ such that $\omega = \delta^x$. Therefore $\omega \in \Delta^x$. Hence Ω is the union of the sets in Ω.

Suppose $\Delta^x \cap \Delta^y \neq \varnothing$. Apply y^{-1} to deduce $\Delta^{xy^{-1}} \cap \Delta \neq \varnothing$. Since Δ is a block, this shows that $\Delta^{xy^{-1}} = \Delta$. Applying y then shows that $\Delta^x = \Delta^y$. We conclude that the members of Σ are either disjoint or equal and it follows that Ω is indeed the disjoint union of the members of Σ. $\square$

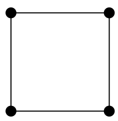
Definition 7.3 Let G be a group that acts transitively on Ω. If Δ is a block for G, we call the set $\Sigma = \{\, \Delta^x \mid x \in G \,\}$ a *system of blocks*.

We have observed that every block gives rise to a system of blocks. In practice, it is often easier to find the system of blocks Σ for a group G (that is, the partition of Ω preserved by G), observe that G permutes the members of this system and hence deduce that each member of Σ is a block.

We now present some trivial examples of blocks and other more interesting examples that illustrate how they arise.

Example 7.4 (i) Let G be any group acting transitively on any set Ω. Observe that Ω is a block since $\Omega^x = \Omega$ for all $x \in G$. If $\omega \in \Omega$, then $\Delta = \{\omega\}$ is a block since if x fixes ω, then $\Delta^x = \Delta$ and if $\omega^x \neq \omega$, then $\Delta^x \cap \Delta = \varnothing$. The whole set Ω and the singletons $\{\omega\}$ are all referred to as *trivial blocks*.

(ii) Let $G = D_8$, the dihedral group of order 8, viewed as a subgroup of S_4 via the following (standard) labelling of the square:



Take $\Delta = \{1, 3\}$. If $x \in D_8$, then x moves the points in Δ to opposite vertices of the square; that is, $\Delta^x = \{1, 3\}$ or $\{2, 4\}$. Thus Δ is a block for this action.

Observe that $\Gamma = \{1, 2\}$ is *not* a block, because upon applying a rotation clockwise through an angle of $\pi/2$ we calculate $\Gamma^{(1\ 2\ 3\ 4)} = \{2, 3\}$ which meets Γ in a single point.

(iii) Let Ω_1 and Ω_2 be sets with $|\Omega_1|, |\Omega_2| > 1$. Let $\Omega = \Omega_1 \times \Omega_2$ and take $G = \mathrm{Sym}(\Omega_1) \times \mathrm{Sym}(\Omega_2)$. We define an action of G on Ω by

$$(\alpha, \beta)^{(x,y)} = (\alpha^x, \beta^y)$$

for $\alpha \in \Omega_1$, $\beta \in \Omega_2$, $x \in \mathrm{Sym}(\Omega_1)$, $y \in \mathrm{Sym}(\Omega_2)$. It is straightforward to verify this is an action.)

Proof (Omitted in lectures): The verification is as follows:

$$\begin{aligned}
\big((\alpha, \beta)^{(x_1,y_1)}\big)^{(x_2,y_2)} &= (\alpha^{x_1}, \beta^{y_1})^{(x_2,y_2)} \\
&= \big((\alpha^{x_1})^{x_2}, (\beta^{y_1})^{y_2}\big) \\
&= (\alpha^{x_1 x_2}, \beta^{y_1 y_2}) \\
&= (\alpha, \beta)^{(x_1 x_2, y_1 y_2)} \\
&= (\alpha, \beta)^{(x_1,y_1)(x_2,y_2)}
\end{aligned}$$

and

$$(\alpha, \beta)^{(1,1)} = (\alpha^1, \beta^1) = (\alpha, \beta).$$

□

This action is transitive since if $\alpha_1, \alpha_2 \in \Omega_1$ and $\beta_1, \beta_2 \in \Omega_2$, there exist $x \in \mathrm{Sym}(\Omega_1)$ and $y \in \mathrm{Sym}(\Omega_2)$ with $\alpha_1^x = \alpha_2$ and $\beta_1^y = \beta_2$ and then $(\alpha_1, \beta_1)^{(x,y)} = (\alpha_2, \beta_2)$.

Fix $\alpha \in \Omega_1$ and define

$$\Delta = \{\alpha\} \times \Omega_2 = \{\, (\alpha, \omega) \mid \omega \in \Omega_2 \,\}.$$

If $(x, y) \in G$, then $\Delta^{(x,y)} = \{\alpha^x\} \times \Omega_2$. Hence $\Delta^{(x,y)} = \Delta$ when x fixes α and $\Delta^{(x,y)} \cap \Delta = \varnothing$ when $\alpha^x \neq \alpha$. This shows that Δ is a block for the direct product G acting on the set $\Omega = \Omega_1 \times \Omega_2$. Our assumption that $|\Omega_1|, |\Omega_2| > 1$ ensures that this Δ is a non-trivial block.

Similarly sets of the form $\Gamma = \Omega_1 \times \{\beta\}$, for $\beta \in \Omega_2$, are non-trivial blocks for this action of G on Ω.

Lemma 7.5 *Let G be a group that acts transitively on a set Ω. If N is a normal subgroup of G and $\omega \in \Omega$, then the orbit $\Delta = \omega^N$ of ω under the action of N is a block for G.*

PROOF: If $x \in G$, then $Nx = xN$ since $N \trianglelefteqslant G$. Hence

$$\Delta^x = \omega^{Nx} = \omega^{xN} = (\omega^x)^N,$$

the orbit of ω^x under the action of N. Since orbits of N are either disjoint or equal, we conclude either $\Delta^x \cap \Delta = \varnothing$ or $\Delta^x = \Delta$. Hence Δ is a block for G. $\square$

Blocks arise in the classification and description of transitive groups in the following definition.

Definition 7.6 Let G be a group that acts transitively on the set Ω. We say that G acts *imprimitively* on Ω if there is some non-trivial block Δ for G. If, on the other hand, the only blocks for G are the trivial ones (as in Example 7.4(i)), then we say that G acts *primitively* on Ω.

Note: We usually only use the terms "primitive" and "imprimitive" when working with transitive groups.

Example 7.7 *Let $n \geqslant 3$. Show that the alternating group A_n and the symmetric group S_n in their natural actions on $\Omega = \{1, 2, \dots, n\}$ are both primitive.*

SOLUTION: Let $G = A_n$ or S_n. Let Δ be a block for G in its action on Ω and assume that $|\Delta| \geqslant 2$. Let $\alpha, \beta \in \Delta$ be distinct points. Suppose that $\gamma \in \Omega \setminus \{\alpha, \beta\}$. Take $x = (\alpha\ \beta\ \gamma) \in A_n \leqslant G$. Then $\beta = \alpha^x \in \Delta^x$ and so $\Delta^x = \Delta$ (since Δ is a block). Hence $\gamma = \beta^x \in \Delta$ also. We conclude that $\Delta = \Omega$.

Hence there are no non-trivial blocks for G and hence this group acts primitively on Ω. $\square$

We shall meet other examples later in this chapter and on Problem Sheet VII.

To work with blocks and to understand primitive groups, we need to observe how blocks relate to the subgroup structure of the permutation group. The following lemma is the fundamental step towards doing this.

Lemma 7.8 *Let G be a group that acts transitively on a set Ω.*

(i) *If H is a subgroup of G that contains the stabilizer G_ω of some point $\omega \in \Omega$, then ω^H is a block for G.*

(ii) *If Δ is a block for G that contains some point ω, then $\Delta = \omega^H$ for some subgroup H of G with $G_\omega \leqslant H$.*

PROOF: (i) Suppose $G_\omega \leqslant H \leqslant G$. Take $\Delta = \omega^H = \{\, \omega^h \mid h \in H \,\}$. Suppose $\Delta^x \cap \Delta \neq \varnothing$ for some $x \in G$. Then there exists $\delta \in \Delta$, say $\delta = \omega^h$ where $h \in H$, such that $\delta^x \in \Delta$. Then $\omega^{hx} = \omega^k$ for some $k \in H$, so $hxk^{-1} \in G_\omega \leqslant H$. Since $h, k \in H$, we deduce $x \in H$. Therefore

$$\Delta^x = \{\, \omega^{hx} \mid h \in H \,\} = \omega^H = \Delta$$

because $Hx = H$. It follows that Δ is a block for G.

(ii) Let Δ be a block for G that contains ω. There is an induced action of G on the set of subsets of Ω (given by $A^x = \{\, \alpha^x \mid \alpha \in A \,\}$ for all $A \subseteq \Omega$ and $x \in G$). Define H to be the stabilizer of Δ in this induced action; that is,

$$H = \{\, x \in G \mid \Delta^x = \Delta \,\}.$$

As this is the stabilizer in an action, certainly H is a subgroup of G. Now if $x \in G_\omega$, then $\omega = \omega^x \in \Delta \cap \Delta^x$. As Δ is a block, this forces $\Delta^x = \Delta$ so $x \in H$. Hence $G_\omega \leqslant H$.

Now by definition $\omega^x \in \Delta^x = \Delta$ for all $x \in H$, so $\omega^H \subseteq \Delta$. Conversely, if $\delta \in \Delta$, then $\delta = \omega^x$ for some $x \in G$ (as G acts transitively). Now $\delta \in \Delta \cap \Delta^x$, so $\Delta^x = \Delta$ and therefore $x \in H$. We conclude that $\Delta = \omega^H$ for the above stabilizer H of Δ in the induced action on subsets. □

This lemma enables us to establish the following theorem.

Theorem 7.9 *Let G be a group that acts transitively on Ω and let $\omega \in \Omega$. Then there is a one-one inclusion-preserving correspondence between the set of subgroups of G that contain the stabilizer G_ω and the set of blocks for G that contain ω. This correspondence is given by*

$$H \mapsto \omega^H \qquad \text{for } H \geqslant G_\omega.$$

PROOF: By Lemma 7.8(i), the map $\phi\colon H \mapsto \omega^H$ is a function from the set of subgroups of G containing G_ω to the set of blocks containing ω. Part (ii) of the lemma tells us that ϕ is surjective.

Suppose that H and K are subgroups of G containing G_ω. If $H \leqslant K$, then by definition $\omega^H \subseteq \omega^K$. Conversely suppose that $H\phi \subseteq K\phi$; that is, $\omega^H \subseteq \omega^K$. If $h \in H$, then $\omega^h \in \omega^H = \omega^K$, so $\omega^h = \omega^k$ for some $k \in K$. Then $\omega^{hk^{-1}} = \omega$, so $hk^{-1} \in G_\omega \leqslant K$. We deduce $h = (hk^{-1})k \in K$. This shows $H \leqslant K$. Hence

$$H \leqslant K \qquad \text{if and only if} \qquad \omega^H \subseteq \omega^K;$$

that is, ϕ preserves inclusions.

In particular, if $H\phi = K\phi$ the above (applied to the inclusions $H\phi \subseteq K\phi$ and $K\phi \subseteq H\phi$) shows that $H = K$. We conclude that ϕ is injective, which completes the proof of the theorem. □

We are now able to characterize primitivity in terms of the stabilizers of points. It will depend upon the following term (which also appeared earlier on Problem Sheet VI in the context of nilpotent groups):

Definition 7.10 A *maximal subgroup* of a group G is a proper subgroup $M < G$ such that there is no subgroup H with $M < H < G$.

Corollary 7.11 *Let G be a group that acts transitively on a set Ω with $|\Omega| > 1$. Then G acts primitively on Ω if and only if every stabilizer G_ω (for $\omega \in \Omega$) is maximal in G.*

To place this corollary in context, recall that Theorem 2.22 tells us that a transitive action is equivalent to the action on the cosets of a subgroup (namely the stabilizer G_ω of any point ω). We are now observing furthermore that it is primitive when this subgroup is maximal.

PROOF: Since G acts transitively, $|G : G_\omega| = |\Omega| > 1$, so G_ω is a proper subgroup of G.

Suppose first that G is imprimitive in its action on Ω. Then there is some non-trivial block Δ for G. Let $\omega \in \Delta$. Then $\{\omega\} \subset \Delta \subset \Omega$ are blocks containing ω, so by Theorem 7.9, Δ corresponds to a subgroup H with $G_\omega < H < G$. In particular, the stabilizer G_ω is not maximal in G. (Since the stabilizers are conjugate, see Proposition 2.7, they are consequently *all* not maximal.)

Conversely, suppose some stabilizer G_ω is not maximal in G. Then there is a subgroup H of G with $G_\omega < H < G$. Applying Theorem 7.9, we conclude $\Delta = \omega^H$ is a block for G with $\{\omega\} \subset \omega^H \subset \Omega$. Hence Δ is a non-trivial block and so G acts imprimitively on Ω.

By taking the contrapositive, we have established the characterization of primitive groups. □

The reason for the presence of the assumption that Ω contains more than one point is to avoid a trivial case. If $\Omega = \{\omega\}$, then all elements of the group G fix ω (as there is only one point!), so the stabilizers G_ω all equal G. The action is primitive here (as there are no non-empty subsets of Ω except for a singleton), but the stabilizer G_ω is not maximal because it is not a *proper* subgroup.

We can now give another example of a primitive group that will be relevant to the discussion of the structure of primitive groups at the end of the chapter. We shall use the characterization within Corollary 7.11 to establish the primitivity of the example. First recall that if V is a vector space of dimension n over a field F, then it is an additive group and the general linear group $\mathrm{GL}_n(F)$ has a natural action upon it. Accordingly, we can make the following definition:

Definition 7.12 Let V be a vector space of dimension n over some field F. The *affine general linear group* $\mathrm{AGL}_n(F)$ is the semidirect product $V \rtimes \mathrm{GL}_n(F)$ determined by the natural action of $\mathrm{GL}_n(F)$ on the vector space V.

Each linear map $h \in \mathrm{GL}_n(F)$ is, in particular, an automorphism of the additive group $(V, +)$. Hence we obtain an associated homomorphism $\mathrm{GL}_n(F) \to \mathrm{Aut}(V, +)$ and so can form the above semidirect product. We shall write elements of $\mathrm{AGL}_n(F)$ as ordered pairs (x, h) where $x \in V$ and $h \in \mathrm{GL}_n(F)$.

Example 7.13 (i) *Show that* $\mathrm{AGL}_n(F)$ *acts transitively on the vector space* V *by the formula*

$$v^{(x,h)} = (v + x)h$$

where $v, x \in V$ *and* vh *denotes the image of* v *under application of the linear map* h.

(ii) *Show that this action is primitive provided* $n \geqslant 1$.

SOLUTION: Throughout this solution, we shall write $G = \mathrm{AGL}_n(F)$.

(i) Since $\mathrm{AGL}_n(F)$ is a semidirect product, the multiplication is given by

$$(x, h)(y, k) = (x + yh^{-1}, hk).$$

Hence, using the fact that h and k induce linear maps on V,

$$\begin{aligned} v^{(x,h)(y,k)} &= v^{(x+yh^{-1},hk)} \\ &= (v + x + yh^{-1})hk \\ &= vhk + xhk + yk \end{aligned}$$

and

$$\begin{aligned}(v^{(x,h)})^{(y,k)} &= \big((v+x)h+y\big)k \\ &= vhk + xhk + yk\end{aligned}$$

so that

$$(v^{(x,h)})^{(y,k)} = v^{(x,h)(y,k)}$$

for all $v \in V$ and $(x,h), (y,k) \in G$. The identity element of G is $(0,1)$ and

$$v^{(0,1)} = (v+0)1 = v \qquad \text{for all } v \in V.$$

(Here 0 is the zero vector in V and 1 is the identity map.) Hence we do indeed have an action of G on V. Finally if $v, w \in V$, then $g = (w-v, 1) \in G$ and

$$v^{(w-v,1)} = v + (w-v) = w.$$

Hence $G = \mathrm{AGL}_n(F)$ acts transitively on V.

(ii) Assume that $\dim V \geqslant 1$, so $V \neq \mathbf{0}$. We shall show that the stabilizer G_0 of the zero vector is maximal in $G = \mathrm{AGL}_n(F)$. First observe that $0^{(x,h)} = xh$ and so, since h is an invertible linear map, it follows that $0^{(x,h)} = 0$ if and only if $x = 0$. Hence

$$G_0 = \{\, (0,h) \mid h \in \mathrm{GL}_n(F) \,\};$$

that is, this stabilizer equals $\mathrm{GL}_n(F)$ under the standard identification of the complement with the corresponding subgroup of the semidirect product. Since $V \neq \mathbf{0}$, G_0 is a proper subgroup of G.

We now show that this stabilizer is a maximal subgroup of G. Suppose that K is a subgroup of G with $G_0 < K \leqslant G$. Hence there exists some $(x,h) \in K$ with $x \neq 0$. Since $\mathrm{GL}_n(F) = G_0 < K$, the subgroup K contains the element

$$(x,h)(0,h^{-1}) = (x + 0h^{-1}, hh^{-1}) = (x,1).$$

If y is a non-zero vector in V, there exists some invertible linear map $k \in \mathrm{GL}_n(F)$ that moves x to y. Hence K also contains

$$(0,k^{-1})(x,1) = (xk, k^{-1}) = (y, k^{-1})$$

and now if h' is any element of $\mathrm{GL}_n(F)$, K also contains

$$(y,k^{-1})(0,kh') = (y,h').$$

Since y is an arbitrary non-zero vector and $h' \in \mathrm{GL}_n(F)$ is arbitrary, we conclude $K = \mathrm{AGL}_n(F)$. (Note we already know that K contains all pairs with $y = 0$ since $G_0 < K$.)

This shows that the stabilizer G_0 is a maximal subgroup of G. Corollary 7.11 tells us that G acts primitively on the vector space V. □

Intransitive groups

In the remainder of the chapter, we shall exploit the nature of the action to describe the general structure of a permutation group. We begin by considering intransitive permutation groups, then move onto transitive groups that are imprimitive, and then finishing by considering primitive groups.

We begin by constructing what will turn out to be a typical intransitive permutation group.

Lemma 7.14 *Let Ω_1 and Ω_2 be disjoint non-empty sets and let $G \leqslant \mathrm{Sym}(\Omega_1)$ and $H \leqslant \mathrm{Sym}(\Omega_2)$. Define an action of the direct product $G \times H$ on the union $\Omega_1 \cup \Omega_2$ by*

$$\alpha^{(x,y)} = \alpha^x \qquad \text{and} \qquad \beta^{(x,y)} = \beta^y$$

for all $\alpha \in \Omega_1$, $\beta \in \Omega_2$, $x \in G$ and $y \in H$. Then

(i) *this defines a faithful action of $G \times H$ on $\Omega_1 \cup \Omega_2$;*

(ii) *if $\alpha \in \Omega_1$ then the orbit of α under $G \times H$ is contained in Ω_1;*

(iii) *if G acts transitively on Ω_1, then Ω_1 is an orbit of $G \times H$ in its action on the union $\Omega_1 \cup \Omega_2$.*

We shall often call this the *standard action* of the direct product $G \times H$ on the disjoint union $\Omega_1 \cup \Omega_2$. Since the action is faithful, we can view the direct product $G \times H$ as a subgroup of the symmetric group $\mathrm{Sym}(\Omega_1 \cup \Omega_2)$ via the permutation representation.

This lemma tells us, in particular, that there is a faithful action of $\mathrm{Sym}(\Omega_1) \times \mathrm{Sym}(\Omega_2)$ on the union $\Omega_1 \cup \Omega_2$ for which the orbits are Ω_1 and Ω_2. Note this action on the union $\Omega_1 \cup \Omega_2$ is different to the action of $\mathrm{Sym}(\Omega_1) \times \mathrm{Sym}(\Omega_2)$ on the Cartesian product $\Omega_1 \times \Omega_2$ that was introduced in Example 7.4(iii).

PROOF: (i) Let $\alpha \in \Omega_1$. If $(x_1, y_1), (x_2, y_2) \in G \times H$, then since $\alpha^{x_1} \in \Omega_1$ by definition,

$$(\alpha^{(x_1,y_1)})^{(x_2,y_2)} = (\alpha^{x_1})^{(x_2,y_2)} = (\alpha^{x_1})^{x_2} = \alpha^{x_1x_2} = \alpha^{(x_1x_2,y_1y_2)} = \alpha^{(x_1,y_1)(x_2,y_2)}.$$

A similar argument shows that $(\beta^g)^h = \beta^{gh}$ for all $\beta \in \Omega_2$ and $g, h \in G \times H$. Also

$$\alpha^{(1,1)} = \alpha^1 = \alpha \qquad \text{and} \qquad \beta^{(1,1)} = \beta^1 = \beta$$

for $\alpha \in \Omega_1$ and $\beta \in \Omega_2$. This demonstrates that the above does define an action of $G \times H$ on $\Omega_1 \cup \Omega_2$.

If $(x, y) \in G \times H$ is not the identity element, then either $x \neq 1$ or $y \neq 1$. If $x \neq 1$, then as G acts faithfully on Ω_1, there exists $\alpha \in \Omega_1$ such that $\alpha^x \neq \alpha$. Therefore

$$\alpha^{(x,y)} = \alpha^x \neq \alpha.$$

We apply a similar argument if $y \neq 1$ to produce $\beta \in \Omega_2$ with $\beta^{(x,y)} \neq \beta$. Hence no non-identity element of $G \times H$ fixes all points of $\Omega_1 \cup \Omega_2$. Therefore the action is faithful.

(ii) This was essentially observed within the proof of the previous part. If $\alpha \in \Omega_1$ and $(x, y) \in G \times H$, then

$$\alpha^{(x,y)} = \alpha^x \in \Omega_1$$

since $G \leqslant \mathrm{Sym}(\Omega_1)$. Hence every point in the orbit $\alpha^{G \times H}$ is contained in Ω_1.

(iii) Suppose now that G acts transitively on Ω_1. Let $\alpha, \alpha' \in \Gamma$. Then there exists $g \in G$ such that $\alpha^g = \alpha'$. Hence

$$\alpha^{(g,1)} = \alpha^g = \alpha'$$

and so α and α' lie in the same orbit of $G \times H$. Hence Ω_1 is an orbit of $G \times H$ in its action on $\Omega_1 \cup \Omega_2$. □

We now observe that the above description is actually typical of intransitive groups. If a group acts intransitively on a set Ω, then it sits inside a permutation group that can be decomposed as a direct product.

Proposition 7.15 *Let G be a subgroup of* $\mathrm{Sym}(\Omega)$ *for some set Ω and suppose that Ω_1 is an orbit of G. Then G is permutation isomorphic to a subgroup of the direct product* $\mathrm{Sym}(\Omega_1) \times \mathrm{Sym}(\Omega \setminus \Omega_1)$ *in its standard action.*

Recall the definition of permutation isomorphism from Definition 2.21. Permutation groups $G \leqslant \mathrm{Sym}(\Omega)$ and $H \leqslant \mathrm{Sym}(\Omega')$ are permutation isomorphic if there is an isomorphism $\theta \colon G \to H$ and a bijection $\phi \colon \Omega \to \Omega'$ such that $(\omega^x)\phi = (\omega\phi)^{x\theta}$ for all $\omega \in \Omega$ and $x \in G$.

Proof: Set $\Omega_2 = \Omega \setminus \Omega_1$. Since Ω is the disjoint union of orbits for the action of G, this complement Ω_2 is a union of orbits. Therefore

$$\alpha^x \in \Omega_1 \qquad \text{and} \qquad \beta^x \in \Omega_2$$

for all $\alpha \in \Omega_1$, $\beta \in \Omega_2$ and $x \in G$. Hence G has an induced action on Ω_1 and on Ω_2. Let $\rho_1 \colon G \to \mathrm{Sym}(\Omega_1)$ and $\rho_2 \colon G \to \mathrm{Sym}(\Omega_2)$ be the associated permutation representations. We may therefore define $\theta \colon G \to \mathrm{Sym}(\Omega_1) \times \mathrm{Sym}(\Omega_2)$ by

$$x \mapsto (x\rho_1, x\rho_2).$$

This is a homomorphism. Furthermore, if $x \in \ker\theta$, then $x\rho_1$ and $x\rho_2$ are the identities in the respective permutation groups; that is,

$$\alpha^x = \alpha \qquad \text{and} \qquad \beta^x = \beta$$

for all $\alpha \in \Omega_1$ and $\beta \in \Omega_2$. Thus x fixes all points of $\Omega = \Omega_1 \cup \Omega_2$, so $x = 1$. We have shown that θ is injective and therefore G is isomorphic to the subgroup $\operatorname{im}\theta$ of $\mathrm{Sym}(\Omega_1) \times \mathrm{Sym}(\Omega_2)$. This is actually a permutation isomorphism: if $\alpha \in \Omega_1$ then

$$\alpha^x = \alpha^{x\rho_1} = \alpha^{(x\rho_1, x\rho_2)} = \alpha^{x\theta}$$

and similarly $\beta^x = \beta^{x\theta}$ for all $\beta \in \Omega_2$. This establishes the formula $\omega^x = \omega^{x\theta}$ for all $\omega \in \Omega$ and $x \in G$, so G is permutational isomorphic to $G\theta \leqslant \mathrm{Sym}(\Omega_1) \times \mathrm{Sym}(\Omega_2)$ (where the associated bijection $\Omega \to \Omega_1 \cup \Omega_2$ is the identity map). □

Let us illustrate the phenomenon described in the above proposition with an easy example.

Example 7.16 Consider $G = \langle (1\ 2)(3\ 4) \rangle \leqslant S_4$ in its natural action on $\Omega = \{1, 2, 3, 4\}$. There are two orbits for G, namely $\Omega_1 = \{1, 2\}$ and $\Omega_2 = \{3, 4\}$. Let $\rho_1 \colon G \to \mathrm{Sym}(\Omega_1)$ and $\rho_2 \colon G \to \mathrm{Sym}(\Omega_2)$ be the permutation representations associated to the induced actions of G on the orbits Ω_1 and Ω_2. By definition, $\big((1\ 2)(3\ 4)\big)\rho_1$ is the permutation on Ω_1 induced by the generator of G. This permutation moves 1 to 2 and 2 to 1, so

$$\big((1\ 2)(3\ 4)\big)\rho_1 = (1\ 2)$$

and similarly $\big((1\ 2)(3\ 4)\big)\rho_2 = (3\ 4)$. Define $G_1 = G\rho_1$ and $G_2 = G\rho_2$, so by our calculation above

$$G_1 = \langle (1\ 2) \rangle \qquad \text{and} \qquad G_2 = \langle (3\ 4) \rangle.$$

Proposition 7.15 asserts that $G = \langle (1\ 2)(3\ 4) \rangle$ is permutation isomorphic to a subgroup of $G_1 \times G_2 = \langle (1\ 2), (3\ 4) \rangle \cong C_2 \times C_2$. Note that G is not itself a direct product but *embeds* in a direct product that reflects the intransitive nature of the action of G.

Imprimitive groups

Our next goal is to describe the structure of transitive groups that are imprimitive. We shall use the presence of a non-trivial block to describe such an imprimitive group in terms of a wreath product, as introduced in Chapter 4. Accordingly, we begin by considering the wreath product of two permutation groups.

Let H be a permutation group on a set Δ and K be a permutation group on a set $\Sigma = \{1, 2, \dots, n\}$; that is, $H \leqslant \mathrm{Sym}(\Delta)$ and $K \leqslant \mathrm{Sym}(\Sigma)$. Set

$$\Omega = \Delta \times \Sigma = \{\, (\delta, i) \mid \delta \in \Delta,\ 1 \leqslant i \leqslant n \,\}.$$

Let $W = H \operatorname{wr}_{\Sigma} K$ be the wreath product of H by K with respect to the action of K on Σ. Thus W is the semidirect product

$$W = B \rtimes K$$

where the base group $B = \{\, (h_1, h_2, \dots, h_n) \mid h_i \in H \,\}$ is a direct product of n copies of H and where the action of K on B is given by permuting the entries:

$$(h_1, h_2, \dots, h_n)^k = (h_{1k^{-1}}, h_{2k^{-1}}, \dots, h_{nk^{-1}})$$

for $(h_1, h_2, \dots, h_n) \in B$ and $k \in K$.

We shall now construct an action of W on Ω. If $g = (h_1, h_2, \dots, h_n)k \in W$ where $h_1, h_2, \dots, h_n \in H$ and $k \in K$, define

$$(\delta, i)^g = (\delta^{h_i}, ik); \tag{7.1}$$

that is, apply the ith element h_i to $\delta \in \Delta$ and apply the permutation k to $i \in \Sigma$. (We are mixing notation by writing the action of H as exponentiation, but that of K as juxtaposition. It will keep later notation cleaner to do so!) If $g = (h_1, h_2, \dots, h_n)k$ and $g' = (h'_1, h'_2, \dots, h'_n)k'$ are elements of W, then

$$\begin{aligned} gg' &= (h_1, h_2, \dots, h_n)k(h'_1, h'_2, \dots, h'_n)k' \\ &= (h_1, h_2, \dots, h_n)(h'_1, h'_2, \dots, h'_n)^{k^{-1}} kk' \\ &= (h_1, h_2, \dots, h_n)(h'_{1k}, h'_{2k}, \dots, h'_{nk})kk' \\ &= (h_1 h'_{1k}, h_2 h'_{2k}, \dots, h_n h'_{nk})kk'. \end{aligned}$$

Then

$$\begin{aligned} \big((\delta, i)^g\big)^{g'} &= (\delta^{h_i}, ik)^{g'} \\ &= (\delta_{h_i h'_{ik}}, ikk') \\ &= (\delta, i)^{gg'}. \end{aligned}$$

The identity element of W is $(1, 1, \dots, 1)1$ and this fixes all elements of Ω. Hence W acts on the set Ω.

If $g = (h_1, h_2, \dots, h_n)k$ is a non-identity element of W, then either there is some i such that $h_i \neq 1$, so h_i moves some point $\delta \in \Delta$, or $k \neq 1$, so k moves some point $i \in \Sigma$. Hence there exists $(\delta, i) \in \Omega$ such that $(\delta, i)^g = (\delta^{h_i}, ik) \neq (\delta, i)$. This show that the action of W on Ω is faithful.

Suppose that H acts transitively on Δ and K acts transitively on Σ. If $(\delta, i), (\delta', j) \in \Omega$, take any $h_i \in H$ with $\delta^{h_i} = \delta'$ and $k \in K$ with $ik = j$. Then $g = (\dots, h_i, \dots)k$ moves (δ, i)

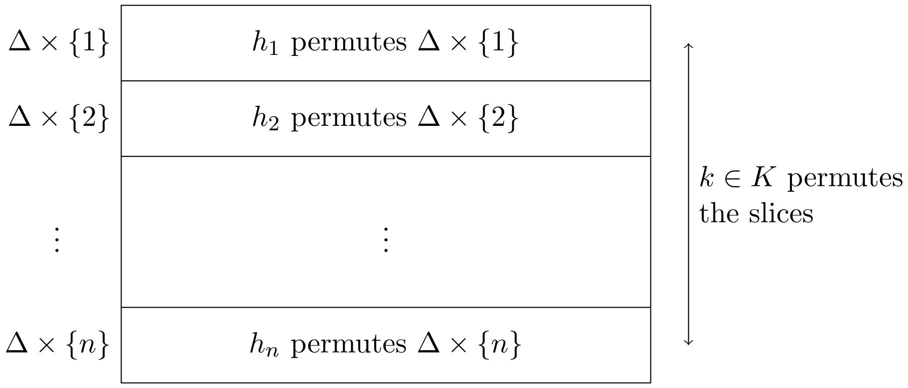


Figure 7.1: The imprimitive action of the wreath product W on $\Omega = \Delta \times \Sigma$

to (δ', j), no matter what we choose for the other elements of H appearing within g. Hence the action of W on Ω is also transitive when H and K are chosen to be transitive.

Finally, if $g = bk$ with $b \in B$ and $k \in K$, then the formula (7.1) for the action tells us that

$$\big(\Delta \times \{i\}\big)^g = \Delta \times \{ik\} = \Delta \times \{j\}.$$

Therefore $\big(\Delta \times \{i\}\big)^g$ is either equal to or is disjoint from $\Delta \times \{i\}$ (depending on whether $ik = i$ or $ik \neq i$). It is a non-trivial block provided both $|\Delta| > 1$ and $n = |\Sigma| > 1$. The corresponding decomposition of Ω as a disjoint union of blocks is

$$\Omega = \bigcup_{i=1}^{n} \big(\Delta \times \{i\}\big).$$

This is a union of n disjoint sets, each of which is naturally in one-one correspondence with the original set Δ.

We record our observations as follows:

Lemma 7.17 *Let H and K be transitive permutation groups on the sets Δ and $\Sigma = \{1, 2, \ldots, n\}$, respectively, where $|\Delta|, |\Sigma| > 1$. Then the wreath product $W = H \operatorname{wr}_\Sigma K$ acts transitively and faithfully, but imprimitively on the set $\Omega = \Delta \times \Sigma$ by the formula*

$$(\delta, i)^g = (\delta^{h_i}, ik)$$

for $\delta \in \Delta$, $i \in \Sigma$ and $g = (h_1, h_2, \ldots, h_n)k \in W$. □

We shall call this the *imprimitive action* of the wreath product W on $\Delta \times \Sigma$. To understand this action, one can think of the index from Σ as providing a way to slice the set Ω into copies of Δ. The entries of each element $b = (h_1, h_2, \ldots, h_n)$ of the base group act independently on each slice: b permutes the entries of $\Delta \times \{i\}$ in the same way as h_i acts on Δ. An element $g = (h_1, h_2, \ldots, h_n)k$ of the wreath product W first shuffles the points in each slice $\Delta \times \{i\}$ according to the element $h_i \in H$ and then k permutes the slices. See Figure 7.1 for an illustration of this action of the wreath product. Since the action is faithful, we can identify W with a subgroup of $\mathrm{Sym}(\Delta \times \Sigma)$ via the associated permutation representation.

We complete our discussion of imprimitive permutation groups by observing that every imprimitive permutation group can be viewed as being contained within a wreath product. Thus wreath products are maximal amongst the *imprimitive* permutation groups.

Theorem 7.18 *Let G be a permutation group that acts transitively on a set Ω and let Δ be a block for G. Assume that the number n of blocks in the system $\Sigma' = \{ \Delta^x \mid x \in G \}$ is finite and set $\Sigma = \{1, 2, \dots, n\}$. Then G is permutation isomorphic to a subgroup of the wreath product* $\mathrm{Sym}(\Delta) \operatorname{wr}_\Sigma \mathrm{Sym}(\Sigma)$ *in its imprimitive action on* $\Delta \times \Sigma$.

The assumption that Σ' contains only finitely many blocks is for notational convenience.

Proof: Let $\Sigma' = \{\Delta_1, \Delta_2, \dots, \Delta_n\}$ for the system of blocks Δ^x (for $x \in G$) determined by Δ. If $\Delta_i = \Delta^y \in \Sigma'$, then $\Delta_i^x = \Delta^{yx} \in \Sigma'$ for all $x \in G$. Moreover, if $\gamma \in \Delta_i$ and $\delta \in \Delta_j$, then by transitivity there exists $x \in G$ such that $\gamma^x = \delta$. Therefore, as these are blocks, $\Delta_i^x = \Delta_j$. We conclude that G acts transitively on Σ'. Hence there is an associated permutation representation $\rho \colon G \to S_n = \mathrm{Sym}(\Sigma)$ satisfying

$$\Delta_i^x = \Delta_{i(x\rho)}$$

for each $i \in \Sigma = \{1, 2, \dots, n\}$ and $x \in G$.

For each $i \in \Sigma$, fix a bijection $\phi_i \colon \Delta_i \to \Delta$ and define $\phi \colon \Omega \to \Delta \times \Sigma$ by

$$\alpha \mapsto (\alpha\phi_i, i) \qquad \text{when } \alpha \in \Delta_i.$$

Since Ω is the disjoint union of the blocks Δ_i, it follows that ϕ is a bijection. If $(\delta, i) \in \Delta \times \Sigma$, then $(\delta, i)^{\phi^{-1}} \in \Delta_i$, so

$$(\delta, i)^{\phi^{-1}x} \in \Delta_i^x = \Delta_{i(x\rho)}$$

for each $x \in G$. Hence $(\delta, i)^{\phi^{-1}x\phi} \in \Delta \times \{i(x\rho)\}$ for all $x \in G$, so there is a permutation $x\beta_i$ in $\mathrm{Sym}(\Delta)$ such that

$$(\delta, i)^{\phi^{-1}x\phi} = (\delta^{x\beta_i}, i(x\rho)) \tag{7.2}$$

for each $\delta \in \Delta$, $i \in \Sigma$ and $x \in G$. We now define $\theta \colon G \to \mathrm{Sym}(\Delta) \operatorname{wr}_\Sigma \mathrm{Sym}(\Sigma)$ by

$$x \mapsto (x\beta_1, x\beta_2, \dots, x\beta_n)(x\rho).$$

We shall show that θ is the required permutation isomorphism.

First let $x, y \in G$. Then, with two applications of Equation (7.2),

$$\begin{aligned}
(\delta, i)^{\phi^{-1}xy\phi} &= (\delta, i)^{(\phi^{-1}x\phi)(\phi^{-1}y\phi)} \\
&= (\delta^{x\beta_i}, i(x\rho))^{\phi^{-1}y\phi} \\
&= (\delta^{(x\beta_i)(y\beta_{i(x\rho)})}, i(x\rho)(y\rho)) \\
&= \big(\delta^{(x\beta_i)(y\beta_{i(x\rho)})}, i((xy)\rho)\big).
\end{aligned}$$

Hence

$$(xy)\beta_i = (x\beta_i)(y\beta_{i(x\rho)})$$

for all $x, y \in G$ and $i \in \Sigma$. This formula enables us to show that θ is a homomorphism:

$$\begin{aligned}
(x\theta)(y\theta) &= (x\beta_1, x\beta_2, \dots, x\beta_n)(x\rho)\,(y\beta_1, y\beta_2, \dots, y\beta_n)(y\rho) \\
&= (x\beta_1, x\beta_2, \dots, x\beta_n)\,(y\beta_1, y\beta_2, \dots, y\beta_n)^{(x\rho)^{-1}}\,(x\rho)(y\rho) \\
&= (x\beta_1, x\beta_2, \dots, x\beta_n)\,(y\beta_{1(x\rho)}, y\beta_{2(x\rho)}, \dots, y\beta_{n(x\rho)})\,(xy)\rho \\
&= \big((x\beta_1)(y\beta_{1(x\rho)}), (x\beta_2)(y\beta_{2(x\rho)}), \dots, (x\beta_n)(y\beta_{n(x\rho)})\big)\,(xy)\rho \\
&= \big((xy)\beta_1, (xy)\beta_2, \dots, (xy)\beta_n\big)\,(xy)\rho \\
&= (xy)\theta
\end{aligned}$$

for $x, y \in G$.

Let $x \in \ker\theta$. Thus $x\beta_1 = x\beta_2 = \cdots = x\beta_n = 1$ (the identity permutation in $\mathrm{Sym}(\Delta)$) and $x\rho = 1$ (the identity permutation in $\mathrm{Sym}(\Sigma)$). Hence substituting into Equation (7.2) gives

$$(\delta, i)^{\phi^{-1}x\phi} = (\delta, i)$$

for all $(\delta, i) \in \Omega' = \Delta \times \Sigma$ and, since $\phi \colon \Omega \to \Delta \times \Sigma$ is a bijection, we deduce

$$\alpha^x = \alpha \qquad \text{for all } \alpha \in \Omega.$$

This shows that $x = 1$ and we have established that $\ker\theta = \mathbf{1}$; that is, θ is injective. Therefore θ determines an isomorphism from G to its image $G\theta$, which is a subgroup of the wreath product $W = \mathrm{Sym}(\Delta) \operatorname{wr}_\Sigma \mathrm{Sym}(\Sigma)$.

Finally, let $x \in G$ and $\alpha \in \Omega$. Define the permutations $x\beta_i$ as above and set $(\delta, i) = \alpha^\phi$. Since $x\theta = (x\beta_1, x\beta_2, \dots, x\beta_n)(x\rho)$, the definition of the imprimitive action of W on $\Delta \times \Sigma$ tells us that

$$(\delta, i)^{x\theta} = (\delta^{x\beta_i}, i(x\rho))$$

(see Equation (7.1)). Comparing with Equation (7.2), we deduce

$$(\delta, i)^{x\theta} = (\delta, i)^{\phi^{-1}x\phi};$$

that is,

$$(\alpha^\phi)^{x\theta} = (\alpha^x)^\phi \qquad \text{for all } \alpha \in \Omega \text{ and } x \in G.$$

This shows that θ is indeed a permutation isomorphism from G to a subgroup of W. $\square$

The structure of primitive groups

We have now shown that an intransitive action of a permutation group essentially corresponds to a direct product decomposition and that a transitive but imprimitive action corresponds to a decomposition as a wreath product. We shall finish the chapter, by examining the structure of a primitive permutation group. The ideas presented here are motivated by an important theorem called the O'Nan–Scott Theorem, but we will not be able to go as far as proving that theorem. Nevertheless, we shall indicate many of the ideas contributing to its proof.

The description of the structure of a primitive group involves use of the centralizer of a subgroup, which is defined as follows:

Definition 7.19 Let G be a group and H be a subgroup of G. The *centralizer* of H in G is

$$\mathrm{C}_G(H) = \{\, x \in G \mid xh = hx \text{ for all } h \in H \,\},$$

the set of all elements that commute with every element of H.

Note that $\mathrm{C}_G(H) = \bigcap_{h \in H} \mathrm{C}_G(h)$ expresses this centralizer as an intersection of centralizers of elements. We already know the latter are subgroups of G and hence $\mathrm{C}_G(H)$ is also a subgroup. It is quite easy (see Problem Sheet VII) to verify that if N is a normal subgroup of G, then $\mathrm{C}_G(N)$ is also a normal subgroup of G.

Lemma 7.20 *Let G be a permutation group on Ω and H be a subgroup of G that acts transitively on Ω. If $C = \mathrm{C}_G(H)$ is the centralizer of H in G, then $C_\omega = 1$ for all $\omega \in \Omega$.*

PROOF: Fix $\omega \in \Omega$. Suppose that $x \in C_\omega$. Let $\alpha \in \Omega$ be arbitrary. Since H acts transitively, there exists some $h \in H$ such that $\alpha = \omega^h$. Note that $xh = hx$ (because $x \in \mathrm{C}_G(H)$) and so

$$\alpha^x = \omega^{hx} = \omega^{xh} = \omega^h = \alpha$$

using the fact that x fixes ω. Hence x fixes all points of Ω and therefore $x = 1$ (since C is faithful on Ω). This shows $C_\omega = \mathbf{1}$ for all $\omega \in \Omega$. □

Lemma 7.21 *Let G be a finite group and let M and N be distinct minimal normal subgroups of G. Then $[M, N] = \mathbf{1}$.*

PROOF: Note that $M \cap N$ is a normal subgroup of G contained in both M and N. If it were non-trivial, then $M = M \cap N = N$ by minimality of the two subgroups, contrary to the assumption that $M \neq N$. Hence $M \cap N = \mathbf{1}$. If $x \in M$ and $y \in N$, then

$$[x, y] = x^{-1}y^{-1}xy \in M \cap N$$

since $y^{-1}xy \in M$ and $x^{-1}y^{-1}x \in N$. Hence $[x, y] = 1$ for all $x \in M$ and $y \in N$; that is, $[M, N] = \mathbf{1}$. □

We can now give the main first step in the description of the structure of a finite primitive permutation group. It is expressed in terms of minimal normal subgroups. Recall from Theorem 3.19 that a minimal normal subgroup of a finite group is a direct product of isomorphic simple groups. It is therefore either an elementary abelian p-group for some prime p (that is, a direct product of cyclic groups of order p) or a direct product of isomorphic non-abelian finite simple groups. In the case of primitive permutation groups, there is actually considerable further constraints on minimal normal subgroups.

Theorem 7.22 *Let G be a non-trivial finite primitive permutation group on a set Ω. Then one of the following holds:*

(i) *G has a unique minimal normal subgroup N which is abelian, N acts regularly on Ω and $\mathrm{C}_G(N) = N$;*

(ii) *G has precisely two minimal normal subgroups M and N, these are both non-abelian, act regularly on Ω with $\mathrm{C}_G(M) = N$ and $\mathrm{C}_G(N) = M$;*

(iii) *G has a unique minimal normal subgroup N that is non-abelian and $\mathrm{C}_G(N) = \mathbf{1}$.*

PROOF: Let N be any minimal normal subgroup of G. If $\omega \in \Omega$, then the orbit ω^N is not a singleton since $N \neq \mathbf{1}$ and G acts faithfully. However, ω^N is a block for G by Lemma 7.5 and hence $\omega^N = \Omega$ since G is primitive. Hence every minimal normal subgroup N of G acts transitively on Ω.

Now suppose that N is an abelian group. Let $C = \mathrm{C}_G(N)$, so that $N \leqslant C$. Certainly then C also acts transitively on Ω. Apply Lemma 7.20 to deduce that $C_\omega = \mathbf{1}$. It follows that $N_\omega = \mathbf{1}$. Hence N and C both act regularly on Ω. Therefore $|N| = |C| = |\Omega|$ by the Orbit-Stabilizer Theorem. We deduce that $C = \mathrm{C}_G(N) = N$. Now if M is any other minimal normal subgroup of G, then M commutes with N by Lemma 7.21. Hence $M \leqslant C = N$, which forces $M = N$, by minimality of N. This establishes that (i) holds when G has an abelian minimal normal subgroup.

For the remainder of the proof, we assume that the minimal normal subgroups of G are non-abelian. Suppose that M and N are two distinct minimal normal subgroups of G. We know these both act transitively on Ω. Let $C = \mathrm{C}_G(N)$. Then $M \leqslant C$ by Lemma 7.21

and so C acts transitively on Ω. Apply Lemma 7.20 to deduce that $C_\omega = \mathbf{1}$. Hence $M_\omega = \mathbf{1}$ and we deduce that both M and C act regularly on Ω. Therefore $|M| = |C| = |\Omega|$ by the Orbit-Stabilizer Theorem and hence $M = C = \mathrm{C}_G(N)$. We can interchange the roles of M and N to conclude N is also regular on Ω and $N = \mathrm{C}_G(M)$. Also there cannot be a further minimal normal subgroup R for the same argument would then show $R = \mathrm{C}_G(N) = M$. This establishes that (ii) holds when G has more than one minimal normal subgroup.

Finally suppose that G has a unique minimal normal subgroup N and this is non-abelian. Let $C = \mathrm{C}_G(N) \trianglelefteq G$. Since N is a direct product of non-abelian simple groups, its centre is trivial, so $N \cap C = \mathbf{1}$. If it were the case that $C \neq \mathbf{1}$, then there would exist some minimal normal subgroup M contained in C, but this could not equal N as $N \nleqslant C$. Hence $C = \mathrm{C}_G(N) = \mathbf{1}$ in this final case and we have established that (iii) holds. □

Note that Theorem 7.22 tells us that if a primitive group $G \leqslant \mathrm{Sym}(\Omega)$ has a minimal normal subgroup N that does not act regularly, then Case (iii) applies: N is the unique minimal normal subgroup of G and is non-abelian. On the other hand, if N acts regularly then by the Orbit-Stabilizer Theorem, there is a bijection from N to the set Ω. Analyzing these situations and the cases within Theorem 7.22 in careful detail yields the full classification of finite primitive permutation groups. We summarize this in the following, but refer to Chapter 4 of Dixon & Mortimer's book (*Permutation Groups*, Springer 1996) for full description of the classes and the proof of the theorem.

Theorem 7.23 (O'Nan–Scott Theorem) *Suppose that G is a non-trivial finite primitive permutation group on a set Ω and let N be a minimal normal subgroup of G. Then*

(i) *either N is regular on Ω and G is of one of the following types:*
  - *affine type,*
  - *regular non-abelian type;*

(ii) *or N is not regular on Ω and G is of one of the following types:*
  - *diagonal type,*
  - *product type,*
  - *an almost simple group.*

We shall only describe three of the "types" appearing in this theorem within these notes.

Affine groups

We defined the affine general linear group in Definition 7.12 and showed in Example 7.13 that $\mathrm{AGL}_n(F)$ acts primitively on the vector space $V = F^n$ whenever $n \geqslant 1$. We shall demonstrate how the groups occurring in Case (i) of Theorem 7.22 occur as subgroups of the affine general linear group. These are the primitive groups of *affine type*.

Let G be a finite primitive permutation group on Ω such that G has a unique minimal normal subgroup N that is abelian, that acts regularly on Ω and that $\mathrm{C}_G(N) = N$. (That is, G occurs as described in Case (i) of Theorem 7.22.) Fix $\omega \in \Omega$ and let $H = G_\omega$. Then $H \cap N = N_\omega = \mathbf{1}$ so, in particular, $N \nleqslant H$. Since G is primitive, H is a maximal subgroup of G by Corollary 7.11 and therefore $G = HN$. Hence G has the structure of a semidirect product $G = N \rtimes_\phi H$ for some $\phi\colon H \to \operatorname{Aut} N$.

Since N is an abelian minimal normal subgroup of G, it is an elementary abelian p-group for some prime p by Theorem 3.19. We shall write the group operation in N additively so that

$$N \cong V = \underbrace{\mathbb{F}_p \oplus \mathbb{F}_p \oplus \cdots \oplus \ldots \mathbb{F}_p}_{d \text{ times}}.$$

This V has the structure of a d-dimensional vector space over the field $\mathbb{F}_p$ of p elements. To simplify notation, we shall identify N and V via this isomorphism. Homomorphisms of N are therefore linear maps and $\operatorname{Aut} N \cong \mathrm{GL}_d(\mathbb{F}_p)$. Hence we can assume $\phi\colon H \to \mathrm{GL}_d(\mathbb{F}_p)$. Since the map ϕ is induced by conjugation of elements of N by elements of H, $h \in \ker\phi$ if and only if h commutes with all elements of N; that is,

$$\ker\phi = \mathrm{C}_H(N) = H \cap \mathrm{C}_G(N) = H \cap N = \mathbf{1}.$$

Hence ϕ is injective and so, again to simplify notation, we shall identify H with its image in $\mathrm{GL}_d(\mathbb{F}_p)$. In conclusion, we have shown $G = N \rtimes_\phi H$ where N has the structure of a vector space of dimension d over $\mathbb{F}_p$ and H is a subgroup of $\mathrm{GL}_d(\mathbb{F}_p)$.

Now let us examine the action of G on Ω. We shall write the elements of G as ordered pairs (x, h) where $x \in N$ and $h \in H$. As N acts regularly, the Orbit-Stabilizer Theorem gives a bijection between N and the elements of Ω and hence every element of Ω is uniquely expressible in the form $\omega^{(v,1)}$ for $v \in N$. Since every element of H fixes ω, we see the action is given by

$$(\omega^{(v,1)})^{(x,h)} = \omega^{(0,h^{-1})(v,1)(x,h)} = \omega^{(0,h^{-1})(v+x,h)} = \omega^{((v+x)h,1)},$$

where we write vh for the image of the vector v under the linear map h (which is how elements of H act by conjugation on N). We conclude that there is a bijection $\psi\colon N \to \Omega$ given by $v\psi = \omega^{(v,1)}$ which has the property that

$$(v\psi)^{(x,h)} = (v^{(x,h)})\psi$$

where the action of $G = N \rtimes H$ on $N = V$ is given by

$$v^{(x,h)} = (v + x)h.$$

The latter is the formula for the action of the affine general linear group $\mathrm{AGL}_d(\mathbb{F}_p)$ considered in Example 7.13. Here $G = N \rtimes H$ has been identified with a subgroup of $\mathrm{AGL}_d(\mathbb{F}_p) = V \rtimes \mathrm{GL}_d(\mathbb{F}_p)$ via our identification of N with V and H with a subgroup of $\mathrm{GL}_d(\mathbb{F}_p)$. It follows that we have a permutation isomorphism between the action of this subgroup of $\mathrm{AGL}_d(\mathbb{F}_p)$ and the action of G on Ω.

We have now described the affine type primitive groups arising in the O'Nan–Scott Theorem:

Theorem 7.24 (Affine Type Primitive Groups) *Let G be a finite primitive permutation group on a set Ω and suppose that G has an abelian minimal normal subgroup N. Let $|N| = p^d$ where p is a prime number. Then there is a subgroup H of $\mathrm{GL}_d(\mathbb{F}_p)$ such that G is permutational isomorphic to the subgroup $V \rtimes H$ of the affine general linear group $\mathrm{AGL}_d(\mathbb{F}_p)$ in its natural action on the vector space V of dimension d over $\mathbb{F}_p$.* □

There is one final thing to note about this theorem. Not all choices of H give rise to primitive actions, but if one assumes the additional condition that H acts *irreducibly* (that is, there is no non-zero proper subspace W of V that is invariant under the action of H) then this completely characterizes the finite primitive groups of affine type. This is discussed in more detail in a question on Problem Sheet VII.

Product type

Let $H \leqslant \mathrm{Sym}(\Delta)$ and $K \leqslant \mathrm{Sym}(\Sigma)$ be permutation groups where, for notational convenience, assume $\Sigma = \{1, 2, \dots, n\}$. Assume that both $|\Delta| > 1$ and $n = |\Sigma| > 1$. There is an action of the wreath product $W = H \operatorname{wr}_\Sigma K$ on the set

$$\Omega = \Delta^n = \{ (\delta_1, \delta_2, \dots, \delta_n) \mid \delta_1, \delta_2, \dots, \delta_n \in \Delta \}$$

by applying elements of the base group $B = H^n$ coordinatewise and allowing elements of K to permute the entries in the same way that K acts on the set Σ. Thus

$$(\delta_1, \delta_2, \dots, \delta_n)^{(h_1, h_2, \dots, h_n)} = (\delta_1^{h_1}, \delta_2^{h_2}, \dots, \delta_n^{h_n})$$

for $(h_1, h_2, \dots, h_n) \in B$ and

$$(\delta_1, \delta_2, \dots, \delta_n)^k = (\delta_{1k^{-1}}, \delta_{2k^{-1}}, \dots, \delta_{nk^{-1}})$$

for $k \in K$. A question on Problem Sheet VII establishes that this action is faithful and determines necessary conditions for the action to be primitive. (It turns out that the conditions listed are sufficient, but the question does not go that far.)

The finite primitive permutations groups of *product type* are primitive subgroups of wreath products acting in the above manner.

Almost simple groups

Finally, we turn to almost simple groups:

Definition 7.25 A finite group G is called *almost simple* if it has a unique minimal normal subgroup N such that N is a non-abelian simple group.

Example 7.26 Let $n \geqslant 5$ and consider the symmetric group S_n acting in the natural way on the set $\Omega = \{1, 2, \dots, n\}$. We observed in Example 7.7 that S_n acts primitively on Ω. This group is almost simple: It has the alternating group $N = A_n$ as its unique minimal normal subgroup.

In the case of an almost simple group G that acts primitively on a set Ω, the O'Nan–Scott Theorem tells us little about the action. These groups arise in Case (iii) of Theorem 7.22 in the situation that the unique minimal normal subgroup N is actually a simple group (rather than a direct product of a number of isomorphic simple groups). Note, however, that Corollary 7.11 tells us that the stabilizer G_ω of a point is maximal in the group G. Consequently, the understanding of primitive groups of almost simple type reduces to the study of finite simple groups S, their automorphism groups $\operatorname{Aut} S$, and the maximal subgroups of subgroups G with $S \leqslant G \leqslant \operatorname{Aut} S$. This all builds on the Classification of Finite Simple Groups and is an area of active research in finite group theory.

Versions

Significant updates to the notes will be listed below. Updates that are merely correcting typographic errors and similar will be indicated by appending a letter to the version number on the front page and will not be listed below.

Version 0.1: New version of lecture notes begun 22nd Aug 2022 in view of change of module title and syllabus.

Version 0.2: Preliminary release version comprising Chapters 1–4.

Version 0.3: Inserted a small example of a stabilizer into Chapter 2. Drafted Chapters 5 & 6. Fixed some typos in earlier chapters.

Version 1.0: First full version of all chapters. Small revisions to the start of Chapter 4.

Version 1.1: Revision (completed 29th Aug 2025) following change of syllabus of *MT4003*.